CURRICULUM

of

Bachelor of Science in Computer Science and Engineering

B.Sc (Engineering) in CSE

For Session 2022-23

Department of Computer Science and Engineering

Faculty of Engineering and Technology

UNDERGRADUATE PROGRAM

JASHORE UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Computer Science and Engineering

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Jashore University of Science and Technology
Department of Computer Science and Engineering

1.0 Introduction

Department of Computer Science and Engineering (CSE) is one of the oldest department at Jashore University of Science and Technology, established in 2008. Currently it is offering B.Sc. (Eng.); MS; PhD Degree. It is the leading educational department in its field which brings together cross-disciplinary, world class researchers to meet the wide range of challenges faced in the field of computer science and engineering. The Department mission is to technically sound man power in the field of computer science and engineering who will be able to lead the nation in the new era of 4th industrial revolution.

1.2 Message from Vice-Chancellor

It is my great pleasure to invite you to explore the Jashore University of Science and Technology (JUST) online through our website. Contribution of science and technology for developing a nation is well known to all. To meet up the diversified demand of people, information and communication technology and biological sciences are playing the key role. University is the most suitable place for education and research. Universities are playing a vital role in building efficient manpower for the development of the country as well as for the global need. With a view to imparting science and technology oriented education in Bangladesh, the Jashore University of Science & Technology was established in 2007 by the Shadhinota Shorok (Independence Road) in Jashore district. The first batch of students was admitted in 2008-2009 session. The honorable Prime Minister of the People's Republic of Bangladesh, Sheikh Hasina, inaugurated the campus of the university on December 27, 2010.

Within a very short period, the JUST already demonstrated its outreach excellence through establishing linkages with the various government and non-government organizations for research, extension and developmental activities. Our honorable Prime Minister has already declared the Vision 2021 for the digitization (advance technology based, economically developed, corruption free) of the country, Bangladesh. I strongly believe that this university will play a significant role to fulfil this vision by producing science and technology based efficient manpower and enlightened citizen. We always adapt the positive changes which is the need of time. Thus, by using this science and technological knowledge we are motivated to boost up the world's technology. As the Vice Chancellor of the Jashore University of Science and Technology, I congratulate all and wish its success in future.

Professor Dr. M. Anwar Hossain
Vice Chancellor
Jashore University of Science and Technology
Jashore, Bangladesh

1.3 Message from Chairman

I warmly welcome you to the Department of Computer Science and Engineering (CSE) at Jashore University of Science and Technology (JUST). JUST started its journey on June 10, 2009 with four departments among which the department of CSE was the first department. Since then, CSE has been widely recognized for its excellent research and teaching capabilities. The department provides an outstanding opportunity to the students to get quality education in Computer Science and Engineering. The graduates from JUST-CSE are heavily recruited by both academia and industry. We are very proud of our graduates and their accomplishments; they are highly skilled IT professionals who pursue successful careers in a wide range of contexts in the competitive global job market.

To know more about the department and its programs, achievements and research activities, I invite you to visit rest of our website at www.cse.just.edu.bd . Last but not the least, ours is a growing family. You are most welcome to be a part of it. The common recruitment procedure is initiated through public advertisement.

Prof. Dr. Syed Md. Galib
Professor and Chairman
email: galib.cse@just.edu.bd

Jashore University of Science and Technology

Department of Computer Science and Engineering

Part A- Introduction

Vision of the University

To emerge as an institute of eminence in the fields of engineering, technology business and management in serving the industry and the nation by empowering students with a high degree of technical, managerial and practical competence.

Mission of the University

1. To strengthen the theoretical, practical and ethical dimensions of the learning process by fostering a culture of research and innovation among faculty members and students.
2. To encourage long-term interaction between the academia and industry through the involvement of the industry in the design of the curriculum and its hands-on implementation
3. To strengthen and mound students in professional, ethical, social and environmental dimensions by encouraging participation in co-curricular and extracurricular activities.

Quality Policy

To provide services of the highest quality both curricular and co-curricular; so that our students can integrate their skills and serve the industry and society equally well at a global level.

Department of Computer Science and Engineering

Vision

Excellence in the state-of-art recent methodologies, techniques and technologies in the field of Computer Science and Engineering through continuous learning and research.

Mission

M1	Providing world class education in the area of computer science at undergraduate and postgraduate level.
M2	Enable the graduates to get their way into higher studies and research across the globe.
M3	Support society by encouraging technology transfer with computers and software.

Program Education Objectives (PEO)

PEO1	Demonstrate proficiency in problem-solving techniques using the computer.
PEO2	Exhibit expertise in high-level programming languages which can be applied to become skilled computer professionals.
PEO3	Show abilities in the analysis of complex problems and the synthesis of solutions to those problems.
PEO4	Demonstrate comprehension of modern software engineering principles.
PEO5	Conduct cutting-edge research in various fields.
PEO6	Apply ethical and social aspects of modern computing technology to the design, development and usage of computing artifacts.
PEO7	Develop leadership and management capabilities.
PEO8	Enhance skills and embrace new computing technologies through self-directed professional development and training or education.

Program Learning Outcome (PLO)

PLO1	An ability to analyze a problem, and identify and define the computing requirements appropriate to its solution
PLO2	An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs

PLO3	An ability to function effectively on teams to accomplish a common goal.
PLO4	An understanding of professional, ethical, legal, security and social issues and responsibilities.
PLO5	An ability to communicate effectively with a range of audiences.
PLO6	An ability to analyze the local and global impact of computing on individuals, organizations, and on society.
PLO7	Recognition of the need for and an ability to engage in continuing professional development.
PLO8	An ability to apply mathematical, physics and chemistry foundations, algorithmic principles, and electrical, electronics and computer science theory in the modeling and design of computer-based systems in a way that demonstrates comprehension of the trade-offs involved in.
PLO9	An ability to apply design and development principles in the construction of software systems of varying complexity.

Generic Skills

GS1	Problem-Solving and Innovation
GS2	Analytical Skill and Creativity
GS3	Teamwork and Leadership
GS4	Research Skill and Awareness
GS5	Adaptation to Tools and Rapid Technological Changes
GS6	Communication and Interpersonal Skills
GS7	Entrepreneurship and Career Management
GS8	Social Responsibility

PEO to Mission Statement Mapping (todo)

PEOs	M1	M2	M3
PEO1	3	1	1
PEO2	2	1	3
PEO3	2	2	2
PEO4	3	1	3
PEO5	1	3	3
PEO6	3	3	1
PEO7	1	1	3
PEO8	3	2	3

Correlation: 3- High, 2-Medium, 1-Low

Mapping of PEOs to PLOs (todo)

PEOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
PEO1	3	3	3	1	1	1	3	1	3
PEO2	2	2	1	1	1	1	1	3	1
PEO3	2	1	1	1	3	2	2	2	3
PEO4	3	1	1	1	2	3	1	3	1
PEO5	2	2	1	1	1	3	3	1	1
PEO6	3	3	3	1	1	1	3	2	2
PEO7	3	2	3	1	1	2	2	1	1
PEO8	3	1	1	2	1	1	1	3	1

Correlation: 3- High, 2-Medium, 1-Low

Part B- Structure

Structure of the curriculum

A. Duration of the program: 4yr (8 bi-yearly semesters)
B. Minimum credit requirements:

BSc (Eng.) in Computer Science and Engineering Program

The students enrolled under BSc (Hons) in CSE degree must complete their studies within 4 (four) years of registration to obtain their degree. The minimum credit to be earned for the degree is 160 (one hundred sixty) along with completion of all core courses. The minimum CGPA requirement is 2.00 to obtain the degree.

C. Course Distribution
   a. General Courses
      i. Arts and Humanities
      ii. Social Science
      iii. Basic science
      iv. Business studies
      v. Ethics
   b. Core courses
      i. Major
      ii. Minor
   c. Elective courses
      i. Major
      ii. Minor
   d. Capstone project/ Thesis/ Internship

D. Curriculum Structure

Fields	Credit Hours	Percentage
Professional Theoretical Subjects (Computer, Software, Networking, Web etc.)	72	44%
Professional Laboratory Subjects	32.5	20%
Field Work/Thesis/Project	17	10%
Related Subjects (both Theory and Lab) (EEE, Others)	9	6%
Mathematics	12	7%
Basic Science (Physics and Chemistry)	4	2%
Humanities and Language	5	3%
Business (Theory and Lab)	8.5	5%
Viva Voce	3	2%
Total	163	100%

Semester/Term/Year/Level wise Courses

First Year First Semester									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 1101	Introduction to Computer Programming	3	0	3	20	8	72	0	100
CSE 1102	Introduction to Computer Programming Lab	0	3	1.5	30	10	0	60	100

CSE 1103	Discrete Mathematics	3	0	3	20	8	72	0	100
EEE 1101	Fundamentals of Electrical and Electronics	3	0	3	20	8	72	0	100
EEE 1102	Fundamentals of Electrical and Electronics Lab	0	3	1.5	30	10	0	60	100
MATH 1101	Differential and Integral Calculus	3	0	3	20	8	72	0	100
CHEM 1101	Chemistry for Engineers	2	0	2	20	8	72	0	100
ENG 1102	English Language Skill Lab	0	2	1	30	10	0	60	100
HUM 1101	Bangladesh Studies	2	0	2	20	8	72	0	100
	Total	**16**	**8**	**20**	**210**	**78**	**432**	**180**	**900**

First Year Second Semester									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 1201	Data Structures	3	0	3	20	8	72	0	100
CSE 1202	Data Structures Lab	0	3	1.5	30	10	0	60	100
CSE 1203	Digital Logic Design	3	0	3	20	8	72	0	100
CSE 1204	Digital Logic Design Lab	0	3	1.5	30	10	0	60	100
CSE 1206	Engineering Drawing	0	2	1	30	10	0	60	100
EEE 1201	Electronic Devices and Circuits	3	0	3	20	8	72	0	100
EEE 1202	Electronic Devices and Circuits Lab	0	3	1.5	30	10	0	60	100
MATH 1201	Differential Equations and Geometry	3	0	3	30	10	0	60	100
PHY 1201	Physics for Engineers	2	0	2	20	8	72	0	100
	Total	**14**	**11**	**19.5**	**230**	**82**	**288**	**300**	**900**

Second Year First Semester

Course code	Course Title	Credit distribution		Total	Marks				Total
		Theory (T)	Lab (Se)		CT	A	SEE	SE	
CSE 2101	Object Oriented Programming	3	0	3	20	8	72	0	100
CSE 2102	Object Oriented Programming Lab	0	3	1.5	30	10	0	60	100
CSE 2103	Algorithm Analysis and Design	3	0	3	20	8	72	0	100
CSE 2104	Algorithm Analysis and Design Lab	0	3	1.5	30	10	0	60	100
CSE 2105	Computer Architecture	3	0	3	20	8	72	0	100
CSE 2107	Numerical Analysis	3	0	3	20	8	72	0	100
CSE 2108	Numerical Analysis Lab	0	3	1.5	30	10	0	60	100
HUM 2101	Sociology and Disaster Management	2	0	2	20	8	72	0	100
MATH 2101	Matrices, Vectors	2	0	2	20	8	72	0	100
	Total	**16**	**9**	**20.5**	**210**	**78**	**432**	**180**	**900**

Second Year Second Semester									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 2201	Data Communication	3	0	3	20	8	72	0	100
CSE 2203	Database Management System	3	0	3	20	8	72	0	100
CSE 2204	Database Management System Lab	0	3	1.5	30	10	0	60	100
CSE 2205	Digital System Design	3	0	3	20	8	72	0	100
CSE 2208	Software Development Project- 1	0	3	1.5	30	10	0	60	100
MATH 2201	Probability and Statistics for Engineers	3	0	3	20	8	72	0	100
MATH 2203	Linear Algebra and Fourier Analysis	2	0	2	20	8	72	0	100
HUM 2201	Business Psychology	2	0	2	20	8	72	0	100
CSE 2200	Viva Voce	0	0	1	0	0	0	100	100

	Total	16	6	20	180	68	432	220	900

Third Year First Semester									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 3101	Operating System and System Programming	3	0	3	20	8	72	0	100
CSE 3102	Operating System and System Programming Lab	0	3	1.5	30	10	0	60	100
CSE 3103	Computer Networks	3	0	3	20	8	72	0	100
CSE 3104	Computer Networks Lab	0	3	1.5	30	10	0	60	100
CSE 3105	Microprocessors and Embedded Systems	3	0	3	20	8	72	0	100
CSE 3106	Assembly Programming and Embedded Systems Lab	0	3	1.5	20	8	72	0	100
CSE 3107	Software Engineering	3	0	3	20	8	72	0	100
CSE 3108	Software Engineering Lab	0	3	1.5	30	10	0	60	100
CSE 3110	Web Development Lab	0	2	1	30	10	0	60	100
HUM 3101	Technology Transfer Policy and Professional Ethics	2	0	2	20	8	72	0	100
	Total	**14**	**15**	**21**	**240**	**88**	**432**	**240**	**1000**

Third Year Second Semester									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 3201	Artificial Intelligence and Machine Learning	3	0	3	20	8	72	0	100

CSE 3202	Artificial Intelligence and Machine Learning Lab	0	3	1.5	30	10	0	60	100
CSE 3203	Compiler Design and Automata Theory	3	0	3	20	8	72	0	100
CSE 3204	Compiler Design and Automata Theory Lab	0	3	1.5	30	10	0	60	100
CSE 3205	Peripherals, Interfacing and IoT	3	0	3	20	8	72	0	100
CSE 3206	Peripherals, Interfacing and IoT Lab	0	3	1.5	30	10	0	60	100
CSE 3208	Software Development Project II	0	3	1.5	30	10	0	60	100
BUS 3202	Business Communication and Technical Writing	0	3	1.5	30	10	0	60	100
CSE 3209	Simulation and Modeling	3	0	3	20	8	72	0	100
AIS 3201	Accounting and Information System	2	0	2	20	8	72	0	100
CSE3200	Viva Voce	0	0	1	0	0	0	100	100
	Total	**14**	**15**	**22.5**	**250**	**90**	**360**	**400**	**1200**

Fourth Year First Semester									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 4101	Computer Graphics and Multimedia	3	0	3	20	8	72	0	100
CSE 4102	Computer Graphics and Multimedia Lab	0	3	1.5	30	10	0	60	100
CSE 4105	Computer Security	3	0	3	20	8	72	0	100
CSE XXXX	Option-I	3	0	3	20	8	72	0	100
CSE XXXX	Option-I Lab	0	3	1.5	30	10	0	60	100

BUS 4101	E-commerce and Management Information System	3	0	3	20	8	72	0	100
CSE 4000	Thesis/Project	0	6	3	30	10	0	60	100
	Total	**12**	**12**	**18**	**170**	**62**	**288**	**180**	**700**

Course code	Course Title	Credit distribution		Total	Marks				Total
		Theory (T)	Lab (Se)		CT	A	SEE	SE	
CSE 4201	Digital Signal and Image Processing	3	0	3	20	8	72	0	100
CSE 4202	Digital Signal and Image Processing Lab	0	3	1.5	30	10	0	60	100
CSE XXXX	Option-II	3	0	3	20	8	72	0	100
CSE 4210	Internship	0	6	3	30	10	0	60	100
CSE 4212	Industrial Tour	0	0	1	30	10	0	60	100
CSE 4000	Thesis/Project	0	6	3	30	10	0	60	100
CSE 4200	Viva Voce	0	0	1	0	0	0	100	100
	Total	**6**	**15**	**15.5**	**160**	**56**	**144**	**340**	**700**

Elective Courses for Option I and Option I Lab (pairwise)									
Course code	Course Title	Credit distribution		Total	Marks				Total
		Theory (T)	Lab (Se)		CT	A	SEE	SE	
CSE 4107	Robotics and Mechatronics	3		3	20	8	72	0	100
CSE 4108	Robotics and Mechatronics Lab	0	3	1.5	30	10	0	60	100
CSE 4109	Data Mining and Warehousing	3		3	20	8	72	0	100
CSE 4110	Data Mining and Warehousing Lab	0	3	1.5	30	10	0	60	100
CSE 4111	Software Testing and Quality Assurance	3		3	20	8	72	0	100
CSE 4112	Software Testing and Quality Assurance Lab	0	3	1.5	30	10	0	60	100

CSE 4113	Digital Forensic Science	3	0	3	20	8	72	0	100
CSE 4114	Digital Forensic Science Lab	0	3	1.5	30	10	0	60	100
CSE 4115	Software Design Pattern and Principles	3	0	3	20	8	72	0	100
CSE 4116	Software Design Pattern and Principles Lab	0	3	1.5	30	10	0	60	100
	Total	**15**	**15**	**22.5**	**250**	**90**	**360**	**300**	**1000**

Elective Courses for Option II									
Course code	**Course Title**	**Credit distribution**		**Total**	**Marks**				**Total**
		Theory (T)	**Lab (Se)**		**CT**	**A**	**SEE**	**SE**	
CSE 4221	Information Control and Cyber Security	3	0	3	20	8	72	0	100
CSE 4223	Fault Tolerant Analysis	3	0	3	20	8	72	0	100
CSE 4225	Introduction to DNA Computing	3	0	3	20	8	72	0	100
CSE 4227	Software Project Management and Maintenance	3	0	3	20	8	72	0	100
CSE 4229	Distributed System and Cloud Computing	3	0	3	20	8	72	0	100
CSE 4231	Parallel and Vector Computing	3	0	3	20	8	72	0	100
CSE 4233	Cryptography	3	0	3	20	8	72	0	100
	Total	**21**	**0**	**21**	**140**	**56**	**504**	**0**	**700**

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Introduction to Computer Programming

Part A- Introduction

. **Course code: CSE1101**

II. **Credit: 3**

1. Course Summary

This is an introductory course on computer programming language. All the basic programming details ranging from variable declaration to exception handling are covered in this course. Primarily it is covered in C.

0. Course Objectives

1. Receive a basic knowledge of programming and the ability of reading with understanding programs.
2. Able to solve basic programming problems using a variety of skills and strategies.
3. Examine working programs to identify their structures and apply appropriate techniques to create entry-level programs

0. Course Learning Outcomes

1. Devise the primary idea to computer programming languages and their applicability
2. Learn to solve computing problems using programming languages.
3. Design the functions to understand reusability and efficiency
4. Explore the Read and Write operations from different I/O options
5. Utilize different data structures

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Programming, Programming Language, Compiler, IDE	Learn how to begin journey into programming	CLO1	Lecture	Quiz
Week 2	Header File, C Printf and Scanf Functions Variables Data Types Constants Comments Operators	Learn C Programming Basics	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 3	If Statement in C If Else Statement in C If and Else If Ladder Statement Switch Statement	Apply control structures	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5	Function In C Introduction with Example Function: Example with Parameters Function : Call by Value Function : Call by Reference Recursion in C	Apply function and recursion	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 6	For loop While loop do-while loop	Implement loop	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8,9	Array in C Array example (Input and Output in an Array) 2D Array example	Evaluate array, string, pointers and memory allocation	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	Passing array to function Pointer to Pointer Pointer Arithmetic Dynamic Memory Allocation Malloc String in C C Gets and Puts Method C String Function C Math Function				
Week 10	User defined data type: structures, unions, enumeration;	Explain user defined data types	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11	Input and output: standard input and output, formatted input and output	Explain input-output	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 12, 13	File Handling Write Data to File File Handling Read Data to File C Preprocessor Directive with Example	Apply file I/O	CLO1, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 14	Variable length argument list; Command line parameters	Explain and apply argument list	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.

Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	**1.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Teach yourself C by Herbert Schildt
2. C: The Complete Reference by Herbert Schildt

Reference Books

1. Computer fundamentals and programming by Reema Thareja
2. C in a Nutshell: The Definitive Reference by Peter Prinz
3. Pointer in C Y. Kanitkar
4. Let us C by Y. Kanitkar
5. The C programming language by Kernighan & Ritchie

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Introduction to Computer Programming Lab

Part A- Introduction

.Course code:
CSE1102

II. Credit: 1.5

1. Course Summary

This is an introductory course on computer programming language. All the basic programming details ranging from variable declaration to exception handling are covered in practical lab sessions in this course. Primarily it is covered in C.

0. Course Objectives

1. Apply the basic knowledge of programming and the ability of reading with understanding coding materials.
2. Solve basic programming problems using a variety of skills and strategies.
3. Illustrate working programs to identify their structures and apply appropriate techniques to create entry-level programs

0. Course Learning Outcomes

1. Solve computing problems using programming languages.
2. Design the functions to understand reusability and efficiency
3. Devise the Read and Write operations from different I/O options
4. Implement different data structures

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Programming, Programming Language, Compiler, IDE	Learn how to begin journey into programming	CLO1	Lecture, Demonstration	Class Participation
Week 2	Header File, C Printf and Scanf Functions Variables Data Types Constants Comments Operators	Learn C Programming Basics	CLO1, CLO2	Lecture Exercise Demonstration	Assignment Practical Exam
Week 3	If Statement in C If Else Statement in C If and Else If Ladder Statement Switch Statement	Apply control structures	CLO1	Lecture Exercise Demonstration	Assignment Practical Exam
Week 4, 5	Function In C Introduction with Example Function: Example with Parameters Function : Call by Value Function : Call by Reference Recursion in C	Apply function and recursion	CLO3	Lecture Exercise Demonstration	Assignment Practical Exam
Week 6	For loop While loop do-while loop	Implement loop	CLO1	Lecture Exercise Demonstration	Assignment Practical Exam
Week 7, 8,9	Array in C Array example (Input and Output in an Array) 2D Array example	Evaluate array, string, pointers and memory allocation	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Assignment Practical Exam

	Passing array to function Pointer to Pointer Pointer Arithmetic Dynamic Memory Allocation Malloc String in C C Gets and Puts Method C String Function C Math Function				
Week 10	User defined data type: structures, unions, enumeration;	Explain user defined data types	CLO5	Lecture Exercise Demonstration	Assignment Practical Exam
Week 11	Input and output: standard input and output, formatted input and output	Explain input-output	CLO4	Lecture Exercise Demonstration	Assignment Practical Exam
Week 12, 13	File Handling Write Data to File File Handling Read Data to File C Preprocessor Directive with Example	Apply file I/O	CLO1, CLO4	Lecture Exercise Demonstration	Assignment Practical Exam
Week 14	Variable length argument list; Command line parameters	Explain and apply argument list	CLO3	Lecture Exercise Demonstration	Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Attendance(Class participation)	Students should regularly attend the classes and actively participate in the class discussion.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.

Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Includes set of programming problems to solve within a timeline

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**1.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**

CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Teach yourself C by Herbert Schildt
2. C: The Complete Reference by Herbert Schildt

Reference Books

1. Computer fundamentals and programming by Reema Thareja
2. C in a Nutshell: The Definitive Reference by Peter Prinz
3. Pointer in C Y. Kanitkar
4. Let us C by Y. Kanitkar
5. The C programming language by Kernighan & Ritchie

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Discrete Mathematics

Part A- Introduction

.Course code:
CSE1103

II. Credit: 3

1. Course Summary

One of the fundamental courses of computer science and engineering. Discrete math covers the logic and their applications which are mapped further into the general mathematics, probability and graph computation problems.

0. Course Objectives

1. Explain mathematical logic and set theory deeply.
2. Be able to apply number theory, functions and relations.
3. Have an in depth understanding of and be able to construct graphs and trees.

0. Course Learning Outcomes

1. Demonstrate logic and logical operations.
2. Measure proofing procedures
3. Illustrate number principals and their applications
4. Organize graph related terminologies
5. Examine group theory and isomorphism

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1, 2	Introduction to logic, proposition and tatology	Describe logic, Identify logical equivalence, Apply logical operations	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 3	Formal proofs and their applications	Evaluate equations	CLO1, CLO2	Lecture, Exercise	Quiz Assignment Exam
Week 4	Predicate and quantifiers, The Universe of Discourse. Rules of Inference: Universal Specification & Generalization.	Convert the natural sentences into logical notions	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 5	Set theory, functions and sequences	Examine set and functional operations	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 6	Elements of Number Theory: Modular Arithmetic, and The Euclidean Algorithm	Apply number theory and modular arithmetics	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 7	The basic of counting and pigeonhole principles, Introduction to discrete probability	Describe counting and probability	CLO3	Lecture Exercise	Quiz Assignment Exam
Week 8	Bayes theorem, relations and their properties	Find the probability of complex events Discover relations from sets	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 9, 10	Introduction to graph and related terminologies	Learn graph and its properties	CLO4, CLO5	Lecture Exercise	Quiz Assignment Exam
Week 11	Paths, Circuits and their applications, Euler path & circuit, Hamilton path and circuit, Dijkstra's algorithm for shortest path	Examine the graph applications	CLO5	Lecture Exercise	Quiz Assignment Exam
Week 12	Tree and related terminologies	Learn tree in graph	CLO4	Lecture Exercise	Quiz Assignment Exam

Week 13, 14	Tree applications, tree traversal, minimum spanning tree	Apply tree in real life applications	CLO4, CLO5	Lecture Exercise	Quiz Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	2			
Apply	5			
Analyze	5			
Evaluate	5			
Create	1			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05

Understand	15
Apply	20
Analyze	17
Evaluate	10
Create	5

Mapping of CLOs to PLOs

CLOs	**1.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	1	0	0	0	1	0	1	0
CLO4	1	1	0	0	0	0	0	0	0
CLO5	1	1	0	0	0	0	0	0	0

Part D- Resources

Textbooks

1. Discrete Mathematics and its Applications by K. H. Rosen
2. Schaum's Outline of Discrete Mathematics

Reference Books

1. Discrete Mathematical Structures for Computer Science by Koleman& Busby
2. Discrete Mathematical Structures with Applications to Computer Science by Trembley & Manohar

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Fundamentals of Electrical and Electronics

Part A- Introduction

I.Course code:
EEE1101

III. Credit: 3

2. Course Summary

To be a computer engineer, one needs to study and apply electricity and electromagnetism in different electrical applications

1. Course Objectives

1. To understand about concept of current, voltage and power.
2. To learn and study of DC network theorems and solve circuits.
3. To know about magnetic circuits and different magnetic theorem.
4. To know about ac fundamentals and solve different ac circuits.

1. Course Learning Outcomes

CLO1) Demonstrate logic and logical operations.
CLO2) Measure proofing procedures
CLO3) Illustrate number principals and their applications
CLO4) Organize graph related terminologies
CLO5) Examine group theory and isomorphism

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2	Fundamental electrical concepts: Different measuring units	Describe logic, Identify logical equivalence, Apply logical operations	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 3	DC voltage, current, resistance and power; Series, networks definitions, mesh and node circuit analysis, reduction of a complicated network, conversion between T and π section.	Evaluate equations	CLO1, CLO2	Lecture, Exercise	Quiz Assignment Exam
Week 4	Networks transformations: Equivalent circuit, Superposition theorem, the reciprocity theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem.	Convert the natural sentences into logical notions	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 5	Introduction to magnetic Circuit: Transformer working principle, construction, and maintenance, transformer's e.m.f equations	Examine set and functional operations	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 6	Transformer regulation and efficiency, different types of transformer.	Apply number theory and modular arithmetics	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 7	Electromagnetic Field: Maxwell's Equation, Electromagnetic wave equation and propagation, Pointing vector,	Describe counting and probability	CLO3	Lecture Exercise	Quiz Assignment Exam
Week 8	Faraday's laws of electromagnetic induction, Lenz's Law, Motional e.m.f, Eddy current. Self and Mutual Inductance.	Find the probability of complex events Discover relations from sets	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 9, 10	Capacitor and Capacitance: Capacitance, Parallel plate	Learn graph and its properties	CLO4, CLO5	Lecture Exercise	Quiz

	capacitor, cylindrical capacitor, spherical capacitor,				Assignment Exam
Week 11	capacitors in series & parallel, energy stored in a capacitor, transformers.	Examine the graph applications	CLO5	Lecture Exercise	Quiz Assignment Exam
Week 12	A.C Circuits: Instantaneous and r.m.s. values of current, voltage and average power,	Learn tree in graph	CLO4	Lecture Exercise	Quiz Assignment Exam
Week 13, 14	Use of complex quantities in AC circuits, resonant circuits, Q value and band width, frequency response.	Apply tree in real life applications	CLO4, CLO5	Lecture Exercise	Quiz Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	2			
Apply	5			
Analyze	5			

Evaluate	5			
Create	1			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	17
Evaluate	10
Create	5

Mapping of CLOs to PLOs

CLOs	**2.**	**1.**	**1.**	**1.**	**1.**	**1.**	**1.**	**1.**	**1.**
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	1	0	0	0	1	0	1	0
CLO4	1	1	0	0	0	0	0	0	0
CLO5	1	1	0	0	0	0	0	0	0

Part D- Resources

Textbooks

1. B.L. Theraja — A text book of Electrical Technology, Volume: I.
2. V. K. Mehta — Principles of Electrical Engineering and Electronics.
3. G. F. Corcoran — Alternating Current Circuits.

4. Corson and Lorrain Introduction to Electromagnetic Field and Waves

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Fundamentals of Electrical and Electronics Lab

Part A- Introduction

I.Course code:
EEE1102

IV. Credit: 1.5

3. Course Summary

To be a computer engineer, one needs to study and apply electricity and electromagnetism in different electrical applications

2. Course Objectives

1. To understand about concept of current, voltage and power.
2. To learn and study of DC network theorems and solve circuits.
3. To know about magnetic circuits and different magnetic theorem.
4. To know about ac fundamentals and solve different ac circuits.

2. Course Learning Outcomes

CLO1) Apply logic and logical operations.
CLO2) Apply proofing procedures
CLO3) Examine number principals and their applications
CLO4) Demonstrate graph related terminologies
CLO5) Examine group theory and isomorphism

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2	Fundamental electrical concepts: Different measuring units	Describe logic, Identify logical equivalence, Apply logical operations	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 3	DC voltage, current, resistance and power; Series, networks definitions, mesh and node circuit analysis, reduction of a complicated network, conversion between T and π section.	Evaluate equations	CLO1, CLO2	Lecture, Exercise	Quiz Assignment Exam
Week 4	Networks transformations: Equivalent circuit, Superposition theorem, the reciprocity theorem, Thevenin's theorem, Norton's theorem, Maximum power transfer theorem.	Convert the natural sentences into logical notions	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 5	Introduction to magnetic Circuit: Transformer working principle, construction, and maintenance, transformer's e.m.f equations	Examine set and functional operations	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 6	Transformer regulation and efficiency, different types of transformer.	Apply number theory and modular arithmetics	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 7	Electromagnetic Field: Maxwell's Equation, Electromagnetic wave equation and propagation, Pointing vector,	Describe counting and probability	CLO3	Lecture Exercise	Quiz Assignment Exam
Week 8	Faraday's laws of electromagnetic induction, Lenz's Law, Motional e.m.f, Eddy current. Self and Mutual Inductance.	Find the probability of complex events Discover relations from sets	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 9, 10	Capacitor and Capacitance: Capacitance, Parallel plate	Learn graph and its properties	CLO4, CLO5	Lecture Exercise	Quiz

	capacitor, cylindrical capacitor, spherical capacitor,				Assignment Exam
Week 11	capacitors in series & parallel, energy stored in a capacitor, transformers.	Examine the graph applications	CLO5	Lecture Exercise	Quiz Assignment Exam
Week 12	A.C Circuits: Instantaneous and r.m.s. values of current, voltage and average power,	Learn tree in graph	CLO4	Lecture Exercise	Quiz Assignment Exam
Week 13, 14	Use of complex quantities in AC circuits, resonant circuits, Q value and band width, frequency response.	Apply tree in real life applications	CLO4, CLO5	Lecture Exercise	Quiz Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2		
Understand	5		
Apply	13		
Analyze	2		

Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**3.**	**2.**	**2.**	**2.**	**2.**	**2.**	**2.**	**2.**	**2.**
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	1	0	0	0	1	0	1	0
CLO4	1	1	0	0	0	0	0	0	0
CLO5	1	1	0	0	0	0	0	0	0

Part D- Resources

Textbooks

5. B.L. Theraja — A text book of Electrical Technology, Volume: I.
6. V. K. Mehta — Principles of Electrical Engineering and Electronics.
7. G. F. Corcoran — Alternating Current Circuits.

8. Corson and Lorrain   Introduction to Electromagnetic Field and Waves

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Differential and Integral Calculus

Part A- Introduction

Course code: MATH 1101

Credit: 3

1. **Course Summary**

To be a computer Engineer one has to have sound knowledge about function, limit, Derivatives, applications of differentiation, integrals and application of integration

2. **Course Objectives**

1. To Learn about various limit problems algebraically and graphically.
2. To Examine and Apply the continuity and differentiability of various types of function.
3. To gain knowledge about Integration and application of Integration.

3. **Course Learning Outcomes**

CLO1) Learn differential mathematics
CLO2) Measure differential mathematics
CLO3) Illustrate differential mathematics
CLO4) Learn Integral mathematics
CLO5) Measure Integral mathematics

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1, 2	Graphing Functions, Mathematical Models and Commonly used Functions (Linear, Polynomial, Power),	Describe logic, Identify logical equivalence, Apply logical operations	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 3	Mathematical Models and Commonly Used Functions (Algebraic, Trigonometric, Exponential, and Logarithmic Functions),	Evaluate equations	CLO1, CLO2	Lecture, Exercise	Quiz Assignment Exam
Week 4	Derivatives and Rate of Change, Derivatives as Functions,	Convert the natural sentences into logical notions	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 5	Rates of Change in Natural and Social Sciences, Exponential Growth and Decay, Linear Approximation and Differentials,	Examine set and functional operations	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 6	Finding Minimum and Maximum Value of Functions and the first and Second Derivative Tests, Indeterminate Forms and L'Hospital's Rule, Curve Sketching.	Apply number theory and modular arithmetics	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 7	Riemann Sum and Definite Integrals, Properties of Integrals, Fundamental Theorem of Calculus, Antiderivative and Indefinite Integral, Net Change Theorem, Substitution Rule.	Describe counting and probability	CLO3	Lecture Exercise	Quiz Assignment Exam
Week 8	Theorem of Calculus, Antiderivative and Indefinite Integral, Net Change Theorem, Substitution Rule.	Find the probability of complex events Discover relations from sets	CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 9, 10	Transformations (Scaling, Reflection, Composition), Inverse of Functions, Growth of Functions.	Learn graph and its properties	CLO4, CLO5	Lecture Exercise	Quiz Assignment Exam

Week 11	Differentiability of Functions, Rules and Techniques of Differentiation.	Examine the graph applications	CLO5	Lecture Exercise	Quiz Assignment Exam
Week 12	Finding Area between Curves, Volumes, Volumes by Cylindrical Shells,	Learn tree in graph	CLO4	Lecture Exercise	Quiz Assignment Exam
Week 13, 14	Average Value of a Function, Mean Value Theorem for Integrals.	Apply tree in real life applications	CLO4, CLO5	Lecture Exercise	Quiz Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	2			
Apply	5			
Analyze	5			
Evaluate	5			
Create	1			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	17
Evaluate	10
Create	5

Mapping of CLOs to PLOs

CLOs	4.	3.	3.	3.	3.	3.	3.	3.	3.
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	1	0	0	0	1	0	1	0
CLO4	1	1	0	0	0	0	0	0	0
CLO5	1	1	0	0	0	0	0	0	0

Part D- Resources

Textbooks

1. Howard Anton Calculus
2. Thomas Finney Calculus and Analytical Geometry
3. E.R. Swokowski Calculus with Analytic Geometry
4. M.R. Spiegel Advanced Calculus
5. S.P. Gordon Calculus and the Computer.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Chemistry for Engineers

Part A- Introduction

.Course code:
CHEM1101

II. Credit: 3

1. Course Summary

A computer scientist or engineer may choose to be expert in cheminformatics or computational chemistry. Therefore, he or she should have a sound knowledge and deep understanding of chemistry.

0. Course Objectives

1. Know the crystal symmetry.
2. Understand the characteristics of solution.
3. Explain chemical equilibrium.
4. Learn basics of nuclear chemistry.

0. Course Learning Outcomes

CLO1) Apply the fundamental concepts of Chemistry.
CLO2) Design solution for chemical problems.
CLO3) Analyze chemical equilibrium.
CLO4) Apply nuclear chemistry.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Different methods for the determination of structure; Structures of the metallic elements and certain	Learn chemistry basisc	CLO 1	Lecture, Demonstration, handouts	Quiz Participation Exam

	compounds with three dimensional lattices; Defects in solid; Semiconductors: Structures of Si, Ge, B, N, P, In. Types of Semiconductor, Electronic and band theory				
Week 2 ,3	Type of solutions, Units of concentration. Solution of gas in liquid,	Learn chemistry basisc	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 4, 5, 6	Henry's law. Solution of solid in liquid, solubility curve. Distribution law and its application. Solvent extraction, Colligative properties of dilute solution.	Identify and explain solution	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 7, 8	Law of mass-action, Chemical equilibrium and Equilibrium Constants, Application of law of mass-action	Identify and explain solution	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 9, 10	Homogeneous and Heterogeneous Equilibrium, Le-Chatelier Principle, Applications of principle of mobile equilibrium to reaction of industrial importance.	Explain chemical equilibrium	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 11, 12	Radioactivity, Patterns of Nuclear Stability, Nuclear Transmutations,	Explain chemical equilibrium	CLO 3	Lecture, Demonstration	Quiz Participation Exam
Week 13	Rates of Radioactive Decay, Detection of Radioactivity, Energy Changes in Nuclear Reactions.	Learn nuclear chemistry	CL03 CLO 4	Lecture, Demonstration, handouts	Quiz Participation Exam

Week 14	Nuclear Fission, Nuclear Fusion, Isotopes, Isobar, Isomers, Methods of Separation of Isotopes, Applications of Radioisotopes, Biological Effects of Radiation.	Learn nuclear chemistry	CLO 4	Lecture, Demonstration	Quiz Participation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assign ment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom’s Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	1.	0.	0.	0.	0.	0.	0.	0.	0.
CLO1	1	0	0	0	0	0	0	0	0
CLO2	1	1	1	0	1	0	0	0	1
CLO3	1	1	1	0	1	0	0	0	1
CLO4	1	1	0	0	0	0	0	0	1

<u>Part D- Resources</u>

Textbooks

Authors	Book Name
1. G.M. Barrow	Physical Chemistry
2. M.M. Haque and M.A.Nawab	Principles of Physical Chemistry
3. B. S. Bhal and G. D. Tuli	Essential of Physical Chemistry
4. S. Z. Haider	Introduction to Modern Inorganic Chemistry
5. R.D. Madan	Modern Inorganic Chemistry
6. S. Gilreath	Fundamental Concepts of Inorganic Chemistry
7. R. M. Felder and R. W. Rousseau	Elementary Principles of Chemical Engineering

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: English language Skills Lab

Part A- Introduction

I. Course code:
ENG1102

II. Credit: 1

1. Course Summary

To be a global IT expert, a student must have the communication skill in an international language and English is the most and widely used language. Therefore, learning Communicative English is a must for a CSE student.

2. Course Objectives

1. Talk about themselves, their families and their weekly schedules.
2. Memorize and use various set phrases for use in all of their classes.
3. Ask questions of other members of the class in English to help carry on a conversation and to expand on a topic.
4. Identify various parts of an English newspaper and predict the topic of the article based on the headline.
5. Make a video of themselves and reflect on their improvement over the course of the semester.
6. Understand the importance of their learning portfolio and demonstrate their knowhow on how to make one for their courses.

1. Course Learning Outcomes

CLO1) Apply the fundamental concepts of English.
CLO2) Learn basic grammar for engineers.
CLO3) Identify article.
CLO4) Understand the importance of learning portfolio.

Part B- Lesson Plan

Course Details Learning plan

Timeline	**Topics /contents**	**Learning Outcomes**	**Mapped CLOs**	**Teaching Strategies**	**Assessment Strategies**
Week 1	Reading Comprehension, vocabulary building	Recognizing communication skills(Reading	CLO 1	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 2 ,3	Précis/summarizing, development of writing skill,	Summarize	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 4, 5, 6	paragraph development, report writing, letter writing (formal/informal), newspaper article writing.	Paragraph writing	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 7, 8	Parts of Speech, Subject-Verb agreement, Articles,	Learn POS	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 9, 10	Tense form, Phrase & Clause, Making Questions, Gerund and Participle,	Learn Tense	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 11, 12	Appropriate preposition, Word formation processes, Verbs & adjectives, Using dictionary,	Place appropriate Preposition	CLO 3	Lecture, Demonstration	Quiz Participation Exam
Week 13	Real life word associated with science & technology,	Place appropriate Preposition	CL03 CLO 4	Lecture, Demonstration, handouts	Quiz Participation Exam

Week 14	Places, clothes, foods, shopping, family, hobbies, health, entertainment, transport and education.	Place appropriate Preposition	CLO 4	Lecture, Demonstration	Quiz Participation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**

Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**2.**	**1.**	**1.**	**1.**	**1.**	**1.**	**1.**	**1.**	**1.**
CLO1	1	0	0	0	0	0	0	0	0
CLO2	1	1	1	0	1	0	0	0	1
CLO3	1	1	1	0	1	0	0	0	1
CLO4	1	1	0	0	0	0	0	0	1

Part D- Resources

Textbooks

	Authors	**Book Name**
1.	Wren & Martin	High school English Grammar" S. Chand & Company
2.	Raymond Murphy	Intermediate English Grammar
3.	Jones Leo	Communicative Grammar Practice
4.	John Eastwood	Oxford Practice Grammar
5.	Sadruddin Ahmed	Learning English The Easy Way.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Bangladesh Studies

Part A- Introduction

.Course code:
HUM1101

II. Credit: 2

1. **Course Summary**
   Each and every individual of a country should have adequate knowledge about the history, culture, economical and sociological aspects of his own country. Bangladesh Studies will provide students an overall view of the Land and People, Emergence of Sovereign State, Political Culture and Administration, Natural Resources, Economy and Sociology of Bangladesh

0. Course Objectives

1. Acquire knowledge about the economy and sociology of Bangladesh.
2. Understand the history and culture of Bangladesh.
3. Gather clear concept about political culture and administration of Bangladesh

0. Course Learning Outcomes

CLO1) Devise the knowledge of Bangladesh
CLO2) Learn the history of Bangladesh
CLO3) Explore the life of Bangobandhu Sheikh Mujibur Rahman
CLO4) Learn the political and cultural administration of Bangladesh.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1	A brief history, Culture, Language and Religion of the land; Population,	Land and People of Bangladesh	CLO 1	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 2 ,3	Ethnological origin and Occupation/Profession of the people.	Land and People of Bangladesh	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 4, 5, 6	Economic factors; Political factors: Language movement (1952), Six Point Movement (1966), Mass upsurge (1969); Bangabandhu and Independence of Bangladesh (1971).	Knowledge about Emergence of Sovereign State of Bangladesh	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 7, 8	Introduction to the Bangladesh Constitution; Forms of Government since independence; Democracy; Administrative System in Bangladesh.	Knowledge about Emergence of Sovereign State of Bangladesh	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 9, 10	Introduction to Bangladesh economy, Major economic Sectors: Agriculture, Industry and Services; Role of Women in national economy.	Knowledge about Political Culture and Administration of Bangladesh	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 11, 12	Scope, Social evolution and techniques of production, Culture and civilization, Social structure of Bangladesh. Population and world resources. Oriental and Occidental societies, History of Nation's father	Knowledge about Political Culture and Administration of Bangladesh	CLO 3	Lecture, Demonstration	Quiz Participation Exam
Week 13	Economics of development and planning, basic concept-saving, investment, GNP, NNP. Fiscal policy, monetary policy and trade policy, some planning tools-capital output ratio, input analysis.	Knowledge about Sociology of Bangladesh	CLO 4	Lecture, Demonstration, handouts	Quiz Participation Exam

Week 14	Industrial revolution. Family - Urbanization and industrialization, Urban Ecology, Co-operative and socialist movements, rural sociology.	Knowledge about Sociology of Bangladesh	CLO 4	Lecture, Demonstration	Quiz Participation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

CIE- Continuous Internal Evaluation (28 Marks):

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	1.	0.	0.	0.	0.	0.	0.	0.	0.
CLO1	0	0	0	0	0	0	1	0	0
CLO2	0	1	1	0	1	0	1	0	1
CLO3	0	1	1	0	1	0	1	0	1
CLO4	0	1	0	0	0	0	1	0	1

<u>Part D- Resources</u>

Textbooks

1. Banglapedia. National Encyclopedia, Asiatic Society of Bangladesh, Dhaka
2. Bangladesh at the Crossroads. University Press Ltd
3. Bangladesh: A Political History since Independence. London: IB Taurus

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Data Structures

Part A- Introduction

I. **Course code: CSE 1201**

II. **Credit: 3**

1.Course Summary

To become a successful computer professional, one needs to have in depth knowledge of data structures to apply them in problem solving and algorithms to analyze different solutions to problems.

2.Course Objectives

1. Apply different data structures to problem solving
2. Explain different algorithms of solutions to problems.
3. Analyze different problem solutions

3.Course Learning Outcomes

1. Implements basic data structures such as Stacks, Queues, Graph and Trees.
2. Uses Big 'O' notation to express algorithmic running time.
3. Describes and analyzes elementary sorting algorithms such as Selection sort, Bubble sort, Insertion sort, and Shell sort.
4. Understands and restates the fundamentals of basic data structures.
5. Designs and analyzes simple algorithms.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1 **Week 2** **Week 3**	**Stacks, Queues and Recursion:** Fundamentals,Different types of stacks and queues: Circular, dequeues etc.; evaluation of expressions, multiple stacks and queues; Recursion: Direct and indirect recursion, depth of recursion; Simulation of Recursion: Removal of Recursion; Towers of Hanoi.	Apply stacks, queues and recursion; Differentiate between stacks and queues; Evaluate expressions using stacks; Analyze recursive functions;	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Elements of Graphs and Trees:** Graph Terminology, Paths and Circuits, Connectedness, Matrix Representation of Graph and Isomorphism of graphs. Trees, Rooted trees, Path Lengths in Rooted Trees.	Construct graphs and trees; Differentiate between graphs and trees	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	**Linked Lists:** Single linked lists, Linked stacks and queues, the storage pool, polynomial addition, equivalence relations, sparse matrices, doubly linked lists and dynamic storage management, generalized lists, garbage collection and compaction.	Construct linked lists; Apply linked lists to stacks and queues; Analyze memory allocation;	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9 **Week 10**	**Extended binary trees:** 2-trees, internal and external path lengths, Huffman codes/algorithm; Threaded binary trees, binary tree representation of trees; Application of Trees: Set representation, decision trees, game trees; Counting binary trees.	Construct and apply extended binary trees	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 11 **Week 12**	**Sorting:** Searching, bubble sort, shell sort, insertion sort, selection sort, quick sort, heap sort, 2-way merge sort, sorting on several keys, practical considerations of internal sorting. searching, hash techniques.	Apply sorting algorithms; Distinguish among sorting algorithms;	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	**Algorithms:** Techniques for analysis of algorithms; Algorithmic Techniques: divide-and-conquer, greedy method, dynamic programming, backtracking, branch and bound; Flow algorithms. Topological sorting; Connected components; spanning trees; Shortest paths; Algebraic simplification and transformations; Lower bound theory; NP-completeness; NP-hard and NP-complete problems; Approximation Algorithms; Introduction to parallel algorithms.	Apply algorithmic techniques; Analyze algorithms;	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE

Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

1. Theory and Problem of Data Structures by S. Lipschutz
2. Data Structure by E. Horowitz

Reference Books

1. The Art of Computer Programming, Vol. 1, Fundamental Algorithms by D. E. Knuth
2. The Art of Computer Programming, Vol. 2, Fundamental Algorithms by D. E. Knuth
3. The Art of Computer Programming, Vol. 3, Fundamental Algorithms by D. E. Knuth

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Data Structures Lab

Part A- Introduction

I. **Course code: CSE 1202**

II. **Credit: 1.5**

1. Course Summary

To become a successful computer professional, one needs to have in depth knowledge of data structures to apply them in problem solving and algorithms to analyze different solutions to problems.

2. Course Objectives

1. Apply different data structures to problem solving
2. Explain different algorithms of solutions to problems.
3. Analyze different problem solutions

3. Course Learning Outcomes

CLO1. Implements basic data structures such as Stacks, Queues, Graph and Trees.

CLO2. Uses Big 'O' notation to express algorithmic running time.

CLO3. Describes and analyzes elementary sorting algorithms such as Selection sort, Bubble sort, Insertion sort, and Shell sort.

CLO4. Understands and restates the fundamentals of basic data structures.

CLO5. Designs and analyzes simple algorithms.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	**Stacks, Queues and Recursion:** Fundamentals,Different types of stacks and queues: Circular, dequeues etc.; evaluation of expressions, multiple stacks and queues; Recursion: Direct and indirect recursion, depth of recursion; Simulation of Recursion: Removal of Recursion; Towers of Hanoi.	Apply stacks, queues and recursion; Differentiate between stacks and queues; Evaluate expressions using stacks; Analyze recursive functions;	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Elements of Graphs and Trees:** Graph Terminology, Paths and Circuits, Connectedness, Matrix Representation of Graph and Isomorphism of graphs. Trees, Rooted trees, Path Lengths in Rooted Trees.	Construct graphs and trees; Differentiate between graphs and trees	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7	**Linked Lists:** Single linked lists, Linked stacks and queues, the storage pool, polynomial addition, equivalence	Construct linked lists; Apply linked lists to stacks and queues;	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 8	relations, sparse matrices, doubly linked lists and dynamic storage management, generalized lists, garbage collection and compaction.	Analyze memory allocation;			
Week 9 **Week 10**	**Extended binary trees:** 2-trees, internal and external path lengths, Huffman codes/algorithm; Threaded binary trees, binary tree representation of trees; Application of Trees: Set representation, decision trees, game trees; Counting binary trees.	Construct and apply extended binary trees	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	**Sorting:** Searching, bubble sort, shell sort, insertion sort, selection sort, quick sort, heap sort, 2-way merge sort, sorting on several keys, practical considerations of internal sorting. searching, hash techniques.	Apply sorting algorithms; Distinguish among sorting algorithms;	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	**Algorithms:** Techniques for analysis of algorithms; Algorithmic Techniques: divide-and-conquer, greedy method, dynamic programming, backtracking, branch and bound; Flow	Apply algorithmic techniques; Analyze algorithms;	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	algorithms. Topological sorting; Connected components; spanning trees; Shortest paths; Algebraic simplification and transformations; Lower bound theory; NP-completeness; NP-hard and NP-complete problems; Approximation Algorithms; Introduction to parallel algorithms.				

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0

CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

1. Theory and Problem of Data Structures by S. Lipschutz
2. Data Structure by E. Horowitz

Reference Books

1. The Art of Computer Programming, Vol. 1, Fundamental Algorithms by D. E. Knuth
2. The Art of Computer Programming, Vol. 2, Fundamental Algorithms by D. E. Knuth
3. The Art of Computer Programming, Vol. 3, Fundamental Algorithms by D. E. Knuth

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

<u>Course Outline: Digital Logic Design</u>

Part A- Introduction

I. **Course code: CSE 1203**

II. **Credit: 3**

1. Course Summary

A computer engineer needs to know about number system, logic design and the basic building blocks used in digital systems, in particular digital computers.

2. Course Objectives

1. To learn about basic concept on number system, universal gates and truth tables.
2. To know about Boolean function and De-Morgan, canonical forms and minimization techniques
3. To learn about combinational and sequential circuits and basic flip flops
4. To know about synchronous and asynchronous counters

3. Course Learning Outcomes

CLO1. Identify and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic components of combinational and sequential circuits

CLO2. Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods

CLO3. Analyze small-scale combinational and sequential digital circuits

CLO4. Design small-scale combinational and synchronous sequential digital circuit using Boolean algebra and K-maps

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	Number systems and codes. Digital logic: Boolean algebra, De-Morgan's law, Logic gates and their truth tables.	Explain number systems Apply De-Morgan law for the Boolean function State and explain De-Morgan	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	Canonical forms, combinational logic circuits, minimization techniques.	Evaluate canonical forms Apply minimization techniques Explain combinational logic circuits	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	Arithmetic and data handling logic circuit, decoders and encoders, Multiplexers and De-multiplexers.	Determine decoder and encoder function Differentiate MUX and DEMUX Draw data handling logic circuit	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 9 **Week 10**	Combinational Circuit design, Flip-flops, race around problems.	Explain flip flops and race problems	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	Counters: Asynchronous and Synchronous counters and their applications. Synchronous and asynchronous logic design: state diagram, Mealy and Moore machine.	Design synchronous and asynchronous circuits Apply state diagram for logic circuits, State and explain Mealy and Moore machine	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	State minimization and assignments. Pulse mode logic, Fundamental mode logic design.	Explain state minimization and assignments Demonstrate pulse mode logic and mode logic design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15

Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Digital logic and Computer Design by M. Morris Mano
2. Digital System Analysis by Tocci

Reference Books

1. Principles of Electronic by D. E. Knuth
2. Electronic Devices and Circuit Theory by R.L. Boylestad

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Digital Logic Design Lab

Part A- Introduction

III. Course code: CSE 1204

IV. Credit: 1.5

4. Course Summary

A computer engineer needs to know about number system, logic design and the basic building blocks used in digital systems, in particular digital computers.

5. Course Objectives

1. To learn about basic concept on number system, universal gates and truth tables.
2. To know about Boolean function and De-Morgan, canonical forms and minimization techniques
3. To learn about combinational and sequential circuits and basic flip flops
4. To know about synchronous and asynchronous counters

6. Course Learning Outcomes

CLO1. Apply and explain fundamental concepts of digital logic design including basic and universal gates, number systems, binary coded systems, basic components of combinational and sequential circuits

CLO2. Implement the acquired knowledge to apply techniques related to the design and analysis of digital electronic circuits including Boolean algebra and multi-variable Karnaugh map methods

CLO3. Analyze small-scale combinational and sequential digital circuits

CLO4. Design small-scale combinational and synchronous sequential digital circuit using Boolean algebra and K-maps

art B- Lesson Plan

Course Details Learning plan

Timeline	**Topics /contents**	**Learning Outcomes**	**Mapped CLOs**	**Teaching Strategies**	**Assessment Strategies**
Week 1 **Week 2** **Week 3**	Number systems and codes. Digital logic: Boolean algebra, De-Morgan's law, Logic gates and their truth tables.	Explain number systems Apply De-Morgan law for the Boolean function State and explain De-Morgan	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	Canonical forms, combinational logic circuits, minimization techniques.	Evaluate canonical forms Apply minimization techniques Explain combinational logic circuits	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	Arithmetic and data handling logic circuit, decoders and encoders, Multiplexers and De-multiplexers.	Determine decoder and encoder function Differentiate MUX and DEMUX Draw data handling logic circuit	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 9 **Week 10**	Combinational Circuit design, Flip-flops, race around problems.	Explain flip flops and race problems	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	Counters: Asynchronous and Synchronous counters and their applications. Synchronous and asynchronous logic design: state diagram, Mealy and Moore machine.	Design synchronous and asynchronous circuits Apply state diagram for logic circuits, State and explain Mealy and Moore machine	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	State minimization and assignments. Pulse mode logic, Fundamental mode logic design.	Explain state minimization and assignments Demonstrate pulse mode logic and mode logic design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test

Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**PLO10**	**PLO11**	**PLO12**	**PLO13**	**PLO14**	**PLO15**	**PLO16**	**PLO17**	**PLO18**
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

3. Digital logic and Computer Design by M. Morris Mano
4. Digital System Analysis by Tocci

Reference Books

3. Principles of Electronic by D. E. Knuth
4. Electronic Devices and Circuit Theory by R.L. Boylestad

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Engineering Drawing

Part A- Introduction

V. **Course code: CSE 1206**

VI. **Credit: 1**

7. **Course Summary**

Capability of computer aided design and drawing is essential for an engineer in order to be able to design objects using some drawing software..

8. **Course Objectives**

1. Understand scaling of drawing different objects.
2. Recognize objects from different views.
3. Design structures using drawing software

9. **Course Learning Outcomes**

CLO5. **Ability** to read basic engineering drawing.
CLO6. **Ability** to draw basic drawing objects on standard drawing sheets.
CLO7. **Apply** engineering drawing skills using Auto CAD tool.
CLO8. **Respond** as instructed while working in groups.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1- **Week 6**	Familiarization with different logic gates and flip-flops.	Apply scales Identify views	CLO1 CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7- **Week 10**	Missing line, Auxiliary view, Detail and assembly drawing.	Find out missing lines. Apply auxiliary view and assembly drawing	CLO2 CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11- **Week 14**	Flip-flops, Sequential circuit, Counter, Combinational Circuit.	Design objects using CAD.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Includes set of programming problems to solve within a timeline

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**

Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**PLO19**	**PLO20**	**PLO21**	**PLO22**	**PLO23**	**PLO24**	**PLO25**	**PLO26**	**PLO27**
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

5. Mastering AutoCAD 2008 and AutoCAD LT 2008 by Wiley
6. Computer Aided Design and manufacturing by M. Groover and E. Zimmers

Reference Books

5. CAD/CAM Robotics and Factories of the Future by S. Narayanan
6. Computer Aided Manufacturing by TienChien Chang

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Electronic Devices and Circuits

Part A- Introduction

I. **Course code: EEE 1201**

II. **Credit: 3**

1. Course Summary

To learn basic concepts about semiconductor physics.

To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.

To know about power electronics devices and its operation.

A computer engineer needs to know about number system, logic design and the basic

building blocks used in digital systems, in particular digital computers.

2. Course Objectives

1. To learn basic concepts about semiconductor physics.
2. To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.
3. To know about power electronics devices and its operation.

3. Course Learning Outcomes

CLO1. learn basic concepts about semiconductor physics
CLO2. Know about formation of semiconductor diodes, transistors
CLO3. know about power electronics devices and its operation
CLO4. Apply amplifier in different circuits

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	**Theory of Semiconductors:** Electronic structure of the elements, crystalline and amorphous solids, different types of crystal, band theory of solids, structure of silicon and germanium.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Intrinsic and Extrinsic Semiconductors:** N and P type semiconductor, carrier densities, generation and recombination of excess carriers, carrier life time, continuity equation.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	**Diode circuits:** The PN junction, biasing and V-I characteristics of diodes, rectifier concept, half wave and full wave rectifiers, Zener diode and voltage	Define diods	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	regulators.				
Week 9 **Week 10**	**Bipolar transistor:** Junction transistors, principles of operation, biasing, characteristics in different configurations (CE, CB & CC), DC and AC load line, transistor equivalent circuits. Gain and impedance, Analysis of small signal low frequency transistor amplifier by using h -parameters.	Draw and describe diode clipping and clamping circuits.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	**Field effect transistor:** Construction of JFET, its parameters, biasing, characteristics and principles of operation, different types of MOSFET and its operation and characteristics. **Amplifier:** Voltage and current amplifiers of different configurations, RC coupled amplifier, operational amplifiers (OPAMPS), linear applications of OPAMPS, gain, input and output	Define and explain Operational Amplifiers and FET	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	impedance.				
Week 13 **Week 14**	**Capacitor and Capacitance:** Capacitance, Parallel plate capacitor, cylindrical capacitor, spherical capacitor, capacitors in series & parallel, energy stored in a capacitor, transformers.	Classify Oscillators.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05

Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO28	PLO29	PLO30	PLO31	PLO32	PLO33	PLO34	PLO35	PLO36
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Basic Electronics (Solid State).
2. Electronic Devices and Circuit Theory.

Reference Books

1. Hand Book of Electronics.
2. Electronic Devices and Circuits.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Electronic Devices and Circuits Lab

Part A- Introduction

I. **Course code: EEE 1202**

II. **Credit: 1.5**

1. Course Summary

To learn basic concepts about semiconductor physics.

To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.

To know about power electronics devices and its operation.

A computer engineer needs to know about number system, logic design and the basic

building blocks used in digital systems, in particular digital computers.

2. Course Objectives

1. To learn basic concepts about semiconductor physics.
2. To know about formation of semiconductor diodes, transistors, field effect transistors, Oscillators and their working principle.
3. To know about power electronics devices and its operation.

3. Course Learning Outcomes

CLO1. Implement basic concepts about semiconductor physics

CLO2. Demonstrate about formation of semiconductor diodes, transistors

CLO3. Apply about power electronics devices and its operation

CLO4. Apply amplifier in different circuits

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	**Theory of Semiconductors:** Electronic structure of the elements, crystalline and amorphous solids, different types of crystal, band theory of solids, structure of silicon and germanium.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Intrinsic and Extrinsic Semiconductors:** N and P type semiconductor, carrier densities, generation and recombination of excess carriers, carrier life time, continuity equation.	Define and discuss semiconductor physics and semiconductors.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	**Diode circuits:** The PN junction, biasing and V-I characteristics of diodes, rectifier concept, half wave and full wave rectifiers, Zener diode and voltage regulators.	Define diods	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9 **Week 10**	**Bipolar transistor:** Junction transistors, principles of operation, biasing, characteristics in different configurations (CE, CB & CC), DC and AC load line, transistor equivalent circuits. Gain and impedance, Analysis of small signal low frequency transistor amplifier by using h -parameters.	Draw and describe diode clipping and clamping circuits.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 11 **Week 12**	**Field effect transistor:** Construction of JFET, its parameters, biasing, characteristics and principles of operation, different types of MOSFET and its operation and characteristics. **Amplifier:** Voltage and current amplifiers of different configurations, RC coupled amplifier, operational amplifiers (OPAMPS), linear applications of OPAMPS, gain, input and output impedance.	Define and explain Operational Amplifiers and FET	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	**Capacitor and Capacitance:** Capacitance, Parallel plate capacitor, cylindrical capacitor, spherical capacitor, capacitors in series & parallel, energy stored in a capacitor, transformers.	Classify Oscillators.	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	PLO37	PLO38	PLO39	PLO40	PLO41	PLO42	PLO43	PLO44	PLO45
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Basic Electronics (Solid State).
2. Electronic Devices and Circuit Theory.

Reference Books

1. Hand Book of Electronics.
2. Electronic Devices and Circuits.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Differential Equations and Geometry

Course Code: MATH 1201 | **Credit: 3.0**
Exam Hours : 03 | **SEE Marks : 72**
CIE Marks : 28

Course Learning Outcomes: ***After the successful completion of the course, students will be able to:***

CLO1	Understand and explain formation and classification of differential equation.
CLO2	Explain basic concepts of first order and first degree differential equation.
CLO3	Understand and explain basic concepts higher order differential equation and their solution process.
CLO4	Learn and explain coordinates system, transformations of coordinates.
CLO5	Understand and explain straight lines, pair of straight lines.
CLO6	Explain and expand General Equation of Second Degree.
CLO7	Learn and determine Circles, System of circles, Tangent, Normal.

SN	**Course Content**	**Hrs**	**CLOs**
1.	Definitions and classifications of differential equations, problems and solutions, Formation of differential equation, Existence and uniqueness theorem (Statement and application only).	**6**	**CLO1**
2.	Separable and homogeneous equations, Exact equation, Integrating factor, Equations made exact by integrating factor, First order linear equation, Bernoulli equation, Riccati equation.	**6**	**CLO2**
3.	First order higher degree equations-solvable for x, y and p, Clairaut's equation, Singular	**6**	**CLO3**

	solutions, Orthogonal and oblique trajectories. Higher order linear homogeneous equation with constant coefficients, Reduction of order, Basic theorems, Linear non-homogeneous equation with constant coefficients, Method of undetermined coefficients.		
4.	Co-ordinates: Coordinates system, Transformations of coordinates	**5**	**CLO4**
5.	Transformations: Change of axes, Translation, Invariants, Analyzing an equation using a rotation (identify and sketch).	**6**	**CLO5**
6.	Straight Lines: Straight lines, Pair of straight lines.	**3**	**CLO5**
7.	General Equation of Second Degree: Reduction to standard form and Identification of conics.	**4**	**CLO6**
8.	Circles: Circles, System of circles, Tangent, Normal.	**3**	**CLO7**

Recommended References:

Anton, H., Bivens, I. and Davis, S. : Calculus

Das and Mukherjee : Differential Calculus

Thomson & Finne. : Calculus

Swokowski. E. W : Calculus

Shanti Narayan : Differential Calculus

ASSESSMENT PATTERN

CIE- Continuous Internal Evaluation (28Marks)

Bloom's Category Marks (out of 20)	Tests (10)	Assignments (05)	Quizzes (05)	External Participation in Curricular/Co-Curricular Activities(08)
Remember			05	
Understand		03		
Apply	04			8
Analyze	03			

Evaluate	03			
Create		02		

SEE- Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	15
Understand	15
Apply	15
Analyze	15
Evaluate	10
Create	02

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Physics for Engineers

Part A- Introduction

I. Course code: PHY 1201

II. Credit: 2

1. Course Summary

Someone wants to develop his/her career as a computer engineer needs to know the basic theories and principles of physics to generate problem solving, analytical, mathematical and solution finding skills; this course will equip him/her with the concepts of Kinetic Theory of Gases, Heat and Thermodynamics, Simple Harmonic Motion, Wave Motion, Sound Waves, Acoustics and Electrostatics.

2. Course Objectives

1. Apply the concepts, ideas and methods of Physics required to solve problems in engineering studies.

2. Acquire knowledge about different laws and models of Physics, which will develop analytical capabilities among them.
3. Apply the laws and skills in higher studies or research areas.
4. Understand the origins in thermodynamics, electronics and Acoustics.
5. Explain everyday things happening around us.

3. Course Learning Outcomes

CLO1. Apply the concepts, ideas and methods of Physics
CLO2. Apply the laws and skills in higher studies
CLO3. Acquire knowledge about different laws and models of Physics
CLO4. Explain everyday things happening around us

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	Kinetic theory of gases: Deduction of gas law, Principle of equipartition of energy, Equation of state – Andrew's Experiment, Vander Waals equation, Critical Constants, Transmission of heat – conduction, convection and radiation.	Define critical constants	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	Heat and Thermodynamics: 1st law of Thermodynamics, Internal energy, Specific heats of gases, Work done by expanding gas, Elasticity of a perfect gas, 2nd law of thermodynamics, Carnot's cycle, Efficiency of heat engines, Absolute scale of temperature, Entropy and its physical concepts, Maxwell's thermodynamic relations, Statistical Mechanics	Explain work done by expanding gas	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	Simple Harmonic Motion: Simple harmonic motion, Combination of S.H.M. and Lissajous figures, Damped Oscillations, Forced oscillations, Resonance, Vibrations of membranes and columns. Electrostatics Charge and matter, Coulomb's Law, The electric field, Gauss's Law, Electrical potential, Capacitance and Resistance, Ohmic and Non ohmic material,	Draw the mismatch between Damped Oscillation and Forced Oscillation	CLO3,	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	Variation of resistance with temperature.				
Week 9 **Week 10**	Wave motion: Travelling waves, Principle of superposition, wave velocity, Group velocity, Phase velocity, Power and intensity in wave motion, Interference, diffraction and transmission of waves, Standing Waves.	List the types of velocity play role in wave motion and interpret each with respective example	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	Sound waves: Audible, Ultrasonic, Infrasonic and Supersonic waves, Propagation and speed of longitudinal waves, Travelling longitudinal waves, Standing longitudinal waves, Vibrating systems and sources of sound, Beats, The Doppler's effect	Differentiate between Ohmic and Non Ohmic material	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	Acoustics: Re-vibration, Noise insulation and reduction, Compound absorption, sound distribution, Room acoustics, Recording.	Explain state minimization and assignments Demonstrate pulse mode logic and mode logic design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test

Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	**PLO46**	**PLO47**	**PLO48**	**PLO49**	**PLO50**	**PLO51**	**PLO52**	**PLO53**	**PLO54**
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0

Part D- Resources

Textbooks

1. Equilibrium Thermodynamics by Adkins
2. Waves and Oscillations by Brijlal
3. Fundamentals of physics by Halliday and Rensick

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Object Oriented Programming

Part A- Introduction

I. **Course code: CSE 2101**

II. **Credit: 3**

1. Course Summary

This course introduces the concepts of object-oriented programming to students with a background in the procedure oriented paradigm. The course begins with a brief review of control structures and data types with emphasis on structured data types and array processing. It then moves on to introduce the object-oriented programming paradigm, focusing on the definition and use of classes along with the fundamentals of object-oriented concepts such as polymorphism, inheritance, data hiding etc. Besides, Template functions and classes; Multi-threaded Programming, abstract classes, function overloading and overriding will be discussed elaborately as well.

2. Course Objectives

1. The primary aim of the module is to enable the students to tackle complex programming problems, making good use of the object-oriented programming paradigm to simplify the design and implementation process.

2. Students should be able to design and implement programs for complex problems, making good use of the features of the language.

3. Teaches students how to design, develop and program computer systems using an object oriented programming language such as C++, Java.

Familiarizes students with the tools that streamline object-oriented development.

I. **Course code: CSE 2101**

II. **Credit: 3**

3. Course Learning Outcomes

CLO1. Understand the basic object oriented paradigm of programming before learning how to use objects and class in the context of programming.

CLO2. Analyze the structure of the new paradigm that helps the students to understand its activities in a logical way that follows a bottom-up process.

CLO3. Remember every single detail such as writing code in a particular language, steps to execute them, translate them for error detection etc.

CLO4. Apply the already learnt things to properly implement the program on various real life problems and learn the correlations between coding and real time solutions.

CLO5. Evaluate numerous possible ways to solve a problem and find the best way using the concepts of object oriented programming.

Part B- Lesson Plan

Course Details Learning plan Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Philosophy of Object Oriented Programming (OOP), features of OOP, advantages of OOP over structured programming	-Details on Object oriented programming, many features and concepts and superiority over structural/ procedural programming.	CLO1, CLO2	Lecture	Quiz
Week 2	Classes and Objects	-Integrating object and class in the structure of this new programming system, how to properly call them and where to use them	CLO1, CLO2, CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 3, 4	Constructors, destructors and copy constructors, array of objects, object references, Single Multidimensional Arrays.	-Discuss about constructor, copy constructor and destructor -demonstrate these features by integrating them in the code. -demonstrating array of objects and how to refer them in the code, Apply single and multidimensional array in development	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment,Exam

Week 5, 6	Function overloading, operator overloading and type conversion of objects.	-discuss on function overloading and operator overloading that are extensively used in OOP. -discussion on how operator overloading is used to convert object types.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 7, 8	Inheritance	-discussion on all kinds of inheritances, their usage, implementing them to solve problems -learning single and multiple inheritance. -involve students to solve expressions.	CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 9, 10	Polymorphism, virtual function run time type identification, exception handling	-discussion and analyzing different types of polymorphism, virtual function/function overriding.	CLO4, CLO5	Lecture, Exercise Demonstration	Quiz Assignment, Exam
Week 11	template functions and classes, namespace, standard template library	-understanding function and template function and class -understanding container, algorithm, iterators	CLO2, CLO3	Lecture Exercise Demonstration	Quiz, Assignment, Exam
Week 12	String manipulation	Create string Manipulate string operation, Create string processing algorithm	CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 13, 14	Introduction to graphical user interface; handling mouse and keyboard events, Multi-Threading; Client Server programming	Create GUI, Apply mouse and keyboard event, Define thread, Design multithreaded program	CLO4, CLO5	Lecture Exercise Demonstration	Quiz Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Attendance	Student's participation in the class lecture, quiz and exam.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3 hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment
CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**

Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	0
CLO3	1	1	1	0	0	0	0	1	1
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	1

<u>Part D- Resources</u>

Textbooks

1. Balagurusamy: Object Oriented Programming with C++

2. Bhushan Trivedi: Programming with ANCI C++

3. Y. D. Liang: Introduction to Java Programming, Comprehensive: International Edition

Reference Books

1. Bjarne Stroustrup: The C++ Programming Language

2. Ashok N. Kamthane: Object-Oriented Programming with ANCI and Turbo C++

3. H. M. Deitel& P. J. Deitel: Java™ How to Program

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

<u>Course Outline: Object Oriented Programming Lab</u>

Part A- Introduction

I. Course code: CSE 2102

II. Credit: 1.5

1. Course Summary

This course introduces the concepts of object-oriented programming to students with a background in the procedure oriented paradigm. The course begins with a brief review of control structures and data types with emphasis on structured data types and array processing. It then moves on to introduce the object-oriented programming paradigm, focusing on the definition and use of classes along with the fundamentals of object-oriented concepts such as polymorphism, inheritance, data hiding etc. Besides, Template functions and classes; Multi-threaded Programming, abstract classes, function overloading and overriding will be discussed elaborately as well.

2. Course Objectives

1. The primary aim of the module is to enable the students to tackle complex programming problems, making good use of the object-oriented programming paradigm to simplify the design and implementation process.

2. Students should be able to design and implement programs for complex problems, making good use of the features of the language.

3. Teaches students how to design, develop and program computer systems using an object oriented programming language such as C++, Java.

Familiarizes students with the tools that streamline object-oriented development.

3. Course Learning Outcomes

CLO1. Understand the basic object oriented paradigm of programming before learning how to use objects and class in the context of programming.

CLO2. Analyze and apply the structure of the new paradigm that helps the students to understand its activities in a logical way that follows a bottom-up process.

CLO3. Remember every single detail such as writing code in a particular language, steps to execute them, translate them for error detection and implementation.

CLO4. Apply the already learnt things to properly implement the program on various real life problems and learn the correlations between coding and real time solutions.

CLO5 Evaluate numerous possible ways to solve a problem and find the best way using the concepts of object oriented programming.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Philosophy of Object Oriented Programming (OOP), features of OOP, advantages of OOP over structured programming	-Details on Object oriented programming, many features and concepts and superiority over structural/ procedural programming.	CLO1, CLO2	Lecture	Quiz, Assignment
Week 2	Classes and Objects	-Integrating object and class in the structure of this new programming system, how to properly call them and where to use them	CLO1, CLO2, CLO3	Lecture, Exercise Demonstration	Quiz, Practical, Exam
Week 3, 4	Constructors, destructors and copy constructors, array of objects, object references, Single Multidimensional Arrays.	-Discuss about constructor, copy constructor and destructor -demonstrate these features by integrating them in the code. -demonstrating array of objects and how to refer them in the code, Apply single and multidimensional array in development	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Practical, Exam

Week 5, 6	Function overloading, operator overloading and type conversion of objects.	-discuss on function overloading and operator overloading that are extensively used in OOP. -discussion on how operator overloading is used to convert object types.	CLO3	Lecture, Exercise Demonstration	Quiz, Practical, Assignment, Exam
Week 7, 8	Inheritance	-discussion on all kinds of inheritances, their usage, implementing them to solve problems -learning single and multiple inheritance. -involve students to solve expressions.	CLO4	Lecture, Exercise Demonstration	Assignment, Practical, Exam
Week 9, 10	Polymorphism, virtual function run time type identification, exception handling	-discussion and analyzing different types of polymorphism, virtual function/function overriding.	CLO4, CLO5	Lecture, Exercise Demonstration	Practical, Assignment Exam
Week 11	Template functions and classes, namespace, standard template library	-understanding function and template function and class -understanding container, algorithm, iterators	CLO2, CLO3	Lecture Exercise Demonstration	Quiz, Practical, Exam
Week 12	String manipulation	Create string Manipulate string operation, Create string processing algorithm	CLO4	Lecture, Exercise Demonstration	Quiz, Practical, Exam

Week 13, 14	Introduction to graphical user interface; handling mouse and keyboard events, Multi-Threading; Client Server programming	Create GUI, Apply mouse and keyboard event, Define thread, Design multithreaded program	CLO4, CLO5	Lecture Exercise Demonstration	Quiz, Practical, Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE.
Attendance	Student's participation in the class lecture, quiz and exam.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Includes set of programming problems to solve within a timeline

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**

Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	0
CLO3	1	1	1	0	0	0	0	1	1
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Balagurusamy: Object Oriented Programming with C++

2. Bhushan Trivedi: Programming with ANCI C++

3. Y. D. Liang: Introduction to Java Programming, Comprehensive: International Edition

Reference Books

1. Bjarne Stroustrup: The C++ Programming Language

2. Ashok N. Kamthane: Object-Oriented Programming with ANCI and Turbo C++

3. H. M. Deitel& P. J. Deitel: Java™ How to Program.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Algorithm Analysis and Design

Part A- Introduction

I. Course code: CSE 2103

II. Credit: 3

1. Course Summary

Algorithm analysis and design provide the theoretical backbone of computer science and are a must in the daily work of the successful programmer. This course applies design and analysis techniques to numeric and nonnumeric algorithms which act on data structures. Design is emphasized so that the student will be able to develop new algorithms. Analysis of algorithms is concerned with the resources an algorithm must use to reach a solution.

2. Course Objectives

1. The goal of this course is to provide a solid background in the design and analysis of the major classes of algorithms.

2. To become familiar with the tools and techniques necessary to propose practical algorithmic solutions to real-world problems which still allow strong theoretical bounds on time and space usage.

3. To introduce a broad variety of important and useful algorithms and data structures in different areas of applications and to concentrate on fundamental algorithms.

4. At the end of the course students will be able to develop their own versions for a given computational task and to compare and contrast their performance.

3. Course Learning Outcomes

CLO1. Understand theoretical and mathematical structures, concepts of computer algorithms

CLO2. Analyze the time complexity, space complexity, asymptotic performance of algorithms, worst, average and best-cases of algorithms. .

CLO3. Demonstrate numerous algorithm design strategies including divide and conquer, greedy method, dynamic programming, backtracking, finding shortest paths, traversing salesmen, and implementing those algorithms.

CLO4. Illustrate and analyze various search and graph algorithms, graph coloring, branch and bound algorithms, Hamiltonian cycle.

CLO5. Evaluate decision problems, the concept of P = NP, NP completeness and Cook's theorem.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	algorithms, basic notations of algorithms, fundamental analysis of algorithms, criteria of algorithms	Discuss about the basic structure of algorithms, notations of algorithms, proof techniques	CLO1	Lecture	Quiz
Week 2	fundamental analysis of complexity, different asymptotic notation, efficiency worst case best case and average case analysis	Solving the problems of asymptotic notation, Apply asymptotic notation to represent algorithm complexity	CLO1, CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 3	Elementary Data Structures Divide and Conquer algorithms merge sort heap sort maxmin algorithm quick sort	Searching strategy using divide and conquer, different types of sorting algorithms with example	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 4, 5	Greedy algorithms, Knapsack Problem, Job Sequencing with deadlines, MST Kruskal’s algorithms, Prim’s algorithms	Discuss about greedy method control abstraction, discuss about the algorithms with example, Demonstrate how to find MST using Prims and Kruskals algorithm.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 6, 7	Dynamic Programming, Matrix Chain Multiplication, Optimal Binary search tree, Multistage Graph, Bellman Ford Algorithm	Discuss about the Dynamic Programming basics, Solve the multistage graph algorithm for showing the application of DP	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 8, 9	Huffman Code Shortest path problem, Floyd's Algorithm, Sequence Alignment Backtracking N Queen Problem, State Space tree	Solve Huffman code and shortest path problem, the process of sequence alignment, understand the concept of backtracking, Understand about the 4-queen problem and state space tree.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment Exam
Week 10, 11	Travelling salesman Problem, Branch and Bound, Sum of subset problem, Hamiltonian cycle, planar graph, graph coloring,	understand sum-of subset method, learn planar graph, graph coloring, Hamiltonian cycle. Understand branch and bound work,	CLO3, CLO4	Lecture Exercise Demonstration	Quiz, Assignment, Exam
Week 12	Traversal and search techniques, BFS, DFS	Understand how traversal techniques work, analyze the BFS and DFS for graph	CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 13, 14	NP-Completeness, Cook's theorem NP- Complete problem	P and NP (Cook's theorem),examples of NP-complete problems; approximate algorithms for NP-hard problems or polynomial algorithms for sub problems of NP-hard problems	CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Attendance	Student's participation in the class lecture, quiz and exam.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment
CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	1
CLO3	1	1	1	0	0	0	0	1	0
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	0

Part D- Resources

Textbooks

1. *Horowitz, Sahni, Rajasekaran: Fundamentals of Computer Algorithms*
2. *Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest: Introduction to Algorithms*

Reference Books

1. *Aho, Hopcroft and Ullman: The design and Analysis of Computer Algorithms*
2. *Sara Baase: Computer Algorithms: Introduction to Design and Analysis*
3. *D. E. Knuth: The Art of Computer Programming, Vol. 1, Fundamental Algorithms*.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Algorithm Analysis and Design Lab

Part A- Introduction

I. Course code: CSE 2104

II. Credit: 1.5

1. Course Summary

Algorithm analysis and design provide the theoretical backbone of computer science and are a must in the daily work of the successful programmer. This course applies design and analysis techniques to numeric and nonnumeric algorithms which act on data structures. Design is emphasized so that the student will be able to develop new algorithms. Analysis of algorithms is concerned with the resources an algorithm must use to reach a solution.

2. Course Objectives

1. The goal of this course is to provide a solid background in the design and analysis of the major classes of algorithms.

2. To become familiar with the tools and techniques necessary to propose practical algorithmic solutions to real-world problems which still allow strong theoretical bounds on time and space usage.

3. To introduce a broad variety of important and useful algorithms and data structures in different areas of applications and to concentrate on fundamental algorithms.

4. At the end of the course students will be able to develop their own versions for a given computational task and to compare and contrast their performance.

3. Course Learning Outcomes

CLO1. Understand theoretical and mathematical structures, concepts of computer algorithms

CLO2. Analyze the time complexity, space complexity, asymptotic performance of algorithms, worst, average and best-cases of algorithms. .

CLO3. Demonstrate and implement numerous algorithm design strategies including divide and conquer, greedy method, dynamic programming, backtracking, finding shortest paths, traversing salesmen, and implementing those algorithms.

CLO4. Illustrate and apply various search and graph algorithms, graph coloring, branch and bound algorithms, Hamiltonian cycle.

CLO5. Evaluate decision problems, the concept of P = NP, NP completeness and Cook's theorem.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	algorithms, basic notations of algorithms, fundamental analysis of algorithms, criteria of algorithms	Discuss about the basic structure of algorithms, notations of algorithms, proof techniques	CLO1	Lecture	Quiz
Week 2	fundamental analysis of complexity, different asymptotic notation, efficiency worst case best case and average case analysis	Solving the problems of asymptotic notation, Apply asymptotic notation to represent algorithm complexity	CLO1, CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 3	Elementary Data Structures Divide and Conquer algorithms merge sort heap sort maxmin algorithm quick sort	Searching strategy using divide and conquer, different types of sorting algorithms with example	CLO3	Lecture, Exercise Demonstration	Practical, Assignment, Exam
Week 4, 5	Greedy algorithms, Knapsack Problem, Job Sequencing with deadlines, MST Kruskal’s algorithms, Prim’s algorithms	Discuss about greedy method control abstraction, discuss about the algorithms with example, Demonstrate how to find MST using Prims and Kruskals algorithm.	CLO3	Lecture, Exercise Demonstration	Practical, Assignment, Exam

Week 6, 7	Dynamic Programming, Matrix Chain Multiplication, Optimal Binary search tree, Multistage Graph, Bellman Ford Algorithm	Discuss about the Dynamic Programming basics, Solve the multistage graph algorithm for showing the application of DP	CLO3	Lecture, Exercise Demonstration	Practical, Assignment, Exam
Week 8, 9	Huffman Code Shortest path problem, Floyd's Algorithm, Sequence Alignment Backtracking N Queen Problem, State Space tree	Solve Huffman code and shortest path problem, the process of sequence alignment, understand the concept of backtracking, Understand about the 4-queen problem and state space tree.	CLO3	Lecture, Exercise Demonstration	Quiz, Practical, Assignment Exam
Week 10, 11	Travelling salesman Problem, Branch and Bound, Sum of subset problem, Hamiltonian cycle, planar graph, graph coloring,	understand sum-of subset method, learn planar graph, graph coloring, Hamiltonian cycle. Understand branch and bound work,	CLO3, CLO4	Lecture Exercise Demonstration	Quiz, Practical, Assignment, Exam
Week 12	Traversal and search techniques, BFS, DFS	Understand how traversal techniques work, analyze the BFS and DFS for graph	CLO4	Lecture, Exercise Demonstration	Quiz, Practical, Assignment, Exam

Week 13, 14	NP-Completeness, Cook’s theorem NP- Complete problem	P and NP (Cook's theorem),examples of NP-complete problems; approximate algorithms for NP-hard problems or polynomial algorithms for sub problems of NP-hard problems	CLO5	Lecture, Exercise Demonstration	Quiz, Practical, Assignment, Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Attendance	Student’s participation in the class lecture, quiz and exam.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	1
CLO3	1	1	1	0	0	0	0	1	0
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	0

Part D- Resources

Textbooks

1. *Horowitz, Sahni, Rajasekaran: Fundamentals of Computer Algorithms*
2. *Thomas H. Cormen, Charles E. Leiserson and Ronald L. Rivest: Introduction to Algorithms*

Reference Books

1. *Aho, Hopcroft and Ullman: The design and Analysis of Computer Algorithms*
2. *Sara Baase: Computer Algorithms: Introduction to Design and Analysis*
3. *D. E. Knuth: The Art of Computer Programming, Vol. 1, Fundamental Algorithms*.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Computer Architecture

Part A- Introduction

I. **Course code: CSE 2105**

II. **Credit: 3**

1. **Course Summary**

   To be a computer engineer one needs to know architectural design, organizational design and computer family and also to learn the performance factors..

2. **Course Objectives**

1. To learn the distinguished features of computer architecture, organization and family.

3. **Course Learning Outcomes**

CLO1. Describe hardware and software
CLO2. Explain stored computer concept
CLO3. Describe of memory organization and design
CLO4. Describe the cache memory system
CLO5. Describe DRAM and SRAM
CLO6. Describe Error Detecting and Correcting Codes
CLO7. Describe External Memory

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	A Brief History of Computers, Designing for Performance, Contemporary hard and software description	Describe the details on hardware and software	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 2 ,3	Computer Functions, Instruction Fetch and Execute, Interrupts, I/O Function, Vonn Neumann Machine, example of how stored program and data executes together	Describe details on stored computer concept, I/O Function and Program execution concepts	CLO 2	Lecture, handouts	Quiz Participation Exam
Week 4, 5	Memory Hierarchy and Characteristics of Memory Systems.	Describe details of the hierarchy of memory systems, their characteristics and organization	CLO 3	Lecture, handouts	Quiz Participation Exam
Week 6, 7, 8	Cache Memory Principles, Elements of Cache Design, Cache Addresses, Cache Size, Mapping Function, Replacement Algorithms, Write Policy, Line Size, Number of Caches	Discuss different aspects of cache memory	CLO 4	Lecture, handouts	Quiz Participation Exam
Week 9, 10	Organization of DRAM and SRAM, Types of ROM, Chip Logic Chip Packaging, Module Organization	Discuss details on different topics related to DRAM and SRAM	CLO 5	Lecture, handouts	Quiz Participation Exam

Week 11, 12	Describe Hard Failure, Soft Failure, Error correcting codes such as Hamming Code.	Describe Error Detecting and Correcting Codes	CLO 6	Lecture, handouts	Quiz Participa tion Exam
Week 13, 14	Magnetic Read and Write Mechanisms, Data Organization and Formatting, Physical Characteristics, Disk Performance Parameters, Different Raid Levels	Describe different characteristics of external memory	CLO 7	Lecture, handouts	Quiz Participa tion Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			

Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. COMPUTER ORGANIZATION. AND ARCHITECTURE. DESIGNING FOR PERFORMANCE: EIGHTH EDITION by William Stallings.

2. Computer Organization and Design: the Hardware/Software Interface: Third Edition by David A Patterson and John L. Hennessy

Reference Books

1. Computer Architecture and Organization by John Hayes

Other resources (Slides)

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

<u>Course Outline: Numerical Analysis</u>

<u>Part A- Introduction</u>

I. **Course code: CSE 2107**

II. **Credit: 3**

1. **Course Summary**

   The goal of the course is to provide the students with a strong background on numerical approximation strategies and a basic knowledge on the theory that supports numerical algorithms.

2. **Course Objectives**

1. Use numerical methods for solving a problem
2. Locate and use good mathematical software,
3. Get the accuracy you need from the computer,
4. Assess the reliability of the numerical results, and
5. Determine the effect of round off error or loss of significance.

3. **Course Learning Outcomes**

CLO1. Recall the algebraic and transcendental equations
CLO2. Illustrate the Eigen value and Eigen matrix
CLO3. Identify the roots using algorithms
CLO4. Devise the matrix calculation algorithms
CLO5. Examine the integration problems
CLO6. Demonstrate the differential equations

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Numerical Solution of Algebraic & Transcendental Equations: Bisection, False Position, Newton-Raphson, and Secant Method.	Explain the Solution of algebraic and transcendental equations	CLO 1	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 2 ,3	Simultaneous Linear Non-Homogeneous Algebraic Equations: Gauss Elimination, Gauss Jordan, Gauss Jacobi, and Gauss Seidel Method.	Compute the solution of Simultaneous Linear Non-Homogeneous Algebraic Equations	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 4, 5, 6	Eigen Values and Eigen Vectors: Introduction and concept of value and eigenvector, Estimation of the size of Eigenvalues.	Calculate Eigen Values and Eigen Vectors	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 7, 8	Interpolation: Newton Forward, Newton Backward, Gauss's Forward, Gauss's Backward, Stirling Interpolation, Bessel's Interpolation, and Laplace-Everett's Interpolation Formula, Maxima-Minima of Tabulated Function.	Discuss different Interpolation method	CLO 3	Lecture, Demonstration, handouts	Quiz Participation Exam

Week 9, 10	Numerical Differentiation: Newton's Forward and Newton's Backward Difference Formula to Compute the Derivatives, Difference Formula to Compute the Derivatives, Maxima-Minima of Tabulated Function.	Compute Numerical Differentiation	CLO 3	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 11, 12	Numerical Integration: Newton-Cotes, Trapezoidal, Simpson's One-Third, Simpson's Three-Eight Formula to Compute Integration. Numerical Double Integration.	Compute Integration	CLO4 CLO5	Lecture, Demonstration	Quiz Participation Exam
Week 13, 14	Numerical Solution of Ordinary Differential Equations: Taylor Series, Taylor Series Method for Simultaneous First-Order and Second-Order Differential Equations, Picard's Method of Successive approximation. Euler's, Modified Euler's, and Improved Euler's Method. Runge-Kutta Methods for Simultaneous First order, Second order, Third order, and Fourth order Differential Equations. Predictor-Corrector Method.	Explain the solution of ordinary differential equations	CLO4 CLO6	Lecture, Demonstration, handouts	Quiz Participation Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment
CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10

Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	0	0	0	0	0	0	0	1	0
CLO2	0	0	0	0	0	0	0	1	0
CLO3	0	1	0	0	0	0	0	1	0
CLO4	0	1	0	0	0	0	0	1	0
CLO5	0	1	0	0	0	0	0	1	0
CLO6	0	1	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Numerical Methods for Engineers by Canale at el.
2. Numerical Methods for Scientists and Engineers by Richard Hamming

Reference Books

1. Interpolation by J.F. Steffensen
2. Numerical Methods by Santosh Gupta

Other resources (Slides)

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Numerical Analysis Lab

Part A- Introduction

I. **Course code: CSE 2108**

II. **Credit: 1.5**

1. Course Summary

The goal of the course is to provide the students with a strong background on numerical approximation strategies and a basic knowledge on the theory that supports numerical algorithms.

2. Course Objectives

1. Use numerical methods for solving a problem
2. Locate and use good mathematical software,
3. Get the accuracy you need from the computer,
4. Assess the reliability of the numerical results, and
5. Determine the effect of round off error or loss of significance.

3. Course Learning Outcomes

CLO1. Recall the algebraic and transcendental equations
CLO2. Illustrate the Eigen value and Eigen matrix
CLO3. Identify the roots using algorithms
CLO4. Devise the matrix calculation algorithms
CLO5. Examine the integration problems
CLO6. Demonstrate the differential equations

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Numerical Solution of Algebraic & Transcendental Equations: Bisection, False Position, Newton-Raphson, and Secant Method.	Explain the Solution of algebraic and transcendental equations	CLO 1	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 2 ,3	Simultaneous Linear Non-Homogeneous Algebraic Equations: Gauss Elimination, Gauss Jordan, Gauss Jacobi, and Gauss Seidel Method.	Compute the solution of Simultaneous Linear Non-Homogeneous Algebraic Equations	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 4, 5, 6	EigenValues and EigenVectors: Introduction and concept of value and eigenvector, Estimation of the size of Eigenvalues.	Calculate Eigen Values and Eigen Vectors	CLO 2	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 7, 8	Interpolation: Newton Forward, Newton Backward, Gauss's Forward, Gauss's Backward, Stirling Interpolation, Bessel's Interpolation, and Laplace-Everett's Interpolation Formula, Maxima-Minima of Tabulated Function.	Discuss different Interpolation method	CLO 3	Lecture, Demonstration, handouts	Quiz Participation Exam

Week 9, 10	Numerical Differentiation: Newton's Forward and Newton's Backward Difference Formula to Compute the Derivatives, Difference Formula to Compute the Derivatives, Maxima-Minima of Tabulated Function.	Compute Numerical Differentiation	CLO 3	Lecture, Demonstration, handouts	Quiz Participation Exam
Week 11, 12	Numerical Integration: Newton-Cotes, Trapezoidal, Simpson's One-Third, Simpson's Three-Eight Formula to Compute Integration. Numerical Double Integration.	Compute Integration	CLO 4 CLO 5	Lecture, Demonstration	Quiz Participation Exam
Week 13, 14	Numerical Solution of Ordinary Differential Equations: Taylor Series, Taylor Series Method for Simultaneous First-Order and Second-Order Differential Equations, Picard's Method of Successive approximation. Euler's, Modified Euler's, and Improved Euler's Method. Runge-Kutta Methods for Simultaneous First order, Second order, Third order, and Fourth order Differential Equations. Predictor-Corrector Method.	Explain the solution of ordinary differential equations	CLO 4 CLO 6	Lecture, Demonstration, handouts	Quiz Participation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	**5**

Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	0	0	0	0	0	0	0	1	0
CLO2	0	0	0	0	0	0	0	1	0
CLO3	0	1	0	0	0	0	0	1	0
CLO4	0	1	0	0	0	0	0	1	0
CLO5	0	1	0	0	0	0	0	1	0
CLO6	0	1	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Numerical Methods for Engineers by Canale at el.
2. Numerical Methods for Scientists and Engineers by Richard Hamming

Reference Books

1. Interpolation by J.F. Steffensen
2. Numerical Methods by Santosh Gupta

Other resources (Slides)

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Sociology and Disaster Management

Part A- Introduction

I. **Course code: HUM 2101**

II. **Credit: 2**

1. Course Summary

This course provides students with an outline of the field of sociology. This course critically and scientifically examines the social forces and processes that shape our personalities, institutions, culture, and society. This course is the conception of *democracy* as a system of governance that inspires power and responsibility in citizens and sociology as a means to understand how society operates and influence it. This course considers general concepts used to interpret government as well as develop a familiarity with the various institutions, political groups, beliefs, and ideas that constitute government. In this course students will learn the basic concepts of the field of Gender, Socialism, Capitalism, Feudalism, Human Civilization, Cultural Lag, Family, Marriage, Crime, deviance, juvenile delinquency disaster management etc.

2. Course Objectives

The main objective of this course is to get students familiar with the process and formula of the sociological imagination to a variety of contemporary social phenomena. The objective also includes explaining or justifying various government structures and procedures, and the political effects of these structures and procedures

3. Course Learning Outcomes

CLO1. Understand sociological terms, theoretical approaches to social phenomena and issues of government and politics.

CLO2. Remember government, sociology, gender, socialism, capitalism, feudalism, human civilization, cultural lag, family, marriage, crime, deviance, juvenile delinquency.

CLO3. Analyze different administrative and social factors and social change in historical and contemporary society.

CLO4. Apply sociological issues in real life or professional field in case of research.

CLO5. Analyze specific sociological topics and utilizing the major theoretical models that are appropriate for a specific topic including disaster management.

CLO6. Evaluate and compare the use of sociological theories and concepts to analyze critically current and historical events, impact of disaster in real life.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction to basic concept of government and politics	- Functions, organs, and forms of modern state and government -discuss about the citizenship; socialism, capitalism, feudalism, political importance of feudalism.	CLO1	Lecture, Exercise Demonstration	Quiz, Assignment,
Week 2	Basic key terms of government and contemporary issues.	-students will develop knowledge about contemporary issues of government	CLO1, CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 3, 4	Administrative functions and various tiers of government and international politics.	students will learn about administrative functions. -Learn about Bangladesh issues -How operates administrative functions of developed countries functions	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 5, 6	Significance of Sociology, Social structure of Bangladesh	-definition and description of sociology -Know about the terms of sociology -Learn about social structure	CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 7, 8	Social research methods and key sociology terms	-students will understand about social research and its methods. -Learn about society, community, association, institution, group -allocate an assignment to develop social research	CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 9, 10	Cultural content and growth of capitalism and development of civilization.	-understand material and non-material issues, cultural lag and civilization. -To learn about growth of capitalism -analyze the features and consequences of socialism .	CLO5	Lecture, Exercise Demonstration	Quiz, Assignment Exam
Week 11, 12	Understand family, marriage issues, industrial society and development of urbanization. Various issues of disaster management	-understand about family, how to forms and turn into industrial society. -understand marriage -Industrial issues -Organizational factors	CLO5 CLO6	Lecture Exercise Demonstration	Quiz, Assignment, Exam

Week 13, 14	Develop the concept of population and migration, youth issues and how technological factors affect in life Impact of disaster management in real life	-Understand development matters and social change issues. -Understand population and emigrational issues. -To leran crime isssues and youth unrest factors. -To understand technological factors on social life.	CLO6	Lecture Exercise Demonstration	Quiz, Assignment, Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Attendance	Student’s participation in the class lecture, quiz and exam.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18
CLO1	1	1	1	0	0	0	0	0	0
CLO2	1	1	1	0	1	1	1	0	0
CLO3	1	1	1	1	0	1	1	0	0
CLO4	1	1	1	0	0	1	0	0	0
CLO5	1	1	1	1	0	1	0	0	0
CLO6	1	1	1	1	0	1	0	0	0

Part D- Resources

Textbooks

1. Turner. J.H. (ed) 2006. Handbook of Sociology
2. Turner, M. and Hulme, D : Governance, administration and development
3.

REFERENCE BOOKS:

1. Lake and Rothchild, Ed : The International Spread of Ethnic Conflict: Fear, Diffusion, and Escalation
2. Giddnes, Anthony. 1984. The Constitution of Society
3. Turner, M. and Hulme, D. (1997) Governance, administration and development, London: Macmillan .
4. Allan, Kenneth D. 2006. Contemporary Social and Sociological Theory.
   Gouldner, Alvin. 1972. The Coming Crisis of Western Sociology

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Matrices and Vectors

Part A- Introduction

I. **Course code: MATH 2101**

II. **Credit: 2**

1. Course Summary

This course covers vector and multi-variable calculus. Topics include vectors and matrices, parametric curves, partial derivatives, double and triple integrals, and vector calculus in 2- and 3-space.

2. Course Objectives

1. Fluency with vector operations, including vector proofs and the ability to translate back and forth among the various ways to describe geometric properties, namely, in pictures, in words, in vector notation, and in coordinate notation.
2. Fluency with matrix algebra, including the ability to put systems of linear equation in matrix format and solve them using matrix multiplication and the matrix inverse.
3. An understanding of a parametric curve as a trajectory described by a position vector; the ability to find parametric equations of a curve and to compute its velocity and acceleration vectors.

3. Course Learning Outcomes

CLO1. A comprehensive understanding of the gradient, including its relationship to level curves (or surfaces), directional derivatives, and linear approximation.

CLO2. The ability to compute derivatives using the chain rule or total differentials.

CLO3. The ability to set up and solve optimization problems involving several variables, with or without constraints.

CLO4. An understanding of line integrals for work and flux, surface integrals for flux, general surface integrals and volume integrals. Also, an understanding of the physical interpretation of these integrals.

CLO5. The ability to set up and compute multiple integrals in rectangular, polar, cylindrical and spherical coordinates.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 Week 2 Week 3	Part A: Vectors, Determinants, and Planes » Session 1: Vectors » Session 2: Dot Products » Session 3: Uses of the Dot Product: Lengths and Angles » Session 4: Vector Components	Vectors, Determinants, and Planes	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 Week 5 Week 6	» Session 5: Area and Determinants in 2D » Session 6: Volumes and Determinants in Space » Session 7: Cross Products » Session 8: Equations of Planes	Vectors, Determinants, and Planes	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 Week 8	Part B: Matrices and Systems of Equations » Session 9: Matrix Multiplication » Session 10: Meaning of Matrix Multiplication	Matrices and Systems of Equations	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 9 **Week 10**	» Session 11: Matrix Inverses » Session 12: Equations of Planes II » Session 13: Linear Systems and Planes » Session 14: Solutions to Square Systems	Matrices and Systems of Equations	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	Part C: Parametric Equations for Curves » Session 15: Equations of Lines » Session 16: Intersection of a Line and a Plane » Session 17: General Parametric Equations; the Cycloid	Parametric Equations for Curves	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	» Session 18: Point (Cusp) on Cycloid » Session 19: Velocity and Acceleration » Session 20: Velocity and Arc Length » Session 21: Kepler's Second Law	Parametric Equations for Curves	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students

	need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Quiz (10%)	**Assignment (8%)**	**Practical (10%)**	**Final (72%)**	**Quiz (10%)**
CLO1	1	1	1	1	1
CLO2	1	1	1	1	1
CLO3	1	1	1	1	1
CLO4	1	1	1	1	1
CLO5	1	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

01. Introduction to Matrices and Vectors (Dover Books on Mathematics) Paperback – November 24, 2011
    by Jacob T. Schwartz

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Data Communication

Part A- Introduction

I. **Course code: CSE 2201**

II. **Credit: 3**

1. Course Summary

This course introduces some basic concepts, models, techniques etc. used in data communication. A computer science engineer needs to know the communication model, different network layer, transmission medium and technique used in digital communication system to fulfill his CSE degree.

2. Course Objectives

1. Build an understanding of the fundamental concepts of Data communication.
2. Learn how computer network hardware and software operate.
3. Investigate the fundamental issues driving network design.
4. Learn about dominant network technologies.

3. Course Learning Outcomes

CLO1. Learn about the basic knowledge of data communication systems
CLO2. Able to understand about various layered model used in data communication
CLO3. Explore different methods and techniques used in data transfer
CLO4. Learn about various Protocols and their packet format
CLO5. Understand the uses and effectiveness of various transmission medium

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Data Communication, Fundamental characteristics, Components, Data representation, Data Flow, Network and its criteria, point-to-point and multipoint connection, different topologies, Network Models, Protocols and standards	Learn about data Communication model, network standards and organizations, different protocol layers	CLO1, CLO4, CLO5	Lecture	Essay Short Question
Week 2	Different tasks in network model, peer-to-peer communication, characteristics and usage area of different layers in OSI model, fundamental properties of different layers of TCP/IP protocol, physical, logical and port address.	Learn about Network Model, Tasks, ISO, OSI Model, TCP/IP Protocol	CLO1, CLO2, CLO4	Lecture	Quiz Assignment Essay Short Question
Week 3,4	Analog and digital data and signal, period, frequency, phase, wavelength, time and frequency domain, bandwidth, transmission of digital signal, attenuation, distortion, noise, SNR, Nyquist bit rate, Shannon's capacity performance parameters.	Learn about physical layer and media, Illustrate Analog and digital data and signal, transmission technique and its impairments, Data rate limit for both noisy and noiseless channel.	CLO1, CLO2, CLO3	Lecture Exercise Demonstration	Quiz Assignment Essay Short Question
Week 5, 6	Line coding techniques NRZ, RZ, Manchester, and differential Manchester encoding, AMI, Block coding, analog to digital	Learn various Digital transmission techniques.	CLO3	Lecture Exercise	Quiz Essay Short

	conversion based on PCM, delta modulation, etc.				Question
Week 7	ASK, FSK, PSK, QPSK, QAM encodings, AM, PM,FM, etc.	Learn various Analog transmission techniques.	CLO3	Lecture Exercise	Quiz Essay Short Question
Week 8, 9	Synchronous and asynchronous data transmission techniques, FDM, international FDM carrier standards, synchronous TDM, international TDM carrier standards, statistical time division multiplexing	Discuss and characterize different data transmission and multiplexing techniques.	CLO3	Lecture Exercise	Assignment Essay Short Question
Week 10	Frequency hopping spread spectrum, direct sequence spread spectrum, code division multiple access.	Discuss and characterize different spread spectrum techniques.	CLO3	Lecture Exercise	Assignment Essay Short Question
Week 11, 12	Characteristics and applications of various types of guided medium, Characteristics and applications of wireless transmission-terrestrial and satellite microwave, radio waves, propagation mechanism, free space propagation, land propagation, path loss, slow fading, fast fading, delay spread, inter symbol interference, VSAT	Discuss and characterize different transmission medium, wireless transmission technique, Compute path loss, learn to distinguish slow and fast fading, Able to define inter symbol interference and VSAT	CLO5	Lecture Exercise	Assignment Essay Short Question

Week 13	Error Detection and Correction; parity check, CRC, forward error correction technique, linear block code, hamming code, etc.	Learn abouterror detection and correction methods, forward and backward detection, checksum	CLO1, CLO3	Lecture Exercise Demonstration	Assignment Essay Short Question
Week 14	Random access, ALOHA, Pure ALOHA, CSMA, CSMA/CD, CSMA/CA, Controlled access, reservation, token passing, polling, poll function, Channelization protocols, FDMA, TDMA and CDMA technology	Discusswireless channel, multiple access protocol, random access and controlled access protocol	CLO3, CLO4	Lecture Exercise Demonstration	Assignment Essay Short Question

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz, Short questions and Essay	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1

CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Mapping of CLOs to PLOs

Part D- Resources

Textbooks

1. Data Communications and networking by Behrouz A. Forouzan.
2. Data and Computer Communication by William Stallings.

Reference Books

1. Data Communication by Hajkins.
2. Data Communication by Taub.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Database Management System

Part A- Introduction

I. **Course code: CSE 2203**

II. **Credit: 3**

1. **Course Summary**

   A computer engineer needs to know the fundamentals of database architecture, database management systems, and database systems, principles and methodologies of database design, and techniques for database application development.

2. **Course Objectives**

1. An understanding of the needs for and uses of database management systems in business;
2. An understanding of the context, phases and techniques for designing and building database information systems in business;
3. An understanding of the components of a computerized database information system (application)
4. An ability to correctly use the techniques, components and tools of a typical database management system -- such as Access 2000 or Oracle 8i -- to build a comprehensive database information system (application);
5. An ability to design a correct, new database information system for a business functional area and implement the design in either Access 2000 or Oracle 8i;
6. An introductory understanding of some advanced topics in database management, e.g., object-relational databases and design, distributed databases, database administration (security, backup and restore, tuning) and data warehousing.

3. **Course Learning Outcomes**

CLO1. Database Concepts
CLO2. Relational Data Model
CLO3. Functional Dependency and Normal Forms
CLO4. Relational Calculus
CLO5. Relational Database Design
CLO6. Entity-Relationship (ER) approach
CLO7. Transformation of the ER model to SQL

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Files and Databases, Database Management systems, Data models	Define Database Explain advantage of Database over file system Describe different data model	CLO 1	Lecture, handouts	Quiz Participation Exam Assignment
Week 2 ,3	Relations, Domains, Attributes and Tuple	Explain Relational Data model Apply RDM in DB design Describe different types of attribute	CLO 2	Lecture, handouts	Quiz Participation Exam Assignment
Week 4, 5	Anomalies, Functional Dependency, First, Second and third normal forms, Boyce-Codd Normal form	Evaluate anomalies in DB design Identify different normal form Apply normalization in DB design Convert one normal form to another normal	CLO 3	Lecture, handouts	Quiz Participation Exam Assignment
Week 6, 7, 8	Relational Calculus Based Languages: SQL, Relational algebra and Set operations	Understanding the concept of SQL	CLO 4	Lecture, handouts	Quiz Participation Exam Assignment
Week 9, 10	Relational design criteria, Lossless decomposition, decomposition algorithms, synthesis algorithms	Explain decomposition and synthesis algorithm Define lossless decomposition	CLO 5	Lecture, handouts	Quiz Participation Exam Assignment

Week 11, 12	The ER model and its constructs, ER modeling in logical database design.	Define entity Relationship, relationship set. Apply E-R approach to DB design	CLO 6	Lecture, handouts	Quiz Particip ation Exam Assign ment
Week 13, 14	Transformation of the ER model to SQL	Explain transformation of ER model to SQL	CLO 7	Lecture, handouts	Quiz Particip ation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment
CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			

Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Database System Concepts by Korth and Silverchatz
2. Principle of Database Systems by O. William

Reference Books

1. Relational Database Management System by Jeffrey Ullman

Other resources (Slides)

Jashore University of Science and Technology
Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Database Management System Lab

Part A- Introduction

I. **Course code: CSE 2204**

II. **Credit: 1.5**

1. Course Summary

A computer engineer needs to know the fundamentals of database architecture, database management systems, and database systems, principles and methodologies of database design, and techniques for database application development.

2. Course Objectives

1. An understanding of the needs for and uses of database management systems in business;
2. An understanding of the context, phases and techniques for designing and building database information systems in business;
3. An understanding of the components of a computerized database information system (application)
4. An ability to correctly use the techniques, components and tools of a typical database management system -- such as Access 2000 or Oracle 8i -- to build a comprehensive database information system (application);
5. An ability to design a correct, new database information system for a business functional area and implement the design in either Access 2000 or Oracle 8i;
6. An introductory understanding of some advanced topics in database management, e.g., object-relational databases and design, distributed databases, database administration (security, backup and restore, tuning) and data warehousing.

3. Course Learning Outcomes

CLO1. Database Concepts
CLO2. Relational Data Model
CLO3. Functional Dependency and Normal Forms
CLO4. Relational Calculus
CLO5. Relational Database Design
CLO6. Entity-Relationship (ER) approach
CLO7. Transformation of the ER model to SQL

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Files and Databases, Database Management systems, Data models	Define Database Explain advantage of Database over file system Describe different data model	CLO 1	Lecture, handouts	Quiz Participation Exam Assignment
Week 2 ,3	Relations, Domains, Attributes and Tuple	Explain Relational Data model Apply RDM in DB design Describe different types of attribute	CLO 2	Lecture, handouts	Quiz Participation Exam Assignment
Week 4, 5	Anomalies, Functional Dependency, First, Second and third normal forms, Boyce-Codd Normal form	Evaluate anomalies in DB design Identify different normal form Apply normalization in DB design Convert one normal form to another normal	CLO 3	Lecture, handouts	Quiz Participation Exam Assignment
Week 6, 7, 8	Relational Calculus Based Languages: SQL, Relational algebra and Set operations	Understanding the concept of SQL	CLO 4	Lecture, handouts	Quiz Participation Exam Assignment
Week 9, 10	Relational design criteria, Lossless decomposition, decomposition algorithms, synthesis algorithms	Explain decomposition and synthesis algorithm Define lossless decomposition	CLO 5	Lecture, handouts	Quiz Participation Exam Assignment

Week 11, 12	The ER model and its constructs, ER modeling in logical database design.	Define entity Relationship, relationship set. Apply E-R approach to DB design	CLO 6	Lecture, handouts	Quiz Participation Exam Assignment
Week 13, 14	Transformation of the ER model to SQL	Explain transformation of ER model to SQL	CLO 7	Lecture, handouts	Quiz Participation Exam

<u>Part C- Assessment and Evaluations</u>
Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment
CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		

Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Database System Concepts by Korth and Silverchatz
2. Principle of Database Systems by O. William

Reference Books

1. Relational Database Management System by Jeffrey Ullman

Other resources (Slides)

Department of Computer Science and Engineering

Course Outline: Digital System Design

Part A- Introduction

I. **Course code: CSE 2205**

II. **Credit: 3**

Register Transfer Logic: Inter Register Transfer, Arithmetic, Logic and Shift Micro-Operations, Conditional Control Statements, Fixed-Point Binary data, Overflow, Arithmetic shifts, Decimal data, Floating-Point data, Non-numeric data, Instruction codes, Design of simple computer.

Processor Logic Design: Processor Organization, Arithmetic Logic Unit, Design of Arithmetic Circuit, Design of Logic Circuit, Design of Arithmetic Logic Unit, Status Register, Design of Shifter, Processor Unit, Design of Accumulator. Design of Multiplier Circuits.

Control Logic Design: Control Organization, Hardwired control, Micro-program Control, Control of Processor Unit, PLA Control, Micro-program Sequencer.

Computer Design: System Configuration, Computer Instructions, Timing and Control, Execution of Instructions, Design of Computer Registers, Design of Control. Register Load and Inter Register Transfer; Bus Buffer and Memory Cycle of Microcomputers.

Computer Design (Simple as Possible): Architecture, Instruction Set, Programming, Fetch Cycle, Execution Cycle.

Recommended Texts:

1. Principle of Digital Electronics. Author: Malvino
2. Digital Principles & System Design, Author: A.P.Godse, D.A.Godse

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

<u>Course Outline: Software Development Project- 1</u>

<u>Part A- Introduction</u>

III. Course code: CSE 2208

IV. Credit: 1.5

4. Course Summary

This course gives students experience designing, implementing, testing, and debugging large programs. This C#.net /ASP.net/PHP/Java Programming Knowledge is valuable to both beginners and advanced developers that already have experience in developing applications software. Students will also get advanced programming experience in C#.net /ASP.net/PHP/Java; covering topics such as inheritance, multithreading, networking, database programming, and web development.

5. Course Objectives

- Will be able to understand how programming solve real life problems using computer
- Learn to develop software projects that support organization's strategic goals
- Match organizational needs to the most effective software development model
- Plan and manage projects at each stage of the software development life cycle (SDLC)
- Create project plans that address real-world management challenges
- Develop the skills for tracking and controlling software deliverable

6. Course Learning Outcomes

CLO1. Able to independently design programs
CLO2. Learn to produce professional-quality code
CLO3. Implement large programs
CLO4. Design and execute tests to identify software bugs
CLO5. Repair software bugs, redesigning and refactoring code when necessary
CLO6. Utilize, analyze, and critique code written by others

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1	Introduction to Software Project, Program execution, Difference between IDE and Compiler, Programming from scratch	Get the idea of the course, Have real life knowledge of program execution and the main difference between IDE and compiler, Learn how to run a program without using any IDE	CLO1, CLO2	Lecture, Demonstration, Practical example	Quiz, Short question, Problem solving
Week 2	Programming and Problem-Solving common sense, Concepts of IDEs (Microsoft Visual Studio, Phpstorm, Eclipse etc.), Giving view to software project	Learn how programming will help you in times of building real life software, Learn how to solve real life problem using computer , Gain the philosophy of programming	CLO1, CLO2, CLO3	Lecture, Demonstration, Practical example	Quiz, Short question, Problem solving
Week 3	Programming and Problem-Solving common sense, Building the base, OOP review	Learn how programming will help you in times of building real life software, Learn how to solve real life problem using computer, Gain the philosophy of programming	CLO1, CLO2, CLO3	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 4	To Apply OOP Knowledge	Data Types, Type Conversion, Boxing & Unboxing, Conditional Statements, Looping, Methods in C#, Properties, Arrays, Indexers, Structures, Enumerations	CLO2, CLO3, CLO4	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories

Week 5	To Apply OOP Knowledge	Garbage Collector, Stack and Heap, System. GC Class Understand the major qualities of a Good Software, Encapsulation, Inheritance, Polymorphism, Class and Object Constructors, Dynamic types, Optional parameters, Names &optional arguments, Covariant generic type parameters, Destructors, Method overloading Method overriding, Early binding, Late Binding, Abstract Classes, Abstract Methods, Interfaces, Multiple Inheritance, Generic classes, Static classes, Static constructors, Object initialize	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 6	To Apply OOP Knowledge	System Defined Exceptions, Custom Exceptions, Try, Catch, Finally, Throwing exceptions, Function Pointers, Multi cast delegates, File Handling System. IO namespace, File stream, Stream Reader, Stream writer, File info, Directory info, Drive Info	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 7,8	To Apply .NET Knowledge	**Developing Microsoft.NET Applications for Windows (Visual C#.NET)** Creating a Form Adding Controls to a Form	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories

		Creating an Inherited Form Organizing Controls on a Form Creating MDI Application **Working with Controls** Creating an Event Handler for Control Using Windows Forms Controls Using Dialog Boxes Application Adding Controls at Run Time Creating Menus Validating User Input **Using Data in Windows Forms Applications** Adding ADO.NET Objects to and Configuring ADO.NET Objects in a Windows Forms Application Accessing and Modifying Data by Using DataSets Binding Data to Controls Overview of XML Web Services Persisting Data			
Week 9	To Apply .NET Knowledge	**Developing Microsoft.NET Applications for Windows (Visual C#.NET)** Printing and Reporting in Windows Forms Applications Lessons Printing From a Windows Forms Application	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories

		Using the Print Preview, Page Setup, and Print Dialogs Constructing Print Document Content by Using GDI+ Creating Reports by Using Crystal Reports Deploying Windows Forms Applications .NET Assemblies			
Week 10	To Apply .NET Knowledge	**Introduction to LINQ and ADO.NET Entity Framework.** LINQ expressions Using via extension methods, Filtering, Sorting, Aggregation, Skip and Take operators, Joins, Query, Lambda expressions. **Data Projection** Single result value, Existing types, Anonymous types,Grouping	CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 11, 12, 13	To Apply .NET Knowledge	**ASP.NET INTRODUCTION** Difference Between ASP and ASP.NET ,Architecture Inline Technique & Code-Behind Technique,Code Render Blocks Server Controls ,Page Basics, Page lifecycle, Post back Request View State, Directives	CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories

		PROGRAMMING WITH SERVER CONTROLS Web Server Controls Basic Web Controls, List Controls, Data Controls,Adv Controls, User Controls, Master Page and Content Page. **Validation Controls** Understanding Validation Client or Server Site Validation Required Filed Validator Rang Validator, Regular Expression Validator, Compare Validator, Custom Validator Validator Summary. **CONTENT Developing Microsoft.NET Applications for Web (ASP.NET using C#.NET)** **STATE MANAGEMENT WITH ASP.NET** Context, View State,Cookie State Session State, Session Tracking Application Object, Session and Application Events **ADO.NET AND ASP.NET** Working with Data Controls, GridView, -Inserting, Updating, Deleting,-Sorting in Data Grid			

		-Paging in Data Grid, DataSource Controls,Dataset,DetailsView FormView,Data List,Repeater Control, Crystal Reports **ADO.NET PROGRAMMING** Architecture, DataReaders and DataSets, Command Object Transaction Programming Procedure Execution Data Adapter and Data Set, Data Tables, Data Relation,Data Views Updating Datase			

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz, Short questions	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.
Implementation	This part is the main development part of the project where students will use different Programming languages and tools to develop software.

Report and Viva	Students have to prepare a document including all the requirements, tools, methodologies, coding samples, output scenario snapshots and raw data used in projects and have to submit that in a proper format. Also, Students have to face a viva where he/she have to answer questions regarding the developments of the project.

Mapping of CLOs to Assessment

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**
Create	**5**

Mapping of CLOs to PLOs

CLOs	**PLO10**	**PLO11**	**PLO12**	**PLO13**	**PLO14**	**PLO15**	**PLO16**	**PLO17**	**PLO18**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1
CLO6	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

3. Online resources: w3schools.com, stackoverflow.com, codeforces.com etc.
4. Online tutorial from udemy, youtube.com etc.

Jashore University of Science and Technology

Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Probability and Statistics for Engineers

Part A- Introduction

I. **Course code: MATH 2201**

II. **Credit: 3**

1. **Course Summary**

This is an introductory course on statistics and probability. All the basic methods of statistics and probability will be covered in this course.

2. **Course Objectives**

1. Receive a basic knowledge of statistics and probability
2. Able to solve basic statistics and probability related problems using a variety of skills and strategies.
3. Able to analyze data and find insights

3. **Course Learning Outcomes**

CLO1. Devise the primary idea to statistics and probability

CLO2. Learn to solve statistics and probability related problems using programming languages.

CLO3. Design the experiment to understand real life problem

CLO4. Explore different theories of statistics and probability

CLO5. Utilize different Statistical Methods

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Preliminaries	Getting familiar with different concepts	CLO1	Lecture	Quiz
Week 2, 3	Measures of Central Tendency	Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Mode, Weighted mean	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5, 6	Measures of Dispersion	Range, Standard Deviation, Mean Deviation, Quartile Deviation, Variance, Moments, Skewness and Kurtosis	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8	Correlation Theory	Linear Correlation - Its measures and significance, Rank Correlation	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9, 10	Regression Analysis	Linear and non-linear regression, Least-square method of curve fittings	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 10, 11	Probability	Elementary concepts, Laws of probability, Conditional Probability and Bay's theorem, Random variables	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 12, 13, 14	Probability Distributions	Binomial distribution, Poisson distribution and Normal distribution	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05

Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Mathematical Statistics, Author: Kapur, J.N. and Saxena, H.C.
2. A First Course in Mathematical Statistics, Author: Weatherburn, C.E.

Reference Books

1. Probability and Mathematical Statistics, Author: MarekFisz
2. Fundamentals of Mathematical Statistics, Author: Gupta, S.C. and Kapoor V.K.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Linear Algebra and Fourier Analysis

Part A- Introduction

III. Course code: MATH 2203

IV. Credit: 2

4. Course Summary

One of the main goals this course is to establish rules for the limiting behavior of functions so that we can deal with functions with as much confidence as we do real or complex numbers. An equally important motivation (that will only become clear in the second half) is that the systematic study of Fourier series requires the Lebesgue integral. The square mean convergence of Fourier series and Parseval's formula cannot be stated accurately in proper generality without the Lebesgue integral and Lebesgue integrable functions.

5. Course Objectives

1. account for basic concept and theorems within the Linear Algebra and Fourier analysis;
2. demonstrate basic numeracy skill concerning the concepts in the previous point;
3. use the numeracy skill at the solution of mathematical and physical problems formulated as ordinary or partial differential equations.

6. Course Learning Outcomes

CLO6. Explain basic concepts of Matrices and their significance in engineering field.
CLO7. Understand and explain basic concepts of systems of linear equations.
CLO8. Learn determinant of a 2x2 Matrix, determinant of a 3x3 matrix.
CLO9. Understand and describe importance of Inverse in solving Linear Systems,
CLO10. Explain scalars and vectors, geometrical representation of Vectors

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	Matrices and their Significance, Matrix Notation, Dimension (Order) of a Matrix, Addressing Elements of a Matrix, Solving Linear Systems in 2 Unknowns, Types of Matrices, Addition and Subtraction of Matrices, Multiplication of Scalars with Matrices, Multiplication of two Matrices	Matrices	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	Systems of Linear Equations Preview, Elementary Row Operations, Row Echelon Form (REF) Determinant of a 2x2 Matrix, Determinant of a 3x3 Matrix, Finding Determinants Quickly Inverse exists only for Square Matrices, Singular Matrices, Importance of Inverse in solving Linear Systems, Inverse of a 2x2 Matrix, Inverse of a 3x3 Matrix - The Two Methods	Linear Equations	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	Scalars and Vectors, Geometrical Representation of Vectors, Vector Addition and Subtraction, Laws of Vector Addition and Head to Tail Rule, Unit Vector Introduction to Vector Spaces, Euclidean Vector Spaces - Part 1, Euclidean Vector Spaces - Part 2, Euclidean Vector Spaces - Part 3, Definition and Closure Properties, Axioms of Vector Spaces, Subspace and Null space.	Vector Spaces	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 9 Week 10	Fourier series, Fourier coefficients, trigonometric polynomials and orthogonality.	Fourier series	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 Week 12	Properties of Fourier coefficients; Bessel's inequality, Parseval's identity and the Riemann-Lebesgue lemma.	Fourier coefficients	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	Various notions of convergence of Fourier series, including pointwise, uniform and mean square convergence. Summability methods, convolution and Young's inequality. - Fourier Analysis in broader contexts; for example, Fourier integrals, Fourier expansions in groups, Schwartz spaces and tempered distributions.	convergence of Fourier series	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Quiz (10%)	Assignment (8%)	Practical (10%)	Final (72%)	Quiz (10%)
CLO1	1	1	1	1	1
CLO2	1	1	1	1	1
CLO3	1	1	1	1	1
CLO4	1	1	1	1	1
CLO5	1	1	1	1	1

Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

H. Anton and C.Rorres : Linear Algebra with Applications.

Brestscher : Linear Algebra with Applications.

Lipschutz, S. : Linear Algebra

Haward A. : Elementary Linear Algebra with Application

Haward A. : Contemporary Linear Algebra

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Business Psychology

Part A- Introduction

I. Course code: HUM 2201

II. Credit: 2

1. Course Summary

This course is designed to provide a broad overview of the field of Psychology. Special attention will be given to helping the student become a better thinker by learning cognitive science, industrial psychology, and using existing knowledge to draw conclusions, make predictions, or construct explanations. The goal of this course is to think consciously, deliberately and skillfully about human behavior and hence, the topics such as physiological psychology, object recognition and language understanding, job analysis, method of section, learning, cognition, emotions, as well as others are included. The concept of behavior-based safety program and insights into accident prevention are included in this course.

2. Course Objectives

The objectives of this course is to give students the idea of the discussing the research and theory as it relates to the following topics such as personnel, employee motivation and satisfaction, group processes and leadership, and organizational change and development, self-efficacy and self-safety and describing the scope of study in the field of industrial and organizational psychology and the explanation of measurement and determinants of job satisfaction.

3. Course Learning Outcomes

CLO1. Analyze the basic theories of motivation, including concepts such as instincts, drive reduction, and self-efficacy, self- safety and apply the nature of cognitive science, introduce business psychology.

CLO2 Understand the theory suggested for organizational development and create the role of learning industrial theory in the workplace.

CLO3. Analyze the types and causes of accidents with diagram and extending cooperation concerning industrial safety measures.

CLO4. Apply different OOP design paradigms like use case diagram, sequence diagram, ER diagram and data flow diagram.

CLO5. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors

CLO6. Evaluate the ability to think critically, to analyze complex and diverse concepts, and to use reason and judgment and explain the measurement and determinants of job satisfaction.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction to Psychology, business psychology.	-Student will able to explain the history and methods of psychology, concepts of business psychology.	CLO1	Lecture	Quiz
Week 2, 3	Cognitive Science.	-students will able to understand the cross disciplinary, historical foundations of cognitive science. -understand the research problems that unify the cognitive sciences, including language, thought, perception, attention, learning, memory, reasoning, problem solving, judgment, and decision-making. -Understand the assumptions, methods and concepts that unify and differentiate the cognitive sciences. -Understand the diversity of theoretical approaches and paradigms across the cognitive sciences, including neural, embodied, social, and/or technological approaches.	CLO1	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 4	Reasoning.	-students will able to explain the three methods of reasoning which are the deductive, inductive, and adductive approaches.	CLO5	Lecture, Exercise Demonstrati on	Quiz, Assignment, Exam
Week 5, 6	Object recognition and language understandi ng. Learning Industrial Psychology	-student will able to give the explanation of object recognition and observe the data spatial resolution. -understand and appreciate the role of human behavior, -Demonstrate knowledge of the biological and conceptual languages of the brain and their potential. -understand and appreciate the connection between scientific inquiry and the creative and artistic dimensions in the field of Psychology	CLO2, CLO4	Lecture, Exercise Demonstratio n	Quiz, Assignment, Exam

Week 7, 8, 9	Introduction to Job and Job analysis, Methods of selection.	-students will able to understand about Job analysis which is a procedure through which they determine the duties and responsibilities, nature of the jobs and finally to decide qualifications, skills and knowledge to be required for an employee to perform particular job. -understand what tasks are important and how they are carried on. Job analysis forms basis for later HR activities such as developing effective training program, selection of employees, setting up of performance standards and assessment of employee's performance appraisal and employee remuneration system or compensation plan.	CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 10	Training in Industry	-student will able to know about coaching, training program implementation, instructional design, and management training.	CLO2, CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 11, 12	Motivation and Work, Job satisfaction	-Students will able to understand about Illustrate intrinsic and extrinsic motivation and describe basic theories of motivation, including concepts such as instincts, drive reduction, and self-efficacy. -explain the measurement and determinants of job satisfaction.	CLO1, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 13, 14	Introduction to Ergonomics, System Engineering, Accident and Safety.	- Students will able to understand the basic physiological systems, nervous system, respiratory system, circulatory system, metabolic system. -analyze the causes of a systems failure. -answer the question 'why is systems engineering important? -explain types and causes of accidents with diagram and extending cooperation concerning industrial safety measures.	CLO1, CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Part C- Assessment and Evaluations

Assessment Procedures

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	1	0	0	0	0	1	1
CLO2	1	1	1	0	0	0	0	1	0
CLO3	1	1	1	0	0	0	0	1	0
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	1
CLO6	1	1	1	0	0	0	0	1	1

Part D- Resources

TEXT BOOKS:

1. A Text-Book of Psychology by Edward Bradford Titchener.
2. **The Social Animal by Elliot Aronson.**

REFERENCE BOOKS:

1. **Authentic Happiness by Martin Seligman**
2. **The Happiness Hypothesis by Jonathan Heidt**
3. **Influence: The Psychology Of Persuasion by Robert B. Cialdini**

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Viva Voce

Part A- Introduction

I. **Course code: CSE 2200**

II. **Credit: 1**

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Operating System and System Programming

Part A- Introduction

I. **Course code: CSE 3101**

II. **Credit: 3**

1. Course Summary	
Computer Engineers should be competent in Operating System. They must be able to apply the basic concepts of operating system, various types of CPU scheduling algorithms, Deadlock problem and some deadlock handling strategies, Paging, segmentation, fragmentation and file-management strategies.	
2. Course Objectives	
1. Identify the role of Operating System. 2. Understanding CPU Scheduling, Synchronization, Deadlock Handling and Comparing CPU Scheduling Algorithms. 3. Solve Deadlock Detection Problems. 4. Describe the role of paging, segmentation and virtual memory in operating systems. 5. Description of protection and security and also the Comparison of UNIX and Windows based OS. 6. Defining I/O systems, Device Management Policies and Secondary Storage Structure and Evaluation of various Disk Scheduling Algorithms. 7. Gather adequate knowledge in system programs (assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems).	
3. Course Learning Outcomes	
CLO1. Learn about the basic knowledge about Operating System. **CLO2.** Able to customize various system functions. **CLO3.** Solve process allocation problems. **CLO4.** Design and implement various scheduling algorithms. **CLO5.** Understand and solve Deadlock Problems. **CLO6.** Understand the role of paging, segmentation and virtual memory in operating system .	

CLO7. Understand I/O systems, Device Management Policies and Secondary Storage Structure
CLO8. Learn adequate knowledge in system programs (assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems).

Part B- Lesson Plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	**Introduction:** Evolution, Goals and Components of OS, Types of OS. Operating System Services	Learn about operating system types and services, Understand the goals of operating system	CLO1, CLO2	Lecture	Essay Short Question
Week 2, 3	**Process management:** Process states and state transition, Process Control Blocks, Job and Process scheduling, Process Communication, Threads	Know different states of a process, Understand process control blocks, Learn about process scheduling, Learn about threads	CLO1, CLO3	Lecture	Quiz Assignment Essay Short Question
Week 4, 5	**CPU Scheduling:** Scheduling levels, Objectives and criteria, CPU scheduling algorithms, Algorithm Evaluation	Understand CPU scheduling and its objectives, Learn and design CPU scheduling algorithms, Compare and analyze different CPU scheduling algorithms	CLO1, CLO4	Lecture Exercise Demonstration	Quiz Assignment Essay Short Question
Week 6, 7	**Process Synchronization:** Process co-ordination, Critical section problems, Semaphores, Monitors, Classical problems of	Learn Process co-ordination, Critical section problems, Understand Semaphores , Know about Classical problems of	CLO1, CLO3	Lecture Exercise	Quiz Essay Short Question

	process synchronization	process synchronization			
Week 8, 9	**Deadlock:** System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock prevention, avoidance, and detection, Recovery form deadlock, Deadlock handling.	Know about deadlock conditions, Understand methods for handling deadlock, Learn recovery from deadlock and prevention of deadlock	CLO1, CLO5	Lecture Exercise	Quiz Essay Short Question
Week 10	**Memory management:** Logical and Physical Address Space, Swapping, Memory allocation schemes, Paging and Segmentation, Segmentation with Paging	Illustrate the memory management scheme with and without swapping, Discuss various types of page replacement algorithm	CLO1, CLO6	Lecture Exercise	Assignme nt Essay Short Question
Week 11	**Virtual memory:** Demand paging, Performance of Demand Paging, Page replacement algorithms, Allocation of frames, Demand Segmentation	Understand demand paging, Learn page replacement algorithm, Know allocation process of frames, Know about demand segmentation	CLO1, CLO6	Lecture Exercise	Assignme nt Essay Short Question
Week 12	**Secondary storage management:** Disk structure; Disk scheduling, Disk management, Swap-space management, Disk reliability, Stable storage implementation	Understand disk structure and management, Know about swap space management, Learn about disk reliability	CLO1, CLO7	Lecture Exercise	Assignme nt Essay Short Question
Week 13	**File-System:** File and Directory concept, File system structure, Allocation method, Free space Management, Directory Implementation.	Learn about file and directory concept, Understand file structure and allocation method	CLO1, CLO7	Lecture Exercise Demonstrati on	Assignme nt Essay Short Question

Week 14	Assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems	Learn adequate knowledge in system programs (assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems).	CLO1, CLO8	Lecture Exercise Demonstration	Assignment Essay Short Question

Course Details Learning plan

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz, Short questions and Essay	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	8
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1
CLO6	1	1	0	0	0	0	0	1	1
CLO7	1	1	0	0	0	0	0	1	1
CLO8	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Operating System Concepts by J. Peterson, A. Silberschatz, and P. Galvin.
2. Introduction to Operating System by William Stalling.
3. Modern Operating Systems by Tanenbaum, Andrew S.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Operating System and System Programming Lab

Part A- Introduction

I. Course code: CSE 3102

II. Credit: 1.5

1. Course Summary	
Computer Engineers should be competent in Operating System. They must be able to apply the basic concepts of operating system, various types of CPU scheduling algorithms, Deadlock problem and some deadlock handling strategies, Paging, segmentation, fragmentation and file-management strategies.	
2. Course Objectives	
1. Identify the role of the Operating System. 2. Understanding CPU Scheduling, Synchronization, Deadlock Handling and Comparing CPU Scheduling Algorithms. 3. Solve Deadlock Detection Problems. 4. Describe the role of paging, segmentation and virtual memory in operating systems. 5. Description of protection and security and also the Comparison of UNIX and Windows based OS. 6. Defining I/O systems, Device Management Policies and Secondary Storage Structure and Evaluation of various Disk Scheduling Algorithms.	

7. Gather adequate knowledge in system programs (assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems).

3. Course Learning Outcomes

CLO1. Apply the basic knowledge about Operating System.
CLO2. Implement various system functions.
CLO3. Solve process allocation problems.
CLO4. Develop various scheduling algorithms.
CLO5. Solve Deadlock Problems.
CLO6. Understand the role of paging, segmentation and virtual memory in operating system .
CLO7. Understand I/O systems, Device Management Policies and Secondary Storage Structure
CLO8. Learn adequate knowledge in system programs (assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems).

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	**Introduction:** Evolution, Goals and Components of OS, Types of OS. Operating System Services	Learn about operating system types and services, Understand the goals of operating system	CLO1, CLO2	Lecture	Essay Short Question
Week 2, 3	**Process management:** Process states and state transition, Process Control Blocks, Job and Process scheduling, Process Communication, Threads	Know different states of a process, Understand process control blocks, Learn about process scheduling, Learn about threads	CLO1, CLO3	Lecture	Quiz Assignment Essay Short Question
Week 4, 5	**CPU Scheduling:** Scheduling levels, Objectives and criteria, CPU scheduling	Understand CPU scheduling and its objectives, Learn and design CPU scheduling	CLO1, CLO4	Lecture Exercise	Quiz Assignme

	algorithms, Algorithm Evaluation	algorithms, Compare and analyze different CPU scheduling algorithms		Demonstration	nt Essay Short Question
Week 6, 7	**Process Synchronization:** Process co-ordination, Critical section problems, Semaphores, Monitors, Classical problems of process synchronization	Learn Process co-ordination, Critical section problems, Understand Semaphores , Know about Classical problems of process synchronization	CLO1, CLO3	Lecture Exercise	Quiz Essay Short Question
Week 8, 9	**Deadlock:** System Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock prevention, avoidance, and detection, Recovery form deadlock, Deadlock handling.	Know about deadlock conditions, Understand methods for handling deadlock, Learn recovery from deadlock and prevention of deadlock	CLO1, CLO5	Lecture Exercise	Quiz Essay Short Question
Week 10	**Memory management:** Logical and Physical Address Space, Swapping, Memory allocation schemes, Paging and Segmentation, Segmentation with Paging	Illustrate the memory management scheme with and without swapping, Discuss various types of page replacement algorithm	CLO1, CLO6	Lecture Exercise	Assignment Essay Short Question
Week 11	**Virtual memory:** Demand paging, Performance of Demand Paging, Page replacement algorithms, Allocation of frames, Demand Segmentation	Understand demand paging, Learn page replacement algorithm, Know allocation process of frames, Know about demand segmentation	CLO1, CLO6	Lecture Exercise	Assignment Essay Short Question
Week 12	**Secondary storage management:** Disk structure; Disk scheduling, Disk management, Swap-space management, Disk	Understand disk structure and management, Know about swap space management, Learn about disk reliability	CLO1, CLO7	Lecture Exercise	Assignment Essay Short

	reliability, Stable storage implementation				Question
Week 13	**File-System:** File and Directory concept, File system structure, Allocation method, Free space Management, Directory Implementation.	Learn about file and directory concept, Understand file structure and allocation method	CLO1, CLO7	Lecture Exercise Demonstration	Assignment Essay Short Question
Week 14	Assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems	Learn adequate knowledge in system programs (assemblers, loaders, linkers, macro-processors, text editors, debuggers, interpreters, compilers, operating systems).	CLO1, CLO8	Lecture Exercise Demonstration	Assignment Essay Short Question

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz, Short questions and Essay	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.

Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	8
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1
CLO6	1	1	0	0	0	0	0	1	1
CLO7	1	1	0	0	0	0	0	1	1
CLO8	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Operating System Concepts by J. Peterson, A. Silberschatz, and P. Galvin.
2. Introduction to Operating System by William Stalling.
3. Modern Operating Systems by Tanenbaum, Andrew S.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Computer Networks

Part A- Introduction

I. **Course code: CSE 3103**

II. **Credit: 3**

1. Course Summary	
This course introduces some basic concepts, protocols, techniques, models etc. used in computer networking. A computer science engineer needs to know the network model, different networking protocols and their formats, transmission medium and technique used in digital communication system to fulfill his Computer Science degree.	
2. Course Objectives	

1. Understand and develop modern network architectures from a design and performance perspective 2. To clarify network terminology. 3. To provide an opportunity to do network programming using TCP/IP. 4. To expose students to emerging technologies and their potential impact. 5. To introduce Internet of Things (IoT) and its protocols.

3. Course Learning Outcomes

CLO1. Learn the basic concepts of networking system
CLO2. Able to understand various protocols used in computer networking
CLO3. Enable to design new networking system
CLO4. Calculate the effectiveness and performance of a network system
CLO5. Troubleshoot problems in networking systems
CLO6. Learn and understand about IoT and its protocols

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Wireless Network concepts: frequency reuse, handoff strategies, system capacity, improving capacity and coverage; Wireless LAN Technology; Ethernet, wireless LAN, broadband wireless.	Illustrate network applications, Identify channel allocation problem, Discuss multiple access protocols, Illustrate WLAN, Ethernet, broadband, Bluetooth technology, Compare broadband and narrowband, Demonstrate data link layer switching,	CLO1, CLO2	Lecture Discussion	Essay Assignment Short Question
Week 2, 3, 4	IPv4 address, IPv6 address, Classfull and classless address,	Identify IP Address, Discuss and understand Classfull and classless address, Illustrate subnetting and	CLO1, CLO2, CLO3,	Lecture Exercise	Quiz Assignment

	subnetting and supernetting, NAT	supernetting concept, Design new networking system	CLO5	Demonstration	Essay Short Question
Week 5, 6	network layer design issues, routing algorithms, congestion control algorithms, quality of service, internetworking, the network layer in the internet	Examine network layer design issues, Analyze routing and congestion control algorithms, Characterize QoS, Implement Internetworking, Discuss the network layer in the internet	CLO1, CLO2, CLO4	Lecture Discussion	Quiz Assignment Essay Short Question
Week 7, 8	The transport layer services, elements of transport protocols, a simple transport protocol, the internet transport protocols: UDP, TCP.	List out transport layer services, Name elements of transport protocols, illustrate SMTP, UDP and TCP, Differentiate UDP and TCP	CLO1, CLO2, CLO3	Lecture Discussion	Quiz Essay Short Question
Week 9, 10	DNS-domain name system, electronic mail, the world wide web, multimedia, Functions, protocols, Manager and agent, management components	List application layer activities, Understand and discuss E-mail, Discuss WWW and multimedia, Define Network management systems and its function and its role	CLO1, CLO2, CLO5	Lecture Discussion	Quiz Essay Short Question
Week 11, 12	Cryptography**,** plaintext, cipher text, cipher, symmetric and asymmetric key, ceaser cipher, substitution, transposition, XOR, shift, modern round cipher, DES, triple DES, RSA algorithm, Diffie Helman algorithm, man in the middle attack, security services, message and digest, Hash function, role of KDC, key management and digital signature	Define cryptography and security; differentiate between symmetric and asymmetric key cryptography, Traditional and modern ciphers, man in the middle attack concept, security services, Hash function, digital signature.	CLO1, CLO2, CLO3, CLO5	Lecture Discussion	Assignment Essay Short Question
Week 13, 14	Introduction to Internet of Things (IoT), IoT protocols, Importance of IoT	Describe what IoT is and how it works today, Discuss IoT protocols, Design and program IoT devices, Know the importance of	CLO3, CLO6	Lecture Discussion	Assignment Essay Short

		IoT			Question

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz, Short questions and Essay	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	8
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1
CLO6	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Data Communications and networking by Behrouz A. Forouzan.
2. Data and Computer Communication by William Stallings.
3. "The Internet of Things" by Samuel Greengard

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Computer Networks Lab

Part A- Introduction

III. Course code: CSE 3104

IV. Credit: 1.5

Course Summary	
This course is designed to impart knowledge about detailed knowledge of Computer Networks, various protocols used in Communication, Managing and configuring Cisco Switches and Routers, and various WAN technologies.	
Course Objectives	
1. Explain network technologies and how devices access local and remote networks. 2. Describe router hardware. 3. Explain how switching operates in a small to a medium-sized business network. 4. Design an IPv4 and IPv6 addressing scheme to provide network connectivity for a small to a medium-sized business network. 5. Configure initial settings on a network device using the Cisco command-line interface (CLI) 6. Implement basic network connectivity between devices.	
Course Learning Outcomes	
CLO1 Demonstrate a broad knowledge of the area of computer networking and its terminology	

CLO2 Design, implement and test the operation of a basic computer network

CLO3 Demonstrate an understanding of the operation of a range of networking protocols and devices

CLO4 Able to create a network for a small organization

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction of the simulator, networking devices, and cabling.	Demonstrate basic knowledge on the simulator, networking devices, and cabling.	CLO1, CLO2	Lecture Discussion	Essay Assignment Short Question
Week 2, 3, 4	● Different topology configuration with multiple networks using basic routing command. ● Introduction of Routing protocols and Configure static routing.	● Design, implement, and test the operation of a basic computer network topology. ● Demonstrate an understanding of the operation of the Static Routing Protocol.	CLO1, CLO2, CLO3,	Lecture Exercise Demonstration	Quiz Assignment Essay Short Question
Week 5, 6	Design, Configuration, and Implementation of Dynamic Routing (RIP).	Demonstrate an understanding of the operation of the Dynamic Routing Protocol (RIP).	CLO1, CLO2, CLO4	Lecture Discussion	Quiz Assignment Essay Short Question
Week 7,	Design, Configuration, and Implementation of Dynamic	Demonstrate an understanding of the	CLO1, CLO2,	Lecture	Quiz

8	Routing (EIGRP)	operation of the Dynamic Routing Protocol (EIGRP).	CLO3	Discussion	Essay Short Question
Week 9, 10	Design, Configuration, and Implementation of Dynamic Routing (OSPF).	Demonstrate an understanding of the operation of the Dynamic Routing Protocol (OSPF).	CLO1, CLO2, CLO5	Lecture Discussion	Quiz Essay Short Question
Week 11, 12	Design, Configuration, and Implementation of Dynamic Host Configuration Protocol (DHCP).	Demonstrate an understanding of the operation of DHCP.	CLO1, CLO2, CLO3, CLO5	Lecture Discussion	Assignment Essay Short Question
Week 13, 14	Design, Configuration, and Implementation of Virtual Local Area Network (VLAN).	Demonstrate an understanding of the operation of VLAN.	CLO3, CLO6	Lecture Discussion	Assignment Essay Short Question

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz, Short questions and Essay	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	8
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbook

Cisco CCNA Study Guide v2.71, Aaron Balchunas

Reference Books

1. CCNA, Study Guide, 6th Edition, TodLammle

2. Cisco website (www.cisco.com) for technical data sheets of devices.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Microprocessors and Embedded System

Part A- Introduction

I. Course code: CSE 3105	
II. Credit: 3	

1. **Course Summary**	
Microprocessors and Microcontrollers course is intended to introduce the architecture, programming of microprocessors and interfacing various hardware circuits to microprocessors. The topics covered are architecture, addressing modes, instruction set of 8086, minimum and maximum mode operation of 8086, 8086 INSTRUCTION SET, Assembly language programming fundamentals, interfacing of static Ram, EPROM, DMA Controller, keyboard, display, stepper motor, A/D and D/A converter, data transmission,8251 USART, 8259 interrupt controller, data transmission, 8251 USART, modes of timer operation of 8051, programming of Real time control by using basic microcontroller, This course analyze the complete architectural, programming, interfacing details of 8086 microprocessor-8051 microcontroller.	
2. **Course Objectives**	
1. To develop an in-depth understanding of the operation of microprocessors. 2. To master the assembly language programming using concepts like assembler directives, procedures, macros, software interrupts etc. 3. To create an exposure to basic peripherals, its programming and interfacing techniques 4. To learn the activities of I/O devices with respect to CPU. 5. To acquire knowledge about microcontroller based system design.	
3. **Course Learning Outcomes**	

CLO1. Understand the architecture and program module of microprocessors and microcontrollers.

CLO2. Analyze Intel 8085/8086 architecture with explanation of internal organization of some popular microprocessors/microcontrollers, addressing modes, coprocessors.

CLO3. Apply knowledge and demonstrate programming proficiency using the various addressing modes and data transfer instructions of the target microprocessor and microcontroller

CLO4. Interface different external peripheral devices with microprocessors and microcontrollers.

CLO5. Develop an assembly language program for specified applications

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 ,2	Introduction and overview of microprocessors	-Students will learn overview of microprocessors, number system -Introduction to assembly language, brief concept of compiler, assembler, linker and debugger -Evolution of computers -ROM and RAM families - Bus structure	CLO1	Lecture	Quiz

Week 3, 4	Microcomputer Structure	-students will learn CPU modules and architecture, Bus configuration, CPU registers. -Interrupt and Interrupt Handling. - Read, Write operation, DMA	CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 5, 6	8085 and 8086 microprocessor	-Introduction to 8085 Architecture, Timing Diagram, Instruction Formats. -Introduction to 8086 Architecture, registers, pin diagram, physical address. -Instruction Formats, Instruction Set, Assembler Directives, Interrupts of 8086. -Features, Signals, I/O & Memory Interfacing	CLO2, CLO4	Lecture, Exercise Demonstration	Quiz, Assignment Exam
Week 7	Addressing modes, Coprocessors	Memory addressing technique of 8086, -learn the examples of addressing modes. -8087 coprocessor structure	CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 8, 9	Introduction to assembly language and machine language programming	-students will learn the basics of the assembly language program. -Branching structure. -Learn about jump, CMP, if-then. -allocate assignment for the course.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 10	Assembly language, Stacks, Procedures.	-understand how looping structures work. -analyze SHIFT and ROTATE instructions. -understand the process of shifting and rotating.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 11, 12	8051 microcontroller architecture	-Hardware features, Architecture, Internal RAM structure. -Special Function Registers, Memory Organization, I/O Ports and Circuits. -Timers, Interrupts, Serial Communication, Interfacing of External Memory. -interfacing LCD & Keyboard, Real Time Clock.	CLO4, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 13, 14	8051 programming, Peripheral devices.	-Understand Addressing Modes, Instruction Set of 8051, Assembly Language Programming and C Programming, -A/D & D/A Interface; -USART (8251), Interrupt Controller (8259) Keyboard and Display Controller.	CLO4, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Attendance	Student’s participation in the class lecture, quiz and exam.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom’s Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	1
CLO3	1	1	1	0	0	0	0	1	1
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	1

<u>Part D- Resources</u>

Textbooks

1. M. Rafiquzzaman, "Microprocessors: Theory and Applications: Intel and Motorola", Revised ed., Prentice Hall, 1992 .
2. Ytha Yu, Charles Marut,"Assembly Language Programming and organization of IBM PC" Mc.Graw Hill International Edition.
3. D. V. Hall: "Microprocessors and Interfacing: Hardware and Software".

Reference Books

1. D. E. Knuth: “The Art of Computer Programming”, Vol. 1, Fundamental Algorithms.
2. Kenneth. J.Ayala.“The 8051 microcontroller”, 3rd edition,Cengage learning,2010.
3. N. Senthil Kumar, M. Saravanan and Jeevananthan (Oxford university press), “Microprocessors and microcontrollers”
4. B.Ram, Dhanpat Rai Publications, “Fundamentals of Microprocessors and Microcontrollers”

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Assembly Programming and Embedded System Lab

Part A- Introduction

I. Course code: CSE 3106	
II. Credit: 1.5	

1. Course Summary	
Microprocessors and Microcontrollers course is intended to introduce the architecture, programming of microprocessors and interfacing various hardware circuits to microprocessors. The topics covered are architecture, addressing modes, instruction set of 8086, minimum and maximum mode operation of 8086, 8086 INSTRUCTION SET, Assembly language programming fundamentals, interfacing of static Ram, EPROM, DMA Controller, keyboard, display, stepper motor, A/D and D/A converter, data transmission,8251 USART, 8259 interrupt controller, data transmission, 8251 USART, modes of timer operation of 8051, programming of Real time control by using basic microcontroller, This course analyze the complete architectural, programming, interfacing details of 8086 microprocessor-8051 microcontroller.	

2. **Course Objectives**	

1. To develop an in-depth understanding of the operation of microprocessors.

2. To master the assembly language programming using concepts like assembler directives, procedures, macros, software interrupts etc.

3. To create an exposure to basic peripherals, its programming and interfacing techniques

4. To learn the activities of I/O devices with respect to CPU.

5. To acquire knowledge about microcontroller based system design.

3. **Course Learning Outcomes**	

CLO1. Understand the architecture and program module of microprocessors and microcontrollers.

CLO2. Analyze Intel 8085/8086 architecture with explanation of internal organization of some popular microprocessors/microcontrollers, addressing modes, coprocessors.

CLO3. Apply knowledge and demonstrate programming proficiency using the various addressing modes and data transfer instructions of the target microprocessor and microcontroller

CLO4. Interface different external peripheral devices with microprocessors and microcontrollers.

CLO5. Develop and implement an assembly language program for specified applications

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 ,2	Introduction and overview of microprocessors	-Students will learn overview of microprocessors, number system -Introduction to assembly language, brief concept of compiler, assembler, linker and debugger -Evolution of computers -ROM and RAM families - Bus structure	CLO1	Lecture	Quiz
Week 3, 4	Microcomputer Structure	-students will learn CPU modules and architecture, Bus configuration, CPU registers. -Interrupt and Interrupt Handling. - Read, Write operation, DMA	CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 5, 6	8085 and 8086 microprocessor	-Introduction to 8085 Architecture, Timing Diagram, Instruction Formats. -Introduction to 8086 Architecture, registers, pin diagram, physical address.	CLO2, CLO4	Lecture, Exercise Demonstration	Quiz, Assignment Exam

		-Instruction Formats, Instruction Set, Assembler Directives, Interrupts of 8086. -Features, Signals, I/O & Memory Interfacing			
Week 7	Addressing modes, Coprocessors	Memory addressing technique of 8086, -learn the examples of addressing modes. -8087 coprocessor structure	CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 8, 9	Introduction to assembly language and machine language programming	-students will learn the basics of the assembly language program. -Branching structure. -Learn about jump, CMP, if-then. -allocate assignment for the course.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 10	Assembly language, Stacks, Procedures.	-understand how looping structures work. -analyze SHIFT and ROTATE instructions. -understand the process of shifting and rotating.	CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 11, 12	8051 microcontroller architecture	-Hardware features, Architecture, Internal RAM structure. -Special Function Registers, Memory Organization, I/O Ports and Circuits. -Timers, Interrupts, Serial Communication, Interfacing of External Memory.	CLO4, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

		-interfacing LCD & Keyboard, Real Time Clock.			
Week 13, 14	8051 programming, Peripheral devices.	-Understand Addressing Modes, Instruction Set of 8051, Assembly Language Programming and C Programming, -A/D & D/A Interface; -USART (8251), Interrupt Controller (8259) Keyboard and Display Controller.	CLO4, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.

Attendance	Student's participation in the class lecture, quiz and exam.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	8
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10

Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	1
CLO3	1	1	1	0	0	0	0	1	1
CLO4	1	1	1	0	0	0	0	1	1
CLO5	1	1	1	0	0	0	0	1	1

Part D- Resources

Textbooks

1. M. Rafiquzzaman,"Microprocessors: Theory and Applications: Intel and Motorola", Revised ed., Prentice Hall, 1992 .
2. Ytha Yu, Charles Marut,"Assembly Language Programming and organization of IBM PC" Mc.Graw Hill International Edition.
3. D. V. Hall: "Microprocessors and Interfacing: Hardware and Software".

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2. Kenneth. J.Ayala."The 8051 microcontroller", 3rd edition,Cengage learning,2010.
3. N. Senthil Kumar, M. Saravanan and Jeevananthan (Oxford university press), "Microprocessors and microcontrollers"
4. B.Ram, Dhanpat Rai Publications, "Fundamentals of Microprocessors and Microcontrollers"

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Engineering

Part A- Introduction

I. **Course code: CSE 3107**

II. **Credit: 3**

Introductory concepts: Classification of software products; Software products attributes; The need of software engineering; System concepts and the information systems environment, information needs, the concept of MIS, the system development life cycle, the role of system analyst.

System analysis and design: System planning and the initial investigation, information gathering, feasibility study, cost/benefit analysis. The process and stages of systems design, input/output design, organization and data base design, SDLC, OO design, system testing.

Software Process: Software engineering process, methods; Generic view of software engineering; Software process models..

Requirements and Specification: Requirements Engineering; Requirements analysis; System models; Requirements definition and specification; Software prototyping; Model-based specification.

Project Management: Overview and importance of project management; Project management activities; Project planning.

Software metrics: Measuring software; Lines of Code and Function Points; Metrics and software quality.

Risk Analysis and Management: Reactive versus proactive risk strategies; Software risks; Risk identification; Risk projection; Risk refinement.

Verification and Validation: Software Quality, Software testing fundamentals; Testing methods and strategies, Version control.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Engineering Lab

Part A- Introduction

I. **Course code: CSE-3108**
II. **Credit: 1.5**

Introductory concepts: Classification of software products; Software products attributes; The need of software engineering; System concepts and the information systems environment, information needs, the concept of MIS, the system development life cycle, the role of system analyst.

System analysis and design: System planning and the initial investigation, information gathering, feasibility study, cost/benefit analysis. The process and stages of systems design, input/output design, organization and data base design, SDLC, OO design, system testing.

Software Process: Software engineering process, methods; Generic view of software engineering; Software process models..

Requirements and Specification: Requirements Engineering; Requirements analysis; System models; Requirements definition and specification; Software prototyping; Model-based specification.

Project Management: Overview and importance of project management; Project management activities; Project planning.

Software metrics: Measuring software; Lines of Code and Function Points; Metrics and software quality.

Risk Analysis and Management: Reactive versus proactive risk strategies; Software risks; Risk identification; Risk projection; Risk refinement.

Verification and Validation: Software Quality, Software testing fundamentals; Testing methods and strategies, Version control.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Web Development Lab

Part A- Introduction

I. **Course code: CSE-3110**

II. **Credit: 1**

Development of a Website based on PHP or ASP.Net.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Technology Transfer Policy and Professional Ethics

Part A- Introduction

I. **Course code: HUM 3101**

II. **Credit: 2**

11.1	**Rationale:**
To work in an organization one has to follow different types of ethical guides, rules, regulations, and policies. An IT expert would face more than theses because there are some additional due to the information technology hardware and software. In this course a computer science student will learn these essential things.	
11.2	**Objectives:**
1.	Understand basics of rules and regulations.
2.	Know the different factors of ethics and morality.
3.	Realize details about professional ethics codes

11.3		**11.4**	**11.5**	**11.6**
Learning Outcomes		**Course Content**	**Teaching Strategy/ Learning Experience**	**Assessment Strategy**
a.	Analyze Technology Transfer & Transmission Process	**Theory and Practice:** Entrepreneurship and Innovation; Technology Transfer & Transmission Process; Technology Commercialization Process; Role of Intellectual Property in Protecting Innovation	Lecture, Demonstration, Exercise, Case Study, IT industry visiting	Exercise, Assignment, Report Writing, Quiz.
b.	Identify Role of Intellectual Property in Protecting Innovation			

a.	Learn Customer Needs Driven Product Specifications	**Technology and Market Assessment:** Customer Needs Driven Product Specifications; Negotiating the Deal and Marketing the Innovation; Financial Plan and Selection of Innovation Projects; Innovation and Risk Management; Technology Valuation and Impact Analysis; Market Assessment and Alignment of Technology.	Lecture, Demonstration, Exercise, Case Study, IT industry visiting	Exercise, Assignment, Report Writing, Quiz.
b.	Learn to Negotiating Deal and Marketing the Innovation			
c.	Analyze Market Assessment and Alignment of Technology			
a.	Understanding Business Plan Science and Technology Policy	**Commercialization Strategy**: Coming Full Circle in the Commercialization Loop; Business Plan Science and Technology Policy; Negotiating and Monitoring the Licensing Agreement; Start-Up and Spin-Off Companies; Joint Venture.	Lecture, Demonstration, Exercise, Case Study, IT industry visiting	Exercise, Assignment, Report Writing, Quiz.
a.	Understanding the facts of rules and regulations of an organization	**Professional Ethics:** Egoism and Relativism, Rationalist Ethics, the Ethics of Character and Virtue, Ethics and Religion, Ethics and Culture, Professional Ethics Codes. Morality and moral thoughts, Responsibility, Interpersonal moral sentiments (anger, blame, shame, guilt and praise), Reason, Emotion, and Intuition in Moral Judgment, Confidentiality, privacy and harassment.	Lecture, Demonstration, Exercise, Case Study, IT industry visiting	Exercise, Assignment, Report Writing, Quiz.
b.	Identifying the impacts of social, religious, family, friends, working environment and etc. on a person's ethics and morality			
c.	Finding reasons of being morally and ethically corrupted in work places			

d.	Gathering knowledge about the differences			

Recommended Books And Periodicals		
	Authors	**Book Name**
1.	George Reynolds	Ethics in Information Technology
2.	Herman T. Tavani and Richard A. Spinello	Readings in Cyberethics
3.	Jacques Berleur	Ethics of Computing: Codes, Spaces for Discussion and Law
4.	Robert Schultz	Contemporary Issues in Ethics and Information Technology
5.	Perry Morrison and Tom Forester	Computer Ethics: Cautionary Tales and Ethical Dilemmas in Computing
6.	Banks McDowell	Ethics and Excuses: The Crisis in Professional Responsibility
7.	Allen, Catherine; Bunting, Robert	A Global Standard for Professional Ethics: Cross-Border Business Concerns
8.	Michael Davis; Andrew Stark	Conflict of Interest in the Professions
9.	Justin Oakley; Dean Cocking	Virtue Ethics and Professional Roles
10.	Richard Rowson	Working Ethics: How to Be Fair in a Culturally Complex World
11.	Pettifor, Jean L.; Paquet, Stephanie	Preferred Strategies for Learning Ethics in the Practice of a Discipline

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Artificial Intelligence and Machine Learning

Part A- Introduction

III. **Course code: CSE 3201**

IV. **Credit: 3**

1. **Course Summary**

To build automated systems and modern IT solutions we need to include artificial intelligence so that it can interact dynamically facilitating customers or optimize the number of employees needed. So every computer science graduate needs sound knowledge in artificial intelligence.

2. **Course Objectives**

1. To understand intelligent agents and environments.
2. To understand and implement informed and uninformed search algorithms.
3. To understand neural networks.
4. To acquire proper knowledge in reasoning.

3. **Course Learning Outcomes**

CLO1. Examine the basic idea of Artificial Intelligence and Machine learning
CLO2. Define intelligent agents
CLO3. Illustrate different search techniques
CLO4. Interpret Knowledge based reasoning and expert systems
CLO5. Apply logic programming

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction: Introduction to AI and intelligent agents.	Describe AI, agent and environments	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 2, 3	Problem Solving: Solving Problems by Searching, Search Strategies, Heuristic	Solve advanced searching problems	CLO1, CLO2	Lecture, Exercise	Quiz Assignment

	search techniques, Game Playing				Exam
Week 4, 5	Knowledge and Reasoning: Building a Knowledge Base Agent, Propositional logic, First order logic, Inference in First order Logic.	Produce reasoning	CLO2, CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 6	Logic Programming: Logic programming using PROLOG, LISP	Examine logical programming	CLO5	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 7,8	Logical Action: Planning, partial order planning, Knowledge Engineering for Planning, Conditional Planning, A Replanning Agent.	Elaborate logical actions	CLO5	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 9	Uncertain Knowledge and Reasoning: Uncertainty, Probabilistic Reasoning Systems, Fuzzy Logic, Making Simple Decisions	Discover knowledge	CLO4	Lecture Exercise	Quiz Assignment Exam
Week 10, 11, 12	Knowledge Acquisition: Overview of different forms of learning, Learning Decision Trees, Neural Networks, Genetic Algorithms, Intelligent Editors, Introduction to Natural Language Processing	Apply new knowledge	CLO4	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 13, 14	Selected topics in AI: Expert consultation, Development of Expert Systems, Pattern recognition, Computer vision, Robotics	Learn expert system	CLO3, CLO4	Lecture Exercise	Quiz Assignment Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	0	0	0	0
CLO2	1	1	0	0	0	0	0	0	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	1	0	0	0	0	0	1	0
CLO5	1	1	0	0	0	0	0	0	0

Part D- Resources

Textbooks

1. Artificial Intelligence: A Modern Approach by Peter Norvig and Stuart Russel

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Artificial Intelligence and Machine Learning Lab

Part A- Introduction

V. **Course code: CSE 3202**

VI. **Credit: 1.5**

4. Course Summary

To build automated systems and modern IT solutions we need to include artificial intelligence so that it can interact dynamically facilitating customers or optimize the number of employees needed. So every computer science graduate needs sound knowledge in artificial intelligence.

5. Course Objectives

5. To understand intelligent agents and environments.
6. To understand and implement informed and uninformed search algorithms.
7. To understand neural networks.
8. To acquire proper knowledge in reasoning.

6. Course Learning Outcomes

CLO6. Examine the basic idea of Artificial Intelligence and Machine learning
CLO7. Define intelligent agents
CLO8. Illustrate different search techniques
CLO9. Interpret Knowledge based reasoning and expert systems
CLO10. Apply logic programming

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1	Introduction: Introduction to AI and intelligent agents.	Describe AI, agent and environments	CLO1, CLO2	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 2, 3	Problem Solving: Solving Problems by Searching, Search Strategies, Heuristic search techniques, Game Playing	Solve advanced searching problems	CLO1, CLO2	Lecture, Exercise	Quiz Assignment Exam
Week 4, 5	Knowledge and Reasoning: Building a Knowledge Base Agent, Propositional logic, First order logic, Inference in First order Logic.	Produce reasoning	CLO2, CLO3	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 6	Logic Programming: Logic programming using PROLOG, LISP	Examine logical programming	CLO5	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 7,8	Logical Action: Planning, partial order planning, Knowledge Engineering for Planning, Conditional Planning, A Replanning Agent.	Elaborate logical actions	CLO5	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 9	Uncertain Knowledge and Reasoning: Uncertainty, Probabilistic Reasoning Systems, Fuzzy Logic, Making Simple Decisions	Discover knowledge	CLO4	Lecture Exercise	Quiz Assignment Exam
Week 10, 11, 12	Knowledge Acquisition: Overview of different forms of learning, Learning Decision Trees, Neural Networks, Genetic Algorithms, Intelligent Editors, Introduction to Natural Language Processing	Apply new knowledge	CLO4	Lecture, Exercise, Handouts	Quiz Assignment Exam
Week 13,	Selected topics in AI: Expert consultation,	Learn expert system	CLO3,	Lecture	Quiz

14	Development of Expert Systems, Pattern recognition, Computer vision, Robotics		CLO4	Exercise	Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	

Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	**PLO10**	**PLO11**	**PLO12**	**PLO13**	**PLO14**	**PLO15**	**PLO16**	**PLO17**	**PLO18**
CLO1	1	0	0	0	0	0	0	0	0
CLO2	1	1	0	0	0	0	0	0	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	1	0	0	0	0	0	1	0
CLO5	1	1	0	0	0	0	0	0	0

Part D- Resources

Textbooks

2. Artificial Intelligence: A Modern Approach by Peter Norvig and Stuart Russel

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Compiler Design and Automata Theory

Part A- Introduction

I. Course code: CSE 3203

II. Credit: 3

1. Course Summary

Computer Engineers should be competent in compiler design and automata theory. They must learn the fundamental concepts of compiler design and automata theory and also various phases in the design of a compiler, how to generate a machine code from a C program statement.

2. Course Objectives

1.To design a LEX compiler

2.To construct a DFA from the NFA

3.To design an NFA for the corresponding regular expressions.

3. Course Learning Outcomes

CLO1. Basic knowledge of compilation steps; ability to apply automata theory and knowledge on formal languages.

CLO2. Ability to design and implement scanner modules in compilers.

CLO3. Ability to identify and select suitable parsing strategies for a compiler for various cases. Knowledge in alternative methods (top-down or bottom-up etc.).

CLO4. Knowledge and ability to devise, select, and use modern techniques and tools needed to design and implement compilers.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	**Introduction:** Phases of a compiler (lexical analyzer, syntax analyzer, semantic analyzer, intermediate code generator, code optimizer, code generator, symbol-table manager & error handler), overview of C, C++, Java, C# compilers.	Explain Phases of a compiler Explain intermediate code generator Overview of compilers	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Lexical analysis:** Role, finite automata, from regular expression to NFA, from NFA to DFA.	Construct NFA from Regular Expression Design DFA from NFA Design of a lexical analyzer generator using LEX	CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	**Syntax analysis:** Role, CFG, writing a grammar, top-down parsing, bottom-up parsing, operator precedence parsing, LR parser, using ambiguous grammar. Symbol table, structure and management.	Classify different types of parsing Design different types of grammar	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9 **Week 10**	**Intermediate code generation:** Intermediate languages, declarations, assignment statement, Boolean	Explain different types of intermediate code generation	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	expression, case statements, back patching, procedure calls.				
Week 11 **Week 12**	**Code generation:** Issues in the design of a code generator, target machine, runtime storage management, basic blocks and flow graphs, register allocation and assignment, dag representation of basic blocks, peephole optimizations, generating code from dags.	Discuss the several issues in the design of a code generator Draw a basic block & corresponding flow graphs for the given three-address statements	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	**Code optimization:** principle of source optimization, optimization of basic blocks, loop in flow graphs, global data flow analysis, iterative solution of data flow equations.	Role of the code optimization in compiler design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.

Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

1. Compilers Principles, Techniques and tools. Third edition by Alfred V.Aho, Ravi Sethi, Jeffrey D.Ullman
2. Introduction to Automata Theory, Languages and Computation by Hopcroft and Ullman

Reference Books

1. Compiler Design Theory by Lewis and Stern

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Compiler Design and Automata Theory Lab

Part A- Introduction

III. Course code: CSE 3204

IV. Credit: 1.5

4. Course Summary

Computer Engineers should be competent in compiler design and automata theory. They must learn the fundamental concepts of compiler design and automata theory and also various phases in the design of a compiler, how to generate a machine code from a C program statement.

5. Course Objectives

1.To design a LEX compiler

2.To construct a DFA from the NFA

3.To design an NFA for the corresponding regular expressions.

6. Course Learning Outcomes

CLO5. Basic knowledge of compilation steps; ability to apply automata theory and knowledge on formal languages.

CLO6. Ability to design and implement scanner modules in compilers.

CLO7. Ability to identify and select suitable parsing strategies for a compiler for various cases. Knowledge in alternative methods (top-down or bottom-up etc.).

CLO8. Knowledge and ability to devise, select, and use modern techniques and tools needed to design and implement compilers.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	**Introduction:** Phases of a compiler (lexical analyzer, syntax analyzer, semantic analyzer, intermediate code generator, code optimizer, code generator, symbol-table manager & error handler), overview of C, C++, Java, C# compilers.	Explain Phases of a compiler Explain intermediate code generator Overview of compilers	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Lexical analysis:** Role, finite automata, from regular expression to NFA, from NFA to DFA.	Construct NFA from Regular Expression Design DFA from NFA Design of a lexical analyzer generator using LEX	CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8**	**Syntax analysis:** Role, CFG, writing a grammar, top-down parsing, bottom-up parsing, operator precedence parsing, LR parser, using ambiguous grammar. Symbol table,	Classify different types of parsing Design different types of grammar	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	structure and management.				
Week 9 **Week 10**	**Intermediate code generation:** Intermediate languages, declarations, assignment statement, Boolean expression, case statements, back patching, procedure calls.	Explain different types of intermediate code generation	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12**	**Code generation:** Issues in the design of a code generator, target machine, runtime storage management, basic blocks and flow graphs, register allocation and assignment, dag representation of basic blocks, peephole optimizations, generating code from dags.	Discuss the several issues in the design of a code generator Draw a basic block & corresponding flow graphs for the given three-address statements	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 **Week 14**	**Code optimization:** principle of source optimization, optimization of basic blocks, loop in flow graphs, global data flow analysis, iterative solution of data flow equations.	Role of the code optimization in compiler design	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

<u>Part C- Assessment and Evaluations</u>

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

3. Compilers Principles, Techniques and tools. Third edition by Alfred V.Aho, Ravi Sethi, Jeffrey D.Ullman
4. Introduction to Automata Theory, Languages and Computation by Hopcroft and Ullman

Reference Books

2. Compiler Design Theory by Lewis and Stern

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Peripherals, Interfacing and IoT

Part A- Introduction

I. **Course code: CSE 3205**

II. **Credit: 3**

1 Course Summary

A computer engineer must have adequate practical knowledge of the various methods and techniques used to interconnect peripheral devices to computers.

2 Course Objectives

1. To understand the principles used in interfacing devices to computers and will gain a practical understanding of how those principles are put to use by manufacturers.

2. To assign I/O addresses, IRQs, DMA channels, and other I/O related parameters in installing real equipment.

3 Course Learning Outcomes

CLO1: Become familiar with the operation of a sophisticated computer system, including high- performance peripheral interfaces, extensive signal processing and graphics software.

CLO2: Be familiar with the different types of interrupt structures.

CLO3: Be able to select appropriate and compatible computer/peripherals combinations.

CLO4: Have a working knowledge of digital communication interface adapters.

CLO5: Be able to interface a microcontroller to various devices.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1 **Week 2** **Week 3**	Design and operation of interface between computer and the outside world	Design and explain the interface between computer and the outside world	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	Sensors, transducers ad signal conditioning circuits, interfacing memory and I/O devices-such as monitors, printers, disc drives, optical displays, some special purpose interface cards, stepper motors and peripheral devices.	Relate the operation of various devices to interfacing	CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 7 **Week 8** **Week 9** **Week 10**	IEEE-488, RS-232 and other buses.	Define and explain IEEE-488, RS-232 and other buses	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 **Week 12** **Week 13** **Week 14**	Study and applications of peripheral chips including 8212, 8155, 8255, 8251, DMA controllers.	Examine various peripheral chips	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO19	PLO20	PLO21	PLO22	PLO23	PLO24	PLO25	PLO26	PLO27
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

Jyoti Snehi	Computer Peripherals and Interfacing
Amit Karma	Computer Peripherals and Interfacing

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Peripherals, Interfacing and IoT Lab

Part A- Introduction

III. Course code: CSE 3206

IV. Credit: 1.5

4 Course Summary

A computer engineer must have adequate practical knowledge of the various methods and techniques used to interconnect peripheral devices to computers.

5 Course Objectives

1. To understand the principles used in interfacing devices to computers and will gain a practical understanding of how those principles are put to use by manufacturers.
2. To assign I/O addresses, IRQs, DMA channels, and other I/O related parameters in installing real equipment.

6 Course Learning Outcomes

CLO1: Become familiar with the operation of a sophisticated computer system, including high- performance peripheral interfaces, extensive signal processing and graphics software.

CLO2: Be familiar with the different types of interrupt structures.

CLO3: Be able to select appropriate and compatible computer/peripherals combinations.

CLO4: Have a working knowledge of digital communication interface adapters.

CLO5: Be able to interface a microcontroller to various devices.

Part B- Lesson Plan

Course Details Learning plan

Timeline	**Topics /contents**	**Learning Outcomes**	**Mapped CLOs**	**Teaching Strategies**	**Assessment Strategies**
Week 1 **Week 2** **Week 3**	Design and operation of interface between computer and the outside world	Design and explain the interface between computer and the outside world	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	Sensors, transducers ad signal conditioning circuits, interfacing memory and I/O devices-such as monitors, printers, disc drives, optical displays, some special purpose interface cards, stepper motors and peripheral devices.	Relate the operation of various devices to interfacing	CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8** **Week 9** **Week 10**	IEEE-488, RS-232 and other buses.	Define and explain IEEE-488, RS-232 and other buses	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 11 **Week 12** **Week 13** **Week 14**	Study and applications of peripheral chips including 8212, 8155, 8255, 8251, DMA controllers.	Examine various peripheral chips	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2	1	0	
Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	

Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO28	PLO29	PLO30	PLO31	PLO32	PLO33	PLO34	PLO35	PLO36
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

Jyoti Snehi	Computer Peripherals and Interfacing
Amit Karma	Computer Peripherals and Interfacing

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Development Project-II

Part A- Introduction

V. Course code: CSE 3208

VI. Credit: 1.5

7 Course Summary

Computer Engineers should be competent in web application software through different web oriented language. This C#.net /ASP.net/Java Programming Knowledge is valuable to both beginners and advanced developers that already have experience in developing applications software.

8 Course Objectives

1. Create and populate Windows Forms.

9 Course Learning Outcomes

CLO1: Become familiar with the operation of a sophisticated computer system, including high- performance peripheral interfaces, extensive signal processing and graphics software.

CLO2: Be familiar with the different types of interrupt structures.

CLO3: Be able to select appropriate and compatible computer/peripherals combinations.

CLO4: Have a working knowledge of digital communication interface adapters.

CLO5: Be able to interface a microcontroller to various devices.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies

Week 1 **Week 2** **Week 3**	**C# .NET Language Basics** Data Types, Type Conversion, Boxing & Unboxing, Conditional Statements, Looping, Methods in C#, Properties, Arrays, Indexers, Structures, Enumerations	To Apply OOP Knowledg e	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4 **Week 5** **Week 6**	**Memory Management:** Garbage Collector, Stack and Heap, System. GC Class.	To Apply OOP Knowledge	CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7 **Week 8** **Week 9** **Week 10**	**Developing Microsoft.NET Applications for Windows (Visual C#.NET)** Creating a Form Adding Controls to a Form Creating an Inherited Form Organizing Controls on a Form Creating MDI Application **Working with Controls** Creating an Event Handler for a Control Using Windows Forms Controls Using Dialog Boxes Application Adding Controls at Run Time Creating Menus Validating User Input **Using Data in Windows Forms Applications** Adding ADO.NET Objects to and Configuring ADO.NET	To Apply .NET Knowledg e	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	Objects in a Windows Forms Application Accessing and Modifying Data by Using Datasets Binding Data to Controls Overview of XML Web Services Persisting Data				
Week 11 **Week 12** **Week 13** **Week 14**	**Developing Microsoft.NET Applications for Windows (Visual C#.NET)** Printing and Reporting in Windows Forms Applications Lessons Printing From a Windows Forms Application Using the Print Preview, Page Setup, and Print Dialogs Constructing Print Document Content by Using GDI+ Creating Reports by Using Crystal Reports Deploying Windows Forms Applications NET Assemblies	To Apply .NET Knowledge	CLO3, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2	1	0	

Understand	5	1	1	
Apply	3	1	1	
Analyze	2	1	1	
Evaluate	4	1	1	
Create	4	0	1	

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	**PLO37**	**PLO38**	**PLO39**	**PLO40**	**PLO41**	**PLO42**	**PLO43**	**PLO44**	**PLO45**
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

Textbooks

W3school.com

On line tutorial

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Linear Algebra

Part A- Introduction

I. **Course code: MATH 3202**

II. **Credit: 3**

Course Learning Outcomes: ***After the successful completion of the course, students will be able to:***

CLO1	Explain basic concepts of Matrices and their significance in engineering field.
CLO2	Understand and explain basic concepts of systems of linear equations.
CLO3	Learn determinant of a 2x2 Matrix, determinant of a 3x3 matrix.
CLO4	Understand and describe importance of Inverse in solving Linear Systems, Inverse of a 2x2 Matrix.
CLO5	Explain scalars and vectors, geometrical representation of Vectors, vector Addition and subtraction.
CLO6	Understand and describe introduction to vector spaces, Euclidean Vector Spaces
CLO7	Explain Linear Dependence.
CLO8	Understand and describe Eigenvalues and Eigenvectors.

Mapping of CLOs with PLOs

CLO/PLO	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7
CLO1			2			3	2
CLO2	3	2	3			3	3
CLO3	3	3	3	3			3
CLO4		3	3		3	3	3
CLO5	3	3			3	3	2
CLO6	2		3		3	2	2
CLO7			3		2	2	3
CLO8	2				2	3	

S N	Course Content	Hrs	CLOs
1.	Matrices and their Significance, Matrix Notation, Dimension (Order) of a Matrix, Addressing Elements of a Matrix, Solving Linear Systems in 2 Unknowns, Types of Matrices, Addition and Subtraction of Matrices, Multiplication of Scalars with Matrices, Multiplication of two Matrices	**6**	**CLO1**
2.	Systems of Linear Equations Preview, Elementary Row Operations, Row Echelon Form (REF)	**6**	**CLO2**
3.	Determinant of a 2x2 Matrix, Determinant of a 3x3 Matrix, Finding Determinants Quickly	**5**	**CLO3,**
4.	Inverse exists only for Square Matrices, Singular Matrices, Importance of Inverse in solving Linear Systems, Inverse of a 2x2 Matrix, Inverse of a 3x3 Matrix - The Two Methods	**5**	**CLO4**
5.	Scalars and Vectors, Geometrical Representation of Vectors, Vector Addition and Subtraction, Laws of Vector Addition and Head to Tail Rule, Unit Vector	**5**	**CLO5**
6.	Introduction to Vector Spaces, Euclidean Vector Spaces - Part 1, Euclidean Vector Spaces - Part 2, Euclidean Vector Spaces - Part 3, Definition and Closure Properties, Axioms of Vector Spaces, Subspace and Null space	**4**	**CLO6**
7.	Linear Dependence - Introduction	**4**	**CLO7**
8.	Introduction to Eigenvalues and Eigenvectors.	**4**	**CLO8**

ASSESSMENT PATTERN

CIE- Continuous Internal Evaluation (28Marks)

Bloom’s Category Marks (out of 20)	Tests (10)	Assignments (05)	Quizzes (05)	External Participation in Curricular/Co-Curricular Activities(08)
Remember			05	
Understand		03		
Apply	04			8
Analyze	03			
Evaluate	03			
Create		02		

SEE- Semester End Examination (72 Marks)

Bloom’s Category	**Test**
Remember	15
Understand	15
Apply	15
Analyze	15
Evaluate	10
Create	02

Recommended References:

H. Anton and C.Rorres	: Linear Algebra with Applications.
Brestscher	: Linear Algebra with Applications.
Lipschutz, S.	: Linear Algebra
Haward A.	: Elementary Linear Algebra with Application

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Business Communication and Technical Writing

Part A- Introduction

I. **Course code: BUS 3202**

II. **Credit: 1.5**

Rationale:

A computer engineer must be able to communicate with others for business purpose. Moreover, to be a successful researcher, a computer engineer must be able to explain, analyze and justify different algorithms developed by practitioners and researchers in writing clearly.

Objectives:

- To make oneself comfortable with the stakeholders
- To communicate with others clearly both in writing and speaking.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Simulation and Modeling

Part A- Introduction

III. **Course code: CSE 3209**

IV. **Credit: 3**

Introduction: Concepts of a system; System Environment; Activities; Continuous and Discrete Systems.

System Modeling: Introduction to modeling; Types of models – Static Physical Models, Dynamic Physical Models, Static Mathematical Models, Dynamical Mathematical Models; Principles used in Modeling.

System Studies: Subsystem concepts; A corporate model – Environment Segment, Production Segment, Management Segment; Types of System Study – System Analysis, System Design, and System postulation.

System Simulation: The Technique of Simulation; Comparison of Simulation and Analytical Methods; Experimental Nature of Simulation; Parallel and Distributed Simulation; Real time Simulation Types of System Simulation;

Probability Concepts in Simulation: Stochastic Variables; Discrete and Continuous Probability Functions; Measures of Probability Functions; The Coefficient of Variation; Generation of Random Variates; Binomial Distribution; Poisson Distribution; Continuous Distribution; Normal Distribution; The Exponential Distribution; The Erlang and Hyper-Exponential Distributions; Uniform Distribution; Beta Distribution.

Random Numbers: Random Numbers Table; Pseudo Random Numbers; Computer Generation of Random Numbers; A Uniform and non-Uniform Continuously Distributed Random Numbers; Qualities of an efficient Random Number Generator.

Arrival Patterns and Service Times: Congestion in Systems; Arrival Patterns; Service Times; Queuing Disciplines, Measure of Queue and Mathematical Solutions of Queuing Problems.

Discrete System Simulation: Discrete Events; Representation of Time; Generation of Arrival Patterns; Simulation of a Telephone System; Simulation.

Analysis of Simulation Output: Nature of the Problem; Verification and Validation of Simulation; Estimation Methods.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Accounting Information and System

Part A- Introduction

I. **Course code: AIS 3201**

II. **Credit: 2**

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Viva-Voce

Part A- Introduction

I. **Course code: CSE-3200**

II. **Credit: 1**

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Computer Graphics and Multimedia

Part A- Introduction

. Course code: CSE4101

II. Credit: 3

1. Course Summary

Introduces the basics of graphics and computer based graphical display. The mathematical basics of the display mechanism and display primitives with their transformation characteristics are elaborately discussed

0. Course Objectives

1. Receive a basic knowledge of computer graphics.
2. Able to apply graphic primitives in different displays.
3. Learn about viewing and positioning in several software and hardware.
4. Apply multimedia system to implements graphics into them.

0. Course Learning Outcomes

1. Devise the primary idea to computer graphics and its primitives
2. Apply different drawing applications.
3. Learn about clipping algorithms
4. Elaborate the idea of transformation and object representations
5. Learn about lights and illumination

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2	**Introduction to Computer Graphics:** Introduction, Presentation graphics, Application Areas, GUI; Graphics Hardware: Display devices Architecture and Input Devices	Learn how graphical system works	CLO1	Lecture	Quiz Assignment Practical Exam
Week 3	**Graphic Primitives:** Drawing Points, Lines, Circles, Ellipse, Rectangles, Arcs; Polygons: Inside-outside tests, polygon fill	Learn and implement primitives	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	algorithms, Character generation.				
Week 4, 5	**Two-dimensional Viewing and Clipping:** Viewing pipeline, Window to view port transformation, Point, Line, Polygon, Curve and Text clipping.	Apply clipping	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 6, 7	**Transformations of Objects:** Basic transformations, Affine Transformations, Translations, Rotations, Scaling, reflection and Shearing, Composite transformations matrices, Transformation of 3D objects (4´4 matrices)	Apply transformation	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 8	**Curve and Surface design:** Interpolation and approximation techniques, B-spline, Bezier curves and Surfaces, Fractal Geometry.	Implement curve	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9, 10,11	**3D Object Representation:** 3D Graphics Pipeline, Projection: Different types of Parallel and Perspective Matrices; B-Rep, Constructive Solid Geometry, BSP tree, Octree, Hidden lines and Surface detection: Back face Detection, Painters algorithm, Z-buffering; light models.	Apply 3d transformation	CLO3, 4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 12, 13	**Illumination Models and Surface-Rendering:** Basic illumination models, ambient light, diffuse reflection, specular reflection, Constant, Goraud and Phong shading; Ray-tracing; Different Types of Color Model, Fractals and Texture Mapping.	Learn light and color	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 14	Variable length argument list; Command line parameters	Explain and apply argument list	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10

Evaluate	10
Create	12

Mapping of CLOs to Assessment
Mapping of CLOs to PLOs

CLOs	**1.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**	**0.**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks
1. Computer Graphics, Author: Donald Hearn and Paullin Baker
2. Computer Graphics, Principle and Practice, Author: Foley, Vandam, Feiner, Hughes
3. Computer Graphics & Geometric Modeling for Engineers, Author: Vera B Anand.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Computer Graphics and Multimedia Lab

Part A- Introduction

1 . Course code: CSE4102

II. Credit: 1.5

1. Course Summary

Introduces the basics of graphics and computer based graphical display in labratory. The mathematical basics of the display mechanism and display primitives with their transformation characteristics are elaborately discussed

0. Course Objectives

1. Receive a basic knowledge of computer graphics.
2. Able to apply graphic primitives in different displays.
3. Learn about viewing and positioning in several software and hardware.
4. Apply multimedia system to implements graphics into them.

0. Course Learning Outcomes

1. Implement the primary idea to computer graphics and its primitives
2. Apply different drawing applications.
3. Experiments the clipping algorithms
4. Elaborate the idea of transformation and object representations
5. Implement about lights and illumination

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2	**Introduction to Computer Graphics:** Introduction, Presentation graphics, Application Areas, GUI; Graphics Hardware: Display devices Architecture and Input Devices	Learn how graphical system works	CLO1	Lecture	Quiz Assignment Practical Exam
Week 3	**Graphic Primitives:** Drawing Points, Lines, Circles, Ellipse, Rectangles, Arcs; Polygons: Inside-outside tests, polygon fill	Learn and implement primitives	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	algorithms, Character generation.				
Week 4, 5	**Two-dimensional Viewing and Clipping:** Viewing pipeline, Window to view port transformation, Point, Line, Polygon, Curve and Text clipping.	Apply clipping	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 6, 7	**Transformations of Objects:** Basic transformations, Affine Transformations, Translations, Rotations, Scaling, reflection and Shearing, Composite transformations matrices, Transformation of 3D objects (4´4 matrices)	Apply transformation	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 8	**Curve and Surface design:** Interpolation and approximation techniques, B-spline, Bezier curves and Surfaces, Fractal Geometry.	Implement curve	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9, 10,11	**3D Object Representation:** 3D Graphics Pipeline, Projection: Different types of Parallel and Perspective Matrices; B-Rep, Constructive Solid Geometry, BSP tree, Octree, Hidden lines and Surface detection: Back face Detection, Painters algorithm, Z-buffering; light models.	Apply 3d transformation	CLO3, 4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 12, 13	**Illumination Models and Surface-Rendering:** Basic illumination models, ambient light, diffuse reflection, specular reflection, Constant, Goraud and Phong shading; Ray-tracing; Different Types of Color Model, Fractals and Texture Mapping.	Learn light and color	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 14	Variable length argument list; Command line parameters	Explain and apply argument list	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Attendance(Class participation)	Students should regularly attend the classes and actively participate in the class discussion.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Includes set of programming problems to solve within a timeline

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	**Test (30)**	**Attendance (10)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	**Test**
Remember	**5**
Understand	**10**
Apply	**25**
Analyze	**10**
Evaluate	**5**

Create	**5**

Mapping of CLOs to PLOs

CLOs	1.	0.	0.	0.	0.	0.	0.	0.	0.
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Computer Graphics, Author: Donald Hearn and Paullin Baker
2. Computer Graphics, Principle and Practice, Author: Foley, Vandam, Feiner, Hughes
3. Computer Graphics & Geometric Modeling for Engineers, Author: Vera B Anand.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Computer Security

Part A- Introduction

I. Course code: CSE 4105

II. Credit: 3

1. Course Summary

To contribute to the ICT based innovative model, one needs to create a random number, justify randomness and correlations in the system and apply probability distribution in the system.

2. Course Objectives

1. To understand and apply simulations in different systems.
2. To create a random number and evaluate the randomness in the system.
3. To understand different modeling methods with its applications.

3. Course Learning Outcomes

CLO1. Describe the system
CLO2. Explain simulation of different systems
CLO3. Introduce various types of simulation
CLO4. Describe different distribution
CLO5. Introduce and generate various types of random numbers
CLO6. Introduce with queuing system
CLO7. Introduce with discrete event system simulation

Part B- Lesson Plan

Course Details Learning plan

Timeline	**Topics /contents**	**Learning Outcomes**	**Mapped CLOs**	**Teaching Strategies**	**Assessment Strategies**
Week 1	Concepts of a system; System Environment; Activities; Continuous and Discrete Systems.	Describe the system and environment	CLO 1 CLO 2	Lecture, handouts	Quiz Participation Exam
Week 2 ,3, 4	The Technique of Simulation; Comparison of Simulation and Analytical Methods; Experimental Nature of Simulation; Parallel and Distributed Simulation; Real time Simulation Types of System Simulation.	Focus on analytical methods of simulation	CLO 3	Lecture, handouts	Quiz Participation Exam
Week 5, 6,7	Stochastic Variables; Discrete and Continuous Probability Functions; Measures of Probability Functions; The Coefficient of Variation; Generation of Random Variates; Binomial Distribution; Poisson Distribution; Continuous Distribution; Normal Distribution; The Exponential Distribution; The Erlang and Hyper-Exponential Distributions; Uniform Distribution; Beta Distribution	Develop solid concept on different types of distribution	CLO 4	Lecture, handouts	Quiz Participation Exam
Week 8,9,10	Random Numbers Table; Pseudo Random Numbers; Computer Generation of Random Numbers; A Uniform and non-Uniform Continuously Distributed Random Numbers; Qualities of an efficient Random Number	Develop solid concept on random numbers	CLO 5	Lecture, handouts	Quiz Participation Exam

Week 11, 12	Congestion in Systems; Arrival Patterns; Service Times; Queuing Disciplines, Measure of Queue and Mathematical Solutions of Queuing Problems.	Develop solid concept on queuing methodologies	CLO 6	Lecture, handouts	Quiz Participation Exam
Week 13, 14	Discrete Events; Representation of Time; Generation of Arrival Patterns; Simulation of a Telephone System; Simulation	Develop solid concept on discrete events methodologies	CLO 7	Lecture, handouts	Quiz Participation Exam

Part C- Assessment and Evaluation

Assessment Procedures

Assessment Name	**Description**
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			

Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

	Attendance (8%)	Quiz (20%)	Final (72%)
CLO1	1	1	1
CLO2	1	1	1
CLO3	1	1	1
CLO4	1	1	1
CLO5	1	1	1
CLO6	1	1	1

CLO7	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Simulation Modeling and Analysis: FIFTH EDITION by Averill M. Law

Reference Books

1. Modelling and Simulation: Exploring Dynamic System Behaviour by Gilbert Arbez and Louis G. Birta

Other resources (Slides)

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Data Mining and Warehousing

Part A- Introduction

I. **Course code: CSE 4109**

II. **Credit: 3**

1. Course Summary

The amount of data collected across a wide variety of domains far exceeds our ability to reduce and analyze without the use of smart and intelligent analysis techniques. It is becoming a challenging task to obtain information from the massive collection of data. Data Science is a set of principles that support and guide the extraction of information and insight from data. Machine learning, on the other hand, is a branch of Artificial Intelligence that deals with the idea that systems can learn from data, identify patterns, and make decisions with minimal human interaction. This course presents the fundamentals of Data Science and Machine Learning. The concept of Data Science will help students to acquire knowledge on various data processing methods, approaches, tools and techniques. The section of Machine Learning will help the students to learn about different algorithms and approaches in order to build models in such a way that the models, when exposed to new data, will be able to produce reliable, repeatable decisions and results by learning from previous computation.

2. Course Objectives

1. To introduce students to data collection and extraction techniques.
2. To develop the mathematical and logical skill sets for data preprocessing.
3. To develop and enhance data visualization techniques.
4. To understand the concepts of machine learning approaches and algorithms.
5. To acquire the skill for building machine learning problems..

3. Course Learning Outcomes

CLO1. Introduction to data science
CLO2. Data preprocessing
CLO3. Data reduction
CLO4. Data transformation, and discretization and visualization
CLO5. Regression and classification analysis
CLO6. Classification
CLO7. Clustering and reinforcement learning

Part B- Lesson Plan

Course Details Learning plan

Timeline	**Topics /contents**	**Learning Outcomes**	**Mapped CLOs**	**Teaching Strategies**	**Assessment Strategies**
Week 1	History of data science, aims and objectives of data science, data, categorize of data, data sources, data collection and extraction.	To become familiar with data science and its applications To learn different data types in data science	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 2 ,3	Data quality, data cleaning techniques, dealing with missing values, handling noisy data, linear regression and non-linear regression..	To know the techniques of dealing with missing and noisy values To know other data preprocessin g techniques	CLO 2	Lecture, handouts	Quiz Participation Exam
Week 4, 5	Wavelet transformation, principal component analysis, attribute subset selection	To understand why is data reduction necessary To become familiar with data	CLO 3	Lecture, handouts	Quiz Participation Exam

		reduction techniques			
Week 6, 7	Overview, normalization, binning, histogram analysis, data visualization tools and techniques	To learn data transformation techniques To know data visualization tools and techniques	CLO 4	Lecture, handouts	Quiz Participation Exam
Week 8, 9	linear regression, logistic regression, maximum likelihood and least square, Bias variance decomposition, over fitting and under filling problems and solutions.	To obtain the fundamental understanding of regression and classification analysis	CLO 5	Lecture, handouts	Quiz Participation Exam
Week 10, 11, 12	Decision tree classifiers, Naïve Bayes classifiers, Neural networks, Support vector machine, Random forest, Nearest neighbor. Evaluation of classifiers. Ensemble methods	To learn the basic concepts and techniques of classification. To understand the mechanism of different classifiers To learn how to evaluate and compare different classifiers	CLO 6	Lecture, handouts	Quiz Participation Exam
Week 13, 14	Cluster analysis, partition method, hierarchical	To learn basic concepts and methods of	CLO 7	Lecture, handouts	Quiz Participation

	method, density based clustering, evaluation of clustering, reinforcement analysis and approaches, model evaluation techniques	cluster analysis. □To learn fundamental clustering techniques □To learn how to evaluate models			Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			

Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

	Attendance (8%)	Quiz (20%)	Final (72%)
CLO1	1	1	1
CLO2	1	1	1
CLO3	1	1	1
CLO4	1	1	1
CLO5	1	1	1
CLO6	1	1	1
CLO7	1	1	1

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9

CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning with application in R"
2. Trevor Hastie, Robert Tibshirani, Jerome Friedman "The elements of statistical learning".

Reference Books

1. Christopher Bishop "Pattern Recognition and Machine Learning"

Other resources (Slides)

1.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Data Mining and Warehousing Lab

Part A- Introduction

III. Course code: CSE 4110

IV. Credit: 3

4. Course Summary

The amount of data collected across a wide variety of domains far exceeds our ability to reduce and analyze without the use of smart and intelligent analysis techniques. It is becoming a challenging task to obtain information from the massive collection of data. Data Science is a set of principles that support and guide the extraction of information and insight from data. Machine learning, on the other hand, is a branch of Artificial Intelligence that deals with the idea that systems can learn from data, identify patterns, and make decisions with minimal human interaction. This course presents the fundamentals of Data Science and Machine Learning. The concept of Data Science will help students to acquire knowledge on various data processing methods, approaches, tools and techniques. The section of Machine Learning will help the students to learn about different algorithms and approaches in order to build models in such a way that the models, when exposed to new data, will be able to produce reliable, repeatable decisions and results by learning from previous computation.

5. Course Objectives

6. To introduce students to data collection and extraction techniques.
7. To develop the mathematical and logical skill sets for data preprocessing.
8. To develop and enhance data visualization techniques.
9. To understand the concepts of machine learning approaches and algorithms.
10. To acquire the skill for building machine learning problems..

6. Course Learning Outcomes

CLO8. Introduction to data science
CLO9. Data preprocessing
CLO10. Data reduction
CLO11. Data transformation, and discretization and visualization
CLO12. Regression and classification analysis
CLO13. Classification
CLO14. Clustering and reinforcement learning

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	History of data science, aims and objectives of data science, data, categorize of data, data sources, data collection and extraction.	To become familiar with data science and its applications To learn different data types in data science	CLO 1	Lecture, handouts	Quiz Participation Exam
Week 2 ,3	Data quality, data cleaning techniques, dealing with missing values, handling noisy data, linear regression and non-linear regression..	To know the techniques of dealing with missing and noisy values To know other data preprocessin g techniques	CLO 2	Lecture, handouts	Quiz Participation Exam
Week 4, 5	Wavelet transformation, principal component analysis, attribute subset selection	To understand why is data reduction necessary To become familiar with data	CLO 3	Lecture, handouts	Quiz Participation Exam

		reduction techniques			
Week 6, 7	Overview, normalization, binning, histogram analysis, data visualization tools and techniques	To learn data transformation techniques To know data visualization tools and techniques	CLO 4	Lecture, handouts	Quiz Participation Exam
Week 8, 9	linear regression, logistic regression, maximum likelihood and least square, Bias variance decomposition, over fitting and under filling problems and solutions.	To obtain the fundamental understanding of regression and classification analysis	CLO 5	Lecture, handouts	Quiz Participation Exam
Week 10, 11, 12	Decision tree classifiers, Naïve Bayes classifiers, Neural networks, Support vector machine, Random forest, Nearest neighbor. Evaluation of classifiers. Ensemble methods	To learn the basic concepts and techniques of classification. To understand the mechanism of different classifiers To learn how to evaluate and compare different classifiers	CLO 6	Lecture, handouts	Quiz Participation Exam
Week 13, 14	Cluster analysis, partition method, hierarchical	To learn basic concepts and methods of	CLO 7	Lecture, handouts	Quiz Participation

	method, density based clustering, evaluation of clustering, reinforcement analysis and approaches, model evaluation techniques	cluster analysis. □To learn fundamental clustering techniques □To learn how to evaluate models			Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			

Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

	Attendance (8%)	Quiz (20%)	Final (72%)
CLO1	1	1	1
CLO2	1	1	1
CLO3	1	1	1
CLO4	1	1	1
CLO5	1	1	1
CLO6	1	1	1
CLO7	1	1	1

Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18

CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

3. Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani. "An introduction to statistical learning with application in R"
4. Trevor Hastie, Robert Tibshirani, Jerome Friedman "The elements of statistical learning".

Reference Books

2. Christopher Bishop "Pattern Recognition and Machine Learning"

Other resources (Slides)

2.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Testing and Quality Assurance

Part A- Introduction

I. **Course code: CSE4111**

II. **Credit: 3**

1. Course Summary

This course provides an elementary introduction to software quality assurance and tests. Upon completion of this course, students will have the ability to:

2. Course Objectives

1. Present effective testing techniques (both black-box and white box) for ensuring high quality software
2. Learn metrics for managing quality assurance
3. Understand capabilities of test tools
4. Establish a testing group and manage the whole testing project;
5. Clearly and correctly report the software defectives;
6. Asses the software product correctly;
7. Distinguish relationship between the software testing and the quality assurance.

3. Course Learning Outcomes

CLO1. Present effective testing techniques (both black-box and white box) for
CLO2. ensuring high quality software
CLO3. Learn metrics for managing quality assurance
CLO4. Understand capabilities of test tools
CLO5. Establish a testing group and manage the whole testing project;
CLO6. Asses the software product correctly;

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Basic Testing Vocabulary Quality Assurance versus Quality Control The Cost of Quality Software Quality Factors How Quality is Defined Why Do We Test Software? What is a Defect? The Multiple Roles of the Software Tester(People Relationships) Scope of Testing When Should Testing Occur? Testing Constraints Life Cycle Testing Independent Testing What is a QA Process? Levels of Testing The "V" Concept of Testing	Know the basic concepts of Software Testing	CLO1	Lecture	Quiz
Week 2	Structural versus Functional Technique Categories Verification versus Validation Static versus Dynamic Testing Examples of Specific Testing Techniques	Apply Testing Techniques	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 3	Test Planning Customization of the Test Process Budgeting Scheduling	Know Test Administration	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5	Prerequisites to test planning Understand the Characteristics of the Software Being Developed Build the Test Plan Write the Test Plan	Create the Test Plan	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 6	Test Cases: Test case Design Building test cases Test data mining Test execution Test Reporting Defect Management Test Coverage – Traceability matrix	Understand Test Metrics – Guidelines and usage	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8,9	Guidelines for writing test reports Test Tools used to Build Test Reports	Apply Test reporting	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 10, 11, 12	Software Configuration Management Change Management	Learn how to Manage Change	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13, 14	Risks – Risk Analysis and Management with examples User Acceptance testing – in	Learn Risk Analysis and Management, User Acceptance	CLO4	Lecture Exercise	Quiz Assignment

	detail explanation with details	testing		Demonstration	Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			
Bloom's Category	**Test**			
Remember	05			

Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

SEE-Semester End Examination (72 Marks)

	Quiz (10%)	**Assignment (8%)**	**Practical (10%)**	**Final (72%)**
CLO1	1	0	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Designing Object-oriented, Author: R. Wirfs-Brock et. al.
2. Software Engineering, Author: Ian Sommerville

Reference Books

1. Software Engineering: A Practitioner's Approach, Author: R.S. Pressman
2. Writing Effective Use Cases, Author: Robert C. Martin

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Testing and Quality Assurance Lab

Part A- Introduction

I. **Course code: CSE4112**

II. **Credit: 1.5**

1. Course Summary

This course provides an elementary introduction to software quality assurance and tests lab. Upon completion of this course, students will have the ability to:

2. Course Objectives

1. Present effective testing techniques (both black-box and white box) for ensuring high quality software
2. Learn metrics for managing quality assurance
3. Understand capabilities of test tools
4. Establish a testing group and manage the whole testing project;
5. Clearly and correctly report the software defectives;
6. Asses the software product correctly;
7. Distinguish relationship between the software testing and the quality assurance.

3. Course Learning Outcomes

CLO1. Present effective testing techniques (both black-box and white box) for
CLO2. ensuring high quality software
CLO3. Learn metrics for managing quality assurance
CLO4. Understand capabilities of test tools
CLO5. Establish a testing group and manage the whole testing project;
CLO6. Asses the software product correctly;

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Basic Testing Vocabulary Quality Assurance versus Quality Control The Cost of Quality Software Quality Factors How Quality is Defined Why Do We Test Software? What is a Defect? The Multiple Roles of the Software Tester(People Relationships) Scope of Testing When Should Testing Occur? Testing Constraints Life Cycle Testing Independent Testing What is a QA Process? Levels of Testing The “V” Concept of Testing	Know the basic concepts of Software Testing	CLO1	Lecture	Quiz
Week 2	Structural versus Functional Technique Categories Verification versus Validation Static versus Dynamic Testing Examples of Specific Testing Techniques	Apply Testing Techniques	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 3	Test Planning Customization of the Test Process Budgeting Scheduling	Know Test Administration	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5	Prerequisites to test planning Understand the Characteristics of the Software Being Developed Build the Test Plan Write the Test Plan	Create the Test Plan	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 6	Test Cases: Test case Design Building test cases Test data mining Test execution Test Reporting Defect Management Test Coverage – Traceability matrix	Understand Test Metrics – Guidelines and usage	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8,9	Guidelines for writing test reports Test Tools used to Build Test Reports	Apply Test reporting	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 10, 11, 12	Software Configuration Management Change Management	Learn how to Manage Change	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13, 14	Risks – Risk Analysis and Management with examples User Acceptance testing – in	Learn Risk Analysis and Management, User Acceptance	CLO4	Lecture Exercise	Quiz Assignment

	detail explanation with details	testing		Demonstration	Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test

Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

	Quiz (10%)	**Assignment (8%)**	**Practical (10%)**	**Final (72%)**
CLO1	1	0	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO10**	**PLO11**	**PLO12**	**PLO13**	**PLO14**	**PLO15**	**PLO16**	**PLO17**	**PLO18**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

3. Designing Object-oriented, Author: R. Wirfs-Brock et. al.
4. Software Engineering, Author: Ian Sommerville

Reference Books

3. Software Engineering: A Practitioner's Approach, Author: R.S. Pressman
4. Writing Effective Use Cases, Author: Robert C. Martin

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Design Patterns and Principles

Part A- Introduction

I. **Course code: CSE 4115**

II. **Credit: 3**

1. **Course Summary**

Students will create project teams of 3 members each. Number of team members can be varied for special cases, decided by the assigned course manager. All the project teams are required to prepare their Software Requirements Specification (SRS) first, and later develop the project accordingly.

2. **Course Objectives**

1. Revision of concepts of OOP
2. Importance of learning design pattern

3. **Course Learning Outcomes**

CLO1. Learn to use design solutions based on design pattern
CLO2. Learn to implement design pattern
CLO3. Identify problems that can be solved using design patterns
CLO4. Becoming familiar with code smells
CLO5. Applying refactoring to the code smells

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Revision of Concepts of OOP	Design and Implement	CLO1	Lecture	Quiz
Week 2	Importance of learning design pattern	Design, Implement and Solve Real-life Problems	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 3	Types of design patterns - Structural, Behavioral and Creational Patterns	Design, Implement and Solve Real-life Problems	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5	Singleton, Factory, Factory Method, Abstract Factory, Builder, Prototype and Object Pool	Design, Implement and Solve Real-life Problems	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 6	Chain of Responsibility, Command, Interpreter, Iterator, Mediator, Memento, Observer, Strategy, Template method, visitor and Null Object	Design, Implement and Solve Real-life Problems	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8,9	Adapter, Bridge, Composite, Decorator, Flyweight and Proxy	Design, Implement and Solve Real-life Problems	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 10	REFACTORING CODE SMELL Introduction	Design, Implement and Solve Real-life Problems	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11, 12	Inappropriate Naming, Comments, Dead Code, Duplicated code, Primitive Obsession, Large Class, Lazy Class, Alternative Class with Different Interface, Long Method, Long Parameter List, Switch Statements, Speculative Generality, Oddball Solution, Feature Envy, Refused Bequest, Black Sheep and Train Wreck	Design, Implement and Solve Real-life Problems	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13, 14	Design Principles (SOLID) - Single responsibility principle, Open Close Principle, Liskov substitution principle, Interface segregation principle, Dependency Inversion principle.	Design, Implement and Solve Real-life Problems	CLO1, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05

Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

	Quiz (10%)	**Assignment (8%)**	**Practical (10%)**	**Final (72%)**
CLO1	1	0	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Gamma, Erich. Design patterns: elements of reusable object-oriented software. Pearson Education, 1995.

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Design Patterns and Principles Lab

Part A- Introduction

I. **Course code: CSE 4116**

II. **Credit: 1.5**

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: E-commerce and Management Information System

I. **Course Code : BUS 4101**

II. **Credit : 3.00**

Mark Distribution:

Semester Final Exam: 72 Marks   Class Test: 20 Marks   Class Attendance: 08 Marks

COURSE OBJECTIVES:

The students will be able

- To evaluate the role of information systems in transforming business and their relationship to globalization.
- To know about the significance of systems in improving organizational performance.
- To understand ethical, social, and political issues are raised by information systems.
- To know the principal tools and technologies for accessing information from databases to improve business performance and decision making.
- To know the functions of Internet and Internet technology and their support in communication, e-commerce and e-business.

COURSE OUTCOMES:

On successful completion of this course, students will be able to:

- Understand management information systems and their role in today's organizations.
- Identify how MIS shapes and controls current business world and improve the performances of organizations.
- Become familiar with the major trends in MIS infrastructures and how these evolutions will affect workplaces and business strategies.

- Understand the different ethical issues raised by information systems and their remedy.

Learning Outcomes		Course Content	Teaching Strategy/ Learning Experience	Assessment Strategy
a	Define Management Information System	Definition-Characteristics-MIS services-Limitations-Types of MIS-importance-Potential risks of information systems-data and Information-Difference between data and information-Quality of good information-The process of converting data into information.	Lecture Visual Presentation, Group Discussion, Assignment, Exercise etc	Quiz, Short Essay, Q/A method, Assignment, Oral Presentation
b	Mention Characteristics and Limitations of MIS, Classify MIS			
c	Describe importance, Potential risks of information systems			
d	Differentiate between data and information, Quality of good information			
e	Explain the process of converting data into information.			
a	Define Information Systems	System concepts-Information systems-Information technology-Differences between IS and IT-The expanding role of	Lecture Visual Presentation, Group Discussion,	Quiz, Short Essay, Q/A method, Assignment, Oral Presentation
b	Differentiate between IS and IT			
c	Explain activities of information systems,			

	Information systems resources	information systems-Activities of information systems-Information systems resources-Classifications of information systems	Assignment, Exercise etc.	
d	Classify information systems			
a	Describe Information systems and organizational structure	Information systems and organizational structure-Roles of IT and IS in organizational decision making-A system approach to problem solving	Lecture Visual Presentation, Group Discussion, Assignment, Exercise etc.	Quiz, Short Essay, Q/A method, Assignment, Oral Presentation
b	Mention roles of IT and IS in organizational decision making			
c	Explain a system approach to problem solving			
a	Define Telecommunication, Information superhighway	Telecommunications-Information superhighway-Components of telecommunications network-Network, protocol-Types of telecommunication signal-Communication channel-	Lecture Visual Presentation, Group Discussion, Assignment, Exercise etc	Quiz, Short Essay, Q/A method, Assignment, Oral Presentation
b	Describe components of telecommunications network-Network, protocol-Types of telecommunication signal			

c	Define Communication channel, Communication processor, Communication software, Network topology, Communication channel characteristics	Communication processor-Communication software-Network topology-Communication channel characteristics-Types of telecommunication network-Telecommunication carriers-Business value of telecommunications		
d	Classify and describe telecommunication network, Telecommunication carriers, Business value of telecommunications			
a	Define management, Mention roles of management information	Information and the roles of management information and the levels of management-Information and decision making-Decision support systems-Different types of business problems-Different types of managerial decision making-Functions, Components and Applications of DSS-	Lecture Visual Presentation, Group Discussion, Assignment, Exercise etc.	Quiz, Short Essay, Q/A method, Assignment, Oral Presentation
b	Describe the levels of management, Information and decision making			
c	Explain types of managerial decision making, Functions, Components and Applications of DSS, GDSS			

d	Executive information systems, functions of an EIS, Rationale for EIS, Characteristics of DSS and EIS	GDSS-Executive information systems-functions of an EIS-Rationale for EIS-Characteristics of DSS and EIS-Critical success factors for DSS/EIS-Difference between TPS, MIS, DSS and EIS.		
e	Describe critical success factors for DSS/EIS-Difference between TPS, MIS, DSS and EIS			
a	Define Information systems in business, Marketing information systems, Manufacturing information systems	Information systems in business-Marketing information systems-Manufacturing information systems-Human resource information system-Accounting information systems-Financial information systems-Strategic information system. Information systems security and control: Computer security-	Lecture Visual Presentation, Group Discussion, Assignment, Exercise etc.	Quiz, Short Essay, Q/A method, Assignment, Oral Presentation
b	Describe Human resource information system, Accounting information systems, Financial information systems, Strategic information system			

c	Define computer security, Reasons of computer systems vulnerability, Types of computer security breaches, Security controls, Audit of information systems	Reasons of computer systems vulnerability-Types of computer security breaches-Security controls-Audit of information systems.		

Text Books:

1.James A. O'Brien : Management Information Systems: Managing Information Technology in the Networked Enterprise

2.Kenneth C. Laudon and Jane P. Laudon: Management Information Systems: Organization and Technology

3.Kenneth C. Laudon and Jane P. Laudon: Management Information Systems: Managing the Digital Firm

4.Uma G. Gupta: Management Information Systems: A Managerial Perspective

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Project/Thesis

Credits: 3.00 **Contact Hours:** 3 hours/week

Continuation of project/thesis topic is undertaken in CSE 4000.

NB: The Courses CSE 4000 of 1st semester and CSE 4000 2nd semester should be evaluated combined after completing CSE 4000 of 2nd semester

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Digital Signal and Image Processing

Part A- Introduction

I. **Course code: CSE 4201**

II. **Credit: 3**

1. **Course Summary**

This course presents the fundamentals of digital image processing. It covers principles and algorithms for processing images. Topics include data acquisition, imaging, filtering, coding, feature extraction, and modeling. The focus of the course is a series of labs that provide practical experience in processing physiological data, with examples from cardiology, speech processing, and medical imaging. The labs are done on the MIT Server in MATLAB® during weekly lab sessions that take place in an electronic classroom. Lectures cover image processing topics relevant to the lab exercise.

2. **Course Objectives**

1. To introduce students to basic image processing techniques.
2. To develop the students mathematical, scientific, and computational skills relevant to the field of digital image processing.
3. To enhance the students ability in formulating problems and designing analysis tools for digital images.
4. To foster effective interaction skills and teamwork communication.

3. **Course Learning Outcomes**

CLO1. Illustrate the concepts of digital images
CLO2. Illustrate the concepts of digital image representations
CLO3. Apply transformation technologies in image
CLO4. Identify the application of Image processing
CLO5. Application of image processing in computer vision

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2, 3, 4	Introduction; Point operations; Line and Edge detection, labeling, Histograms; Spatial operations; Affine transformations; Image Segmentation, Image Representation and Modelling, Image rectification; Interpolation and other transformations; Contrast enhancement; Convolution operation, Magnification and Zooming; Fourier transform; Edge detection; Boundary extraction and representation; Mathematical morphology.	Explain histograms Discuss spatial operations Illustrate image segmentation, image representation ,Image rectification, Image interpolation and other transformations Illustrate edge detection,boundary extraction and representation.	CLO1	Lecture	Quiz
Week 5, 6, 7	Point processing; Histogram Processing – Normalization, Matching, Equalization; Average Filter, Weighted Average Filter, Median Filter, Gaussian Filters; 1st and 2ndDerivative, Laplacian; Sobel operator	Describe, apply and analyze different image filtering techniques and image enhancement operators and algorithms in the spatial domain	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 8, 9, 10	Frequency domain of an image; Fourier transform, 1D and 2D Discrete Fourier transform, smoothing frequency domain filter - Ideal, Butterworth and Gaussian low pass filters; Sharpening frequency domain filter - Ideal, Butterworth and Gaussian high pass filters;	Describe, apply and analyze different image filtering techniques and image enhancement operators and algorithms in the frequency domain	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11, 12, 13, 14	Pattern Recognition: Statistical, Structural, Neural and Hybrid Techniques, Document Analysis and Optical Character Recognition, Object Recognition, Scene Matching and Analysis.	Discuss different types of pattern recognition techniques Define optical character recognition Explain scene matching and analysis	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3 hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10

Create	12

SEE-Semester End Examination (72 Marks)

	Quiz (10%)	Assignment (8%)	Practical (10%)	Final (72%)
CLO1	1	0	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO 1	1	1	0	0	0	0	0	1	0
CLO 2	1	1	0	0	0	0	0	1	1
CLO 3	1	1	0	0	0	0	0	1	1
CLO 4	1	1	0	0	0	0	0	1	1
CLO 5	1	1	0	0	0	0	0	1	1

<u>Part D- Resources</u>

Textbooks

1. R. A. Plastock& G. Kalley :"Theory and Problems of Computer Graphics"
2. Gonzalez :"Pattern Recognition Principles"

Reference Books

1. Steven Harrington :"Computer Graphics : A Programming Approach"
2. NewmannSprocell :"Principles of Interactive Computer Graphics"

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Digital Signal and Image Processing Lab

Part A- Introduction

I. **Course code: CSE 4202**

II. **Credit: 1.5**

1. Course Summary

This course presents the fundamentals of digital image processing. It covers principles and algorithms for processing images. Topics include data acquisition, imaging, filtering, coding, feature extraction, and modeling. The focus of the course is a series of labs that provide practical experience in processing physiological data, with examples from cardiology, speech processing, and medical imaging. The labs are done on the MIT Server in MATLAB® during weekly lab sessions that take place in an electronic classroom. Lectures cover image processing topics relevant to the lab exercise.

2. Course Objectives

1. To introduce students to basic image processing techniques.
2. To develop the students mathematical, scientific, and computational skills relevant to the field of digital image processing.
3. To enhance the students ability in formulating problems and designing analysis tools for digital images.
4. To foster effective interaction skills and teamwork communication.

3. Course Learning Outcomes

CLO1. Implement the concepts of digital images
CLO2. Experiment the concepts of digital image representations
CLO3. Apply transformation technologies in image
CLO4. Install the application of Image processing
CLO5. Experiment of image processing in computer vision

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2, 3, 4	Introduction; Point operations; Line and Edge detection, labeling, Histograms; Spatial operations; Affine transformations; Image Segmentation, Image Representation and Modelling, Image rectification; Interpolation and other transformations; Contrast enhancement; Convolution operation, Magnification and Zooming; Fourier transform; Edge detection; Boundary extraction and representation; Mathematical morphology.	Explain histograms Discuss spatial operations Illustrate image segmentation, image representation ,Image rectification, Image interpolation and other transformations Illustrate edge detection,boundary extraction and representation.	CLO1	Lecture	Quiz
Week 5, 6, 7	Point processing; Histogram Processing – Normalization, Matching, Equalization; Average Filter, Weighted Average Filter, Median Filter, Gaussian Filters; 1st and 2ndDerivative, Laplacian; Sobel operator	Describe, apply and analyze different image filtering techniques and image enhancement operators and algorithms in the spatial domain	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 8, 9, 10	Frequency domain of an image; Fourier transform, 1D and 2D Discrete Fourier transform, smoothing frequency domain filter - Ideal, Butterworth and Gaussian low pass filters; Sharpening frequency domain filter - Ideal, Butterworth and Gaussian high pass filters;	Describe, apply and analyze different image filtering techniques and image enhancement operators and algorithms in the frequency domain	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11, 12, 13, 14	Pattern Recognition: Statistical, Structural, Neural and Hybrid Techniques, Document Analysis and Optical Character Recognition, Object Recognition, Scene Matching	Discuss different types of pattern recognition techniques Define optical character recognition Explain scene matching	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	and Analysis.	and analysis			

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3 hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Quiz (10%)	**Assignment (8%)**	**Practical (10%)**	**Final (72%)**
CLO1	1	0	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO10**	**PLO11**	**PLO12**	**PLO13**	**PLO14**	**PLO15**	**PLO16**	**PLO17**	**PLO18**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. R. A. Plastock& G. Kalley :"Theory and Problems of Computer Graphics"
2. Gonzalez :"Pattern Recognition Principles"

Reference Books

1. Steven Harrington :"Computer Graphics : A Programming Approach"
2. NewmannSprocell :"Principles of Interactive Computer Graphics"

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

<u>Course Outline: Information Control and Cyber Security</u>

Part A- Introduction

I. **Course code: CSE4221**

II. **Credit: 3**

1. Course Summary

The accelerated expansion of security issues in computing products means students need to learn the basics of information security, in both management and technical aspects. Students must understand various types of security incidents and attacks, and learn methods to prevent, detect and react to incidents and attacks. Students also need to learn the basics of application of cryptography which are one of the key technologies to implement security functions.

2. Course Objectives

1. Explain various Information security threats and controls for it
2. Analyze security incidents and design countermeasures
3. Apply information security incident response
4. Implement the algorithms of common key cryptography and public key cryptography
5. Evaluate the mechanism to protect confidentiality and completeness of data

3. Course Learning Outcomes

CLO1. Articulate the concepts of Information security
CLO2. Detect the security threats in cyber spaces
CLO3. Detect the security threats in devices
CLO4. Evaluate risk managements of the systems
CLO5. Employ security preserving solutions in the information system

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Information Security, Examples of Information Security Incidents, Information Security Management	Understand the goal and scope of this course through several examples of security incidents.	CLO1	Lecture	Quiz
Week 2	1. The three concepts of Information Security (Confidentiality, Integrity, Availability) 2. Basic terminologies in Information Security 3. Human Aspect of Information Security 4. Social Engineering	Learn the three concepts of information security and other basic concepts.	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 3	1. Attacks to Server Systems connected to the Internet and countermeasures 2. Attacks to Web Servers and countermeasure 3. Denial of Service Attack 4. Attacks to Network Systems	Explain Security Attacks for Server systems and discuss counter measure for attacks	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5	1. Attacks for Personal Computers and Smartphones, and countermeasure 2. How the malicious software intrude the device 3. What the malicious software does to the system	Explain Information Security for Client devices	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

	4. Stolen and Lost Devices				
Week 6	1. What is Risk Management process 2. Identifying Information Assets 3. Identifying Security Risk and evaluation 4. Risk Treatment	Learn Risk Management process for Information Systems	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8,9	1. Information Security Governance 2. Information Security Management System (ISMS) 3. Information Security Policy, Standards and Procedures 4. Information Security Evaluation	Learn how an organization manages security risk, including, establishing policy, building organization and internal rules.	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 10	1. What is Security Incident response 2. Computer Security Incident response team 3. Incident response exercise	Learn organization to handle security incidents and how to react to security incidents through exercise	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11	1. Requirements for Secure Communication 2. What is Cryptography? 3. Classic Cryptography 4. Modern Cryptography	Learn about basic concept of cryptography	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 12, 13	1. Common Key Cryptography algorithms: DES, Triple DES, AES 2. Encryption modes 3. Exercise on Common Key Cryptography	Apply Common Key Cryptography	CLO1, CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Week 14	1. Exercise of Public Key Cryptography 2. Exercise of Hybrid encryption	Apply Public Key Cryptography and Hybrid encryption.	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
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Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
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Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

	Quiz (10%)	**Assignment (8%)**	**Practical (10%)**	**Final (72%)**
CLO1	1	0	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1

CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Teach yourself C by Herbert Schildt
2. C: The Complete Reference by Herbert Schildt

Reference Books

1. Computer fundamentals and programming by Reema Thareja
2. C in a Nutshell: The Definitive Reference by Peter Prinz
3. Pointer in C Y. Kanitkar
4. Let us C by Y. Kanitkar
5. The C programming language by Kernighan & Ritchie

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Fault Tolerant Analysis

Part A- Introduction

I. **Course code: CSE4223**

II. **Credit: 3**

1. Course Summary

Fault-tolerant systems are used in applications that require high dependability, such as safety-critical control systems in vehicles and airplanes, or business-critical systems for e-commerce, automatic teller machines and financial transactions. This is an introductory course that covers basic techniques for design and analysis of fault-tolerant systems, as well as project management and development processes for safety-critical systems. The course covers techniques for tolerating hardware and software faults, analysis of fault-tolerant systems, project management and development processes for safety-critical systems.

2. Course Objectives

1. Learn the basics of fault-tolerance in a computer system
2. Devise the attributes of fault tolerance
3. Identify different systems to examine
4. Learn the designs of fault tolerance

3. Course Learning Outcomes

CLO1. Adopt the basics of Fault Tolerant System (FLS)
CLO2. Examine the attributes of FLS
CLO3. Explore the countables of FLS
CLO4. Design fault tolerances of the systems

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Definition of fault tolerance, Redundancy, Applications of fault-tolerance, Fundamentals of dependability	Understand the fault tolerance basics and applications	CLO1	Lecture	Quiz Assignment Exam
Week 2, 3, 4	Reliability, availability, safety, Impairments: faults, errors and failures, Means: fault prevention, removal and forecasting	Understand fault tolerant system attributes like reliability, safety, errors etc	CLO2	Lecture Exercise Demonstration	Quiz Assignment Project Task Exam
Week 5, 6	Common measures: failures rate, mean time to failure, mean time to repair, etc. Reliability block diagrams, Markov processes	Ability to design and develop dependable systems for mission critical applications	CLO3	Lecture Exercise Demonstration	Quiz Assignment Exam
Week 7, 8	Canonical and Resilient Structures; Reliability Evaluation Techniques and Models; Processor-level Fault Tolerance; Byzantine Failures and Agreements.	Ability to know about the faults of various systems	CLO1 CLO2	Lecture Exercise Demonstration	Quiz Exam
Week 9, 10, 11	Error Detection/Correction Codes (Hamming, Parity, Checksum, Berger, Cyclic, Arithmetic); Encoding/ Decoding circuits; Resilient Disk Systems (RAID) Various checkpoints and shared memory systems.	Ability to know about different types of codes	CLO3	Lecture Exercise Demonstration	Quiz Exam
Week 12,13	Fault tolerant circuit design: Adder, subtractor, multiplicator etc. Defect-tolerance in VLSI Designs; Fault Detection in Cryptographic Systems.	Apply Fault Tolerant in various design	CLO4	Lecture Exercise Demonstration	Project Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Project	Evaluate the systems using gained knowledge from the course
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

M**apping of CLOs to Assessment**

	Participation (8%)	**Quiz (5%)**	**Assignment (5%)**	**Project (10%)**	**Final (72%)**
CLO1	1	1	0	1	1
CLO2	1	1	1	1	1
CLO3	1	1	1	1	1
CLO4	1	1	1	1	1

Mapping of CLOs to PLOs

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	0	1	0	0	0	0	0	1	0
CLO2	0	1	0	0	0	0	0	1	0
CLO3	0	1	0	0	0	0	0	1	0
CLO4	0	1	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Fault-Tolerant Systems by Israel Koren
2. Fault-Tolerant Design by Elena Dubrova

Reference Books

1. Reliability of Computer Systems and Networks: Fault Tolerance, Analysis, and Design by Martin L. Shooman

Jashore University of Science and Technology
Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Introduction to DNA Computing

Part A- Introduction

I. **Course code: CSE4225**

II. **Credit: 3**

1. **Course Summary**

This course is designed to introduce bioinformatics tools and analysis methods. Upon completion of the course, students should be more comfortable working with the vast amounts of biomedical and genomic data and online tools that will be relevant to their work in future.

2. **Course Objectives**

1. Define computational genomics and phylogenetics concepts.
2. Apply common bioinformatics tools and techniques effectively.
3. Implement basic algorithms such as sequence alignment.
4. Perform independent genome comparisons and assemblies using gained knowledge.
5. Evaluate on your own the promise and challenges for computing on biological datasets.

3. **Course Learning Outcomes**

CLO1. Introduce basic bioinformatics
CLO2. Discover chain and sequences
CLO3. Perform genome comparison
CLO4. Validate genome assembly

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	DNA, genes and the genome.	Explain introductory concepts of bioinformatics	CLO1	Lecture	Quiz Assignment Exam
Week 2	Sequence statistics.	Knowledge about Sequence statistics	CLO2	Lecture Exercise Demonstration	Quiz Assignment Project Task Exam
Week 3, 4	Sequence alignment.	Implement basic algorithms on Sequence alignment	CLO2	Lecture Exercise Demonstration	Quiz Assignment Exam
Week 5	Variation and natural selection	Learn about variation and natural selection	CLO2	Lecture Exercise Demonstration	Quiz Exam
Week 6	Hidden Markov Models	Learn and apply Hidden Markov Models	CLO2	Lecture Exercise Demonstration	Quiz Exam
Week 7, 8	Ab initio gene finding	Learn gene finding	CLO3	Lecture Exercise Demonstration	Quiz Exam
Week 9, 10	Whole genome comparisons	Perform independent genome comparisons and assemblies using	CLO3	Lecture Exercise Demonstration	Quiz Exam

		gained knowledge			
Week 11, 12	Genome assembly and validation	Learn about Genome assembly and validation	CLO4	Lecture Exercise Demonstration	Quiz Exam
Week 13, 14	Phylogenetic analysis	Define phylogenetics concepts	CLO3	Lecture Exercise Demonstration	Quiz Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Participation **(8%)**	**Quiz** **(20%)**	**Final** **(72%)**
CLO1	1	1	1
CLO2	1	1	1
CLO3	1	1	1

CLO4	1	1	1

Mapping of CLOs to PLOs

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**

CLO1	0	1	0	0	0	0	0	1	0
CLO2	0	1	0	0	0	0	0	1	0
CLO3	0	1	0	0	0	0	0	1	0
CLO4	0	1	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Statistical Methods in Bioinformatics by W. Ewens and G. Grant
2. Neural Networks and Genome Informatics by C.H. Wu

Reference Books

1. Bioinformatics; The Machine Learning Approach by Pierre Baldi and SÿrenBrunak

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Software Project Management and Maintenance

Part A- Introduction

I. **Course code: CSE4227**

II. **Credit: 3**

1. **Course Summary**

The course provides an in depth examination of project management principles and modern software project management practices. Methods for managing and optimizing the software development process are discussed along with techniques for performing each phase of the systems development lifecycle. Portfolio management and the use and application of software project management tools are also discussed.

2. **Course Objectives**

1. Understand the effectively strategies of testing, the methods and technologies of software testing;
2. Design test plan and test cases;
3. Do automatic testing;
4. Establish a testing group and manage the whole testing project;
5. Clearly and correctly report the software defectives;
6. Asses the software product correctly;
7. Distinguish relationship between the software testing and the quality assurance.

3. Course Learning Outcomes

CLO1. Devise the basics of Project Management (PM)
CLO2. Develop project management
CLO3. Assess risk of projects
CLO4. Manage delivery of the system

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction of software project management	Learn basics of project management	CLO1	Lecture	Quiz Exam Project
Week 2, 3	Business Case Development, Project Initiation and Process Management	Learn project development life cycle	CLO1	Lecture Exercise Demonstrat ion	Quiz Exam Project
Week 4	Software Project Management Methodology and planning	Implement basic algorithms on Sequence alignment	CLO2	Lecture Exercise Demonstrat ion	Quiz Exam Project
Week 5, 6	Work Activities and the Project Schedule and estimation	Design schedule	CLO2	Lecture Exercise Demonstrat ion	Quiz Exam Project
Week 7, 8	Project risk assessment	Build risk matrix	CLO3	Lecture Exercise Demonstrat ion	Quiz Exam Project
Week 9, 10, 11	Project release and maintenance	Learn delivery	CLO4	Lecture Exercise Demonstrat ion	Quiz Exam Project
Week 12, 13, 14	Change request management	Assess middleware changes	CLO2	Lecture Exercise Demonstrat ion	Quiz Exam Project

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Project	On-hand exercise to learn PM by developing a full phase project
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Participation (8%)	**Quiz (10%)**	**Project (10)**	**Final (72%)**
CLO1	1	1	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1

Mapping of CLOs to PLOs

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			

Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	0	1	0	1	0	1	1	0	1
CLO2	0	1	0	1	0	1	1	0	1
CLO3	0	1	0	1	0	1	1	0	1
CLO4	0	1	0	1	0	1	1	0	1

Part D- Resources

Textbooks

1. A Brief Guide to MS Project 2010 by Kathy Scwhalbe
2. PMP in Depth, 2nd Edition by Paul Sanghera
3. Microsoft Project Standard

Reference Books

1. Leadership: Theory and Practice 5th Edition by Peter Guy Northouse

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Distributed System and Cloud Computing

Part A- Introduction

I. **Course code: CSE4229**

II. **Credit: 3**

1. Course Summary

The syllabus below describes a recent offering of the course, but it may not be completely up to date. For current details about this course, please contact the course coordinator. Course coordinators are listed on the course listing for undergraduate courses and graduate courses.

2. Course Objectives

1. To give students a basic grounding in designing and implementing distributed and cloud systems.
2. The global consensus and Paxos, and their application in building cloud systems
3. The advantages and disadvantages of using distributed NoSQL stores
4. The "CAP Theorem," and its implication for building highly available services
5. Roles of REST, Websockets and stream processing in cloud applications
6. Tools and principles of building distributed and cloud applications
7. Cloud support for batch processing, such as the Hadoop and Pig frameworks, and their use with NoSQL data stores such as Cassandra.

3. Course Learning Outcomes

CLO1. Devise the basics of cloud computing and distributed process
CLO2. Learn architecture of cloud system
CLO3. Implement algorithms into cloud
CLO4. Examining cloud and distributed database

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2	Introduction to distributed systems and cloud computing. Cloud architectures: SaaS, PaaS, IaaS. End-to-end system design. Networks and protocol stacks.	Learn basics of distributed system	CLO1	Lecture Exercise Demonstration Handout	Quiz Exam Project
Week 3, 4	Distributed file systems and cache consistency. NFS, AFS. Storage in the Cloud: Google/ Hadoop file system.	Experimenting cloud file system	CLO4	Lecture Exercise Demonstration Handout	Quiz Exam Project
Week 5	Web services and REST. Example: Amazon S3. The JAX-RS API. Persistent cloud services. Failure models and failure detectors.	Understanding requests	CLO2	Lecture Exercise Demonstration Handout	Quiz Exam Project
Week 6, 7, 8	Batch cloud computing: map-reduce and Hadoop. Domain-specific languages for cloud data processing: Pig and Hive.	Implement cloud data processing	CLO2	Lecture Exercise Demonstration Handout	Quiz Exam Project
Week 9, 10, 11	Transactions. Atomic commitment protocols: 2PC and 3PC, Highly available services. Replicated services and quorum consensus. The CAP Theorem.	Executing CAP theorem	CLO3	Lecture Exercise Demonstration Handout	Quiz Exam Project
Week 12, 13, 14	NoSQL data stores. Table-based (Google BigTable), key-based (Amazon Dynamo), and Cassandra. The Hector API. Query processing with Map-reduce.	Discover NoSQL	CLO4	Lecture Exercise Demonstration Handout	Quiz Exam Project

P**art C- Assessment and Evaluations**

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Project	On-hand exercise to learn PM by developing a full phase project
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Participation (8%)	**Quiz (10%)**	**Project (10)**	**Final (72%)**
CLO1	1	1	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1

Mapping of CLOs to PLOs

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			

Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	0	0	1	0	1	0	1
CLO2	1	1	0	0	1	0	1	0	1
CLO3	1	1	0	0	1	0	1	0	1
CLO4	1	1	0	0	1	0	1	0	1

Part D- Resources

Textbooks

1. Distributed and Cloud Computing From Parallel Processing to the Internet of Things by Kai Hwang, Jack Dongarra, Geoffrey Fox
2. Cloud Computing: Principles and Paradigms by Rajkumar Buyya, James Broberg, Andrzej M. Goscinski
3. Distributed Computing: Principles, Algorithms, and Systems by Ajay D. Kshemkalyani and Mukesh Singhal
4. Distributed Computing: Fundamentals, Simulations and Advanced Topics by Hagit Attiya and Jennifer Welch

Reference Books

1. Distributed Algorithms by Nancy Lynch
2. Cloud Computing Bible by Barrie Sosinsky
3. Cloud Computing: Principles, Systems and Applications by Nikos Antonopoulos, Lee Gillam

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Parallel and Vector Computing

Part A- Introduction

I. **Course code: CSE4231**

II. **Credit: 3**

1. Course Summary

To understand the architectures for parallel processing and to learn the concepts of pipelining and multithreading

2. Course Objectives

1. Introduce to the basic concepts of parallel processing
2. Multithread program and their applications
3. Instructions and their properties
4. introduce students to the fundamentals of vector and tensor algebra
5. Vector computation and their advantages over traditional programs
6. Develop algorithms of parallel processing

3. Course Learning Outcomes

CLO1. To understand the architectures for parallel processing.
CLO2. To learn the concepts of parallelism and Data Flow Machine
CLO3. Understanding the concepts of instruction level of parallelism
CLO4. Understanding the parallel algorithms

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1, 2	Parallel computer models- the state of computing, Multiprocessors and multicomputers and Multivectors and SIMD computers, PRAM and VLSI models, Architecture development tracks, Program and network properties.	Learn basics of parallel processing	CLO1	Lecture Exercise Demonstration	Quiz Exam Project
Week 3, 4, 5	Conditions of parallelism, Program partitioning and scheduling, Program flow mechanisms, system interconnect architectures. Principles of scalable performance, performance metrics and Measures, Data Flow Machine Language- Architecture of Data Flow Machines.	Preparing parallel system	CLO2	Lecture Exercise Demonstration	Quiz Exam Project
Week 6, 7	Processor and memory hierarchy- advanced processor technology, superscalar and vector processors, memory hierarchy technology, virtual memory technology, bus cache and shared memory, backplane bus systems, cache memory organizations, shared memory Organizations,sequential and weak consistency models.	Understanding the vector memory	CLO2	Lecture Exercise Demonstration	Quiz Exam Project

Week 8, 9, 10	Pipelining in processing elements- delays in Pipeline execution- difficulties in PipeliningSuperscalar Processors- Vector Processor _ Very Long Instruction Word Processor (VLIW)- Commercial Processor- Power PC 620 RISC Processor- Two Instruction Superscalar RISCProcessor- Multithreaded Processors- Future Processor Architecture- Trace Processor, MultiscalarProcessor, Superflow Architecture.	Design instruction	CLO3	Lecture Exercise Demonstration	Quiz Exam Project
Week 11, 12	Classification of Parallel Algorithms: Synchronized and Asynchronized parallel algorithms,Performance of Parallel algorithms- Elementary parallel algorithms: Searching, Sorting, Matrix Operations	Simulating parallel algorithms	CLO4	Lecture Exercise Demonstration	Quiz Exam Project
Week 13, 14	Vector in three non-coplanar directions, vector in two non-coplanar directions, magnitude of vector, addition and subtraction of vectors, scalar multiplication and projection, Tensor Analysis	Learn tensor and vector analysis	CLO3	Lecture Exercise Demonstration	Quiz Exam Project

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Project	On-hand exercise to learn PM by developing a full phase project
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Participation (8%)	**Quiz (10%)**	**Project (10)**	**Final (72%)**
CLO1	1	1	1	1
CLO2	1	1	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1

Mapping of CLOs to PLOs

Bloom's Category Marks (out of 20)	**Test (10)**	**Assignment (5)**	**Quizzes (5)**	**External Participation in Curricular/Co-Curricular Activities**
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			

Create	4			

CIE- Continuous Internal Evaluation (28 Marks):

SEE-Semester End Examination (72 Marks)

Bloom's Category	**Test**
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	0	0	0	0	0	1	1
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Advanced Computer Architecture by Kai Hwang Tata
2. Computer Architecture and Parallel Processing by Kai Hwang and FA Briggs
3. Advanced Engineering Mathematics by Palgrave Macmillan

Reference Books

1. Advanced Vector Analysis with Application to Mathematical Physics by Weatherburn C. E

Jashore University of Science and Technology

Faculty of Science and Engineering

Department of Computer Science and Engineering

Course Outline: Cryptography

Part A- Introduction

I. Course code: CSE 4233	
II. Credit: 3	

1. Course Summary	
Security mechanism is designed to detect, prevent or recover from a security attack and the security service enhances the security of data processing systems and information transfers. A security service makes use of one or more security mechanisms. Threat analysis is a process where all possible threats to a system are identified. It is an important aid for defining the security policy. Security policy is a set of rules stating what is permitted and what is not permitted in a system during normal operation. It is written in general terms and describes the security requirements for a system. Security mechanisms implement functions that prevent, detect and respond to recovery from security attacks. Cryptography underlies many security mechanisms. At the end of each chapter, the assignment is compulsory to make students involved to use knowledge of this course to implement programs and solve practical problems which they will use for their research study in the future.	
2. Course Objectives	
1. The objective of this course is to give students the idea of Information Security and its components. 2. Analyze security incidents and design countermeasures. 3. Apply information security incident response 4. Implement the algorithms of Common Key cryptography and Public Key cryptography. 5. Evaluate the mechanism to protect confidentiality and completeness of data.	
3. Course Learning Outcomes	

CLO1. Understand internet, information security, network security, vulnerability, threat, brute-force attack, stream cipher, block cipher, DES, AES, Merkle-Hellman Cryptosystem, RSA cryptosystem, El Gamal Algorithm, Cryptographic Hash Function, MD5 algorithm, Message Authentication Code (MAC), key management, authentication application, Kerberos, network access control, cloud computing.
CLO2. Analyze attack, control, services, mechanisms, security services, Cryptanalysis, symmetric cryptosystem, asymmetric cryptosystem, signature verification, properties and application of hash function, types of MACs, HMAC, wireless security and pretty good privacy.
CLO3. Remember computer security, internet security, security goals, how to make a system trustworthy, distribution of public keys, public key certificate and secret key distribution.
CLO4. Apply cryptography, key management, Diffie-Hellman Key Exchange.
CLO5 Evaluate the security life cycle, digital signature.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction	-acquire knowledge about internet, security, threat know about different types of controls, services, mechanisms and attacks learn about security goals	CLO1, CLO2, CLO3	Lecture	Quiz

Week 2, 3	Cryptography	-students will understand about cryptography in details -acquire knowledge about symmetric cryptography and asymmetric cryptography -Common Key Cryptography algorithms: DES, Triple DES, AES - Encryption modes -Exercise on Common Key Cryptography -Exercise of Public Key Cryptography -Exercise of Hybrid encryption -illustration on Diffie-Hellman key exchange	CLO1, CLO2, CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 4, 5	Explain Security Attacks for Server systems and discuss counter measure for attacks Explain Information Security for Client devices	-Attacks to Server Systems connected to the Internet and countermeasures -Attacks to Web Servers and countermeasure -Denial of Service Attack -Attacks to Network Systems -Attacks for Personal Computers and Smartphones, and countermeasure -How the malicious software intrudes the device -What the malicious software does to the system -Stolen and Lost Devices	CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 6	Digital Signature	students will learn about digital signature -learn about signature verification and encrypted signature -acquire knowledge about the algorithm used in digital signature	CLO1, CLO2, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 7, 8	Hash Function	-discuss on cryptographic hash function -discuss on the properties and application of hash function - students will understand how to check integrity and authenticity using has function -students will learn cryptographic hash algorithm	CLO1, CLO2	Lecture, Exercise Demonstr ation	Quiz, Assignment, Exam
Week 9	Message Authentic ation Code (MAC)	-students will learn about message authentication code -acquire knowledge about how to check integrity and authenticity using MAC -students will know about different types of MAC including HMAC (Hash function based MAC)	CLO1, CLO2	Lecture, Exercise Demonstr ation	Quiz Assignment Exam
Week 10	User Authentic ation	-understand remote user authentication principles -students will know about Kerberos, remote user authentication using symmetric encryption and asymmetric encryption	CLO2, CLO3	Lecture, Exercise Demonstr ation	Quiz, Assignment, Exam

Week 11, 12	Network Access Control and Cloud Security	-student will know about network access control -acquire knowledge about cloud computing and cloud security as a service -learn about authentication protocol	CLO1, CLO2	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 13, 14	Wireless Network Security, Electronic Mail Security	-Understand wireless security, mobile device security and wireless LAN security -acquire knowledge about Email security -understand Pretty Good Privacy (PGP)	CLO1, CLO2	Lecture, Exercise Demonstration	Quiz, Assignment Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	**Description**
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Attendance	Student’s participation in the class lecture, quiz and exam
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students

	need to answer at most 6 questions. Every question may include sub questions

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to Assessment

	Quiz (10%)	Assignment (10%)	Attendance (8%)	Exam (72%)
CLO1	0	1	1	1

CLO2	1	0	1	1
CLO3	1	1	1	1
CLO4	1	1	1	1
CLO5	1	1	1	1

Mapping of CLOs to PLOs

CLOs	**PLO1**	**PLO2**	**PLO3**	**PLO4**	**PLO5**	**PLO6**	**PLO7**	**PLO8**	**PLO9**
CLO1	1	1	1	0	0	0	0	1	0
CLO2	1	1	1	0	0	0	0	1	0
CLO3	1	1	1	0	0	0	0	1	0
CLO4	1	1	1	0	0	0	0	1	0
CLO5	1	1	1	0	0	0	0	1	0

Part D- Resources

Textbooks

1. *William Stallings: Cryptography and Network Security, Principles and Practice*
2. *Charles P. Pfleeger: Security in Computing*
3. Christof Paa and Jan Pelzl: To understand cryptography in depth

Reference Books

1. *Dhillon: Managing Information System Security*
2. Michael E. Whitman and Herbert J. Mattord: Principles of Information Security

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Internship

Part A- Introduction

I. **Course code: CSE 4210**

II. **Credit: 3.0**

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Industrial Tour

Part A- Introduction

I. **Course code: CSE 4212**

II. **Credit: 1.0**

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Thesis/Project

Part A- Introduction

I. **Course code: CSE 4000**

II. **Credit: 3.0**

NB: The Courses CSE 4000 of 1st semester and CSE 4000 2nd semester should be evaluated combined after completing CSE 4000 of 2nd semester

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Viva Voce

Part A- Introduction

I. **Course code: CSE 4200**

II. **Credit: 1.0**

Part - D

Grading Policy

Grading Scale, Grades, Grade Point Average:

Each course, irrespective of the credit hours attributed to it, will be graded at a scale of 4.00 (four). Initially the courses will be assessed in 100 to calculate percentage of marks obtained by the students. Letter Grades and corresponding Grade Points are awarded to the students (B.Sc.) in accordance with provision shown below in Table 1.

T**able 1.** The grading system consists of Letter Grading Point Average (GPA), Letter Grade, corresponding Grade Point of JUST of Science and Technology.

Numerical Grade	Letter Grade	Grade Point
80% and above	A+	4.00
75% to less than 80%	A	3.75
70% to less than 75%	A-	3.50
65% to less than 70%	B+	3.25
60% to less than 65%	B	3.00
55% to less than 60%	B-	2.75
50% to less than 55%	C+	2.50
45% to less than 50%	C	2.25
40% to less than 45%	C-	2.00
Less than 40%	F	0.00

*Source: Controller Section of JUST

1. **GPA:** Grade Point Average (GPA) is the weighted average of the grade points obtained in all the courses completed by a student in a semester.
2. **CGPA:** Cumulative Grade Point Average (CGPA) will be calculated by the weighted average of previous CGPA and current GPA.
3. S**GPA:** Semester Grade Point Average (SGPA) is the weighted average of the grade points obtained in all the courses completed by a student in a semester.
4. **YGPA: Y**early Grade Point Average (YGPA) is the weighted average of the grade points obtained in all the courses completed by a student in a year (1st and 2nd semesters).
5. **F Grades:** If a student obtains an 'F' grade his grade will not be counted for GPA and s/he has to repeat the course. If the same course is not available then the Head of the department will assign an equivalent course. An 'F' grade will be in his/her record permanently and s/he will not be eligible for honors or Distinction.

Course Withdrawal

A student can withdraw a course by a written application to the Chairman of the department through his/her Advisor on or before the last date of instruction. The Chairman of the department will inform it to the Chairman of the Examination Committee and the Controller of Examinations. The Controller of Examinations will send the revised registration list to the department before the examination.

Withdrawal from a Semester

If a student is unable to complete the Semester Final Examination due to illness, accident or any other valid reason etc., he/she may apply to the Controller of Examination through his/her Chairman of the Department for total withdrawal from the semester within a week after the end of the Semester Final Examination. However, he/she may choose not to withdraw any laboratory/ sessional / design course if the grade obtained in such a course is 'D' or better and that he/she has to indicate clearly in his/her withdrawal application. The withdrawal application must be supported by a medical certificate from University Medical Officer. The Academic Council will take final decision about such an application.

Retake:

(a) If a student obtains "F" grade(s) in theory course(s), this grade(s) will not be counted for SGPA calculation and s/he can repeat the course(s), which will be termed as "RETAKE". When the student passes the RETAKE course(s) with grade higher than or equal to "A-", s/he will be awarded a grade of "A-". If s/he obtains grade(s) lower than "A-", s/he will be awarded her/his obtained grade(s). This RETAKE course (theory course) will be conducted as per the rules.

If a student obtains ‘F’ Grade in one or more courses, he/she can participate in re-exam through any of the following three systems:

i) retake with regular students by paying TK 2,000 per course

ii) Through special examination by paying TK 10,000 per course but the student shall have to apply within one week of the publication of results

iii) Using special semester after the results of the Final Semester by paying TK 5,000 for a single course and Tk.15,000 for more than one course.

(b) If a student obtains “F” grade in Sessional/Practical course, s/he will apply for RETAKE of that course to the Chairman of the concerned Department. The Chairman will take recommendation from the Academic Committee of the Department. After recommendation of those committees, the student will pay the required fees (TK 10,000/=) to the Bank (JUST branch). The Chairman will forward all these documents to the Controller of Examinations for course registration. Thus the student can appear for the sessional/practical examination. It is to be mentioned that a student will have the opportunity to take “RETAKE” in sessional/practical course only for two times in his/her academic tenure. Any student having "RETAKE" in theory and/or sessional/practical will not be considered for "Distinction".

(c) “F” grade(s) will be mentioned in the transcript unless the "Retake" course(s) has been passed. After passing "Retake" course(s), the indication of “F” grade(s) will be removed from the previous transcripts and in the fresh transcript ‘R’ will not be included along with the obtained grade(s) of that/those particular ‘Retake’ course(s).

Special Semester Examination (Incomplete course):

A special semester examination will be conducted for final year students only with RETAKE. This will be a non-taught semester (Theory /Sessional/Practical courses). The Chairman may allow a student to register for the SPECIAL SEMESTER credits. The office of the Controller of Examinations will attach a list of names of the unsuccessful students along with their RETAKE courses (from 1st year 1st semester to 4th year 2nd semester) during publishing the final result (i.e. 4th year 2nd semester). At the same time RETAKE course(s) registration notice will be announced by the office of the Controller of Examinations. This course(s) registration must be completed within a week after publishing the final result (4th year 2nd semester) and the examination of RETAKE courses should be started within 4 weeks. Once the examination begins, the examination will be completed within 25 days. The class attendance marks of 8% will be carried over from previously attended theory courses. On the basis of application of the student(s), the Academic Committee may consider him/her to take part to improve marks of Class Test / Assignment / Spot Test / Quiz etc. The special semester examination will carry the remaining 72% marks. The student will have to pay the fee as follows:

For theory course	: TK 5,000 for one course
For Sessional/Practical course	: TK 10,000 for each course

It is to be mentioned that a student will have the opportunity to appear in the Special Semester for subsequent two times in his/her academic tenure i.e. within 12 semesters from his/her first admission (first year first semester).

(e) Examination Committee for the Special Semester Examination: The existing 4th year Examination Committee will conduct the Special Semester Examination. In case of any vacancy, absence or inability on the part of any one of the members of the Examination Committee, the examination work shall not be invalidated. The Examination Committee will act according to the instructions mentioned in Sub-section 3.4.6 of this ordinance.